

1204

INTERNAL REPORT

USER MANUAL FOR THE
COMPUTER PROGRAM CAP

52406-47

1st OCTOBER, 1985

INTERNAL REPORT

USER MANUAL FOR THE
COMPUTER PROGRAM CAP

52406-47 1st OCTOBER, 1985



<u>CONTENTS</u>	<u>PAGE</u>
RUN DESCRIPTION	2
DESCRIPTION OF INPUT	2
PROGRAM "FLOW CHART" FOR CAP	9
SUBROUTINES	10
ALPHABETIC LIST OF SYMBOLS USED IN CAP	14
DRAWING - STANDARD INPUT FORM	17

A P P E N D I X II: EXAMPLES

A P P E N D I X IV: PARAMETER STUDY

A P P E N D I X V: COMPARISON STUDY

A P P E N D I X VI: STABILITY CALCULATIONS FOR OFFSHORE GRAVITY
STRUCTURES, OTC 1976

A P P E N D I X VII: A REVISED AND EXTENDED FORMULA FOR BEARING
CAPACITY, J. BRINCH HANSEN, 1968

A P P E N D I X VIII: SOME RECENT RESEARCH ON THE BEARING CAPACITY
OF FOUNDATIONS, G.G. MEYERHOF, 1963.

Prepared by Arne Kvitrud, Peter To and Rolf Lauritzsen

Norwegian Geotechnical Institute

Address:
P.O.Box 40 Tåsen
0801 Oslo 8, Norway

Telephone:
(02) 23 03 88

Telex:
19787 ngi n

Telegrams:
GEOTEKNIKK

RUN DESCRIPTION

The CAP program can be found on NGI's PRIME computer, using:

SEG PROGLIB>CAP

The program will ask for the names of the input data file and the output listing file.

The program will automatically open and close the necessary work files used.

DESCRIPTION OF INPUT

The standard input form is shown on Drawing 52406-20-1.

1. LINE (PROJECT) FORMAT 14A4, A1
01-57 - Heading of run, project, case, operator's name, etc.
2. LINE (UNITS USED) FORMAT 7A4, A2
01-30 - length and force units used. Any set of units may be chosen, but it is important that input values are given with consistent units. Output units will be the same as input units.
3. LINE (PLATFORM BASE GEOMETRY, ETC) B, AL, ZSS, ZSF, ZSSTEP, GAMS, GAMW, PS. FORMAT 8F10.0
01-10 B - Width of foundation, measured parallel to the direction of the horizontal force.
11-20 AL - Length of foundation, measured normal to the direction of the horizontal force.

- 21-30 ZSS - Depth of embedment, start value for ZS.
- 31-40 ZSF - Depth of embedment, final value for ZS.
- 41-50 ZSSTEP - Depth of embedment, step value for ZS. If calculations are performed for one value only, ZSSTEP = 0.0 and ZSS = ZSF. A maximum of 20 embedment depths can be analysed in each run.
- 51-60 GAMS - Total unit weight of soil (Force/length³)
- 61-70 GAMW - Unit weight of water (Force/length³)
- 71-80 PS - Selection of parameter study.
0 < PS < 1.0: A selection of variables will be increased and decreased by the percentage indicated by PS (PS = 0.1 means a variation of $\pm 10\%$). These variables are: Area (both B and AL are varied), depth of embedment (only one value should be given, see columns 21-50 above), horizontal force, moment, excess water pressure and load factor. Only one value of submerged weight is used, PVS. The only output made available for parameter study is "summary of results". If a regular run is wanted, PS should be left blank.

4. LINE (FORCES ON PLATFORM) PVS, PVF, PVSTEP, PH, AM, DELPH, DELPT, FAC. FORMAT 8F10.0

- 01-10 PVS - Submerged weight of structure, start value.
Includes weight of structure above mudline and possible environmental vertical forces on structure (multiplied by the appropriate load factor).
- 11-20 PVF - Submerged weight of structure, final value.
- 21-30 PVSTEP - Submerged weight of structure, step value.
If calculations are performed for one value only, PVSTEP = 0.0 and PVS = PVF. A maximum of 20 submerged weight values can be analysed in each run.
- 31-40 PH - Horizontal load at mudline. Value is disregarded

if NPR5 = 1.

- 41-50 AM - Overturning moment at mudline.
- 51-60 DELPH - Excess water pressure on sea floor adjacent to platform outside the heel, loward side.
- 61-70 DELPT - Excess water pressure on sea floor adjacent to platform outside the toe, leeward side.
- 71-80 FAC - Load factor(s) to be used for horizontal force (PH), moment (AM) and excess water pressure (DELPH, DELPT). Three options are available:
 FAC = 1.0, only load factor 1.0 will be used
 FAC > + 1.0, load factor 1.0 and the given
 value of FAC will be used
 FAC < - 1.0, only one load factor, | FAC |
 will be used

5. LINE (SLIP SURFACE PARAMETERS) ZMAX, ZSTEP, REDBS, REDSS, CRACK, PASTYP, ZSPEC1, ZSPEC2. FORMAT 8F10.0

- 01-10 ZMAX - The maximum depth from mudline to which the program should search for the minimum safety factor. The program will not accept any value of ZMAX greater than $(B \cdot \sin 45^\circ) + ZS$. If ZMAX is given a greater value it will be reduced to $(B \cdot \sin 45^\circ) + ZS$. The reason is that the slip surface angle ALPHA should be kept below 45° .
- 11-20 ZSTEP - The depth Z to the lowest point on the slip surface is varied in steps. The first depth is the depth of embedment, ZS, and the last ZMAX. A maximum of 200 slip surfaces can be analysed in each run.
- 21-30 REDBS - Reduction factor for side shear on embedded base area. The horizontal resistance of the side base area is calculated as the in-situ shear strength of the soil, SUD, times the side area times REDBS. Due to soil disturbance during the penetration phase, REDBS is nor-

mally chosen to be 0.5.

- 31-40 REDSS - Reduction factor for the resistance of the soil side areas, which is calculated as the in-situ shear strength of the soil, SUD, times soil side area times REDSS. A factor of 0.4 has been proven to give results in good agreement with Brinch Hansen's formula for homogeneous soil conditions, and this value is therefore recommended (see Appendix IV).
- 41-50 CRACK - The active earth pressure is not allowed to be negative. In such case an open crack should be assumed between the skirt and the soil.
CRACK = 0.0 means open crack.
CRACK = 1.0 means closed crack.
- 51-60 PASTYP - An opportunity is given to choose between two types of passive failure zones. If the soil from mudline down to the skirt tip is significantly weaker than the soil below the skirt tip, PASTYP = 0.0, i.e. small area, should be chosen. If no change in material is present around skirt tip level, PASTYP = 1.0, large area, should be chosen.
- 61-70 ZSPEC1 - In special cases when output of forces are
71-80 ZSPEC2 wanted for failure surfaces other than the critical one, one or two specified depths, ZSPEC1 and ZSPEC2, may be given. If such output is not necessary, these fields should be left blank.

6. LINE (PRINT OPTIONS, AREA VARIATIONS, ETC.) NSU, NPR1, NPR2, NPR3, NPR4, NPR5, BMAX, DELB, ALMAX, DELAL.

FORMAT F5.0, 5F3.0, 4F10.0

(The first six variables must be given as integers, right justified).

- 01-05 NSU - total number of shear strength input data lines

that follows, i.e. total number of line 7.

NSU $\leq$ 50.

- 06-08 NPR1 - print option for output of table with calculated safety factor versus depth.
NPR1 = 1; tables are given for each combination of ZS, PV and LF
NPR1 = 0; tables are not given.
- 09-11 NPR2 - print option for output of table with all calculated forces for the critical slip surface.
NPR2 = 1; tables are given for each combination of ZS, PV and LF.
NPR2 = 0; tables are not given.
- 12-14 NPR3 - print option which is dummy in CAP.
The number will be read but not used. If ZSPEC1 (and ZSPEC2) is greater than 0, NPR3 will automatically be set equal to 1 in the program.
- 15-17 NPR4 - flag value used internally in CAP. The input number will be read but not used.
- 18-20 NPR5 - option to calculate minimum horizontal force when SF = 1.0.
NPR5 = 0; searching for minimum safety factor (PH = fixed)
NPR5 = 1; searching for minimum horizontal force (SF = 1.0)
- 21-30 BMAX - Area may be changed in steps from (B $\cdot$ AL)
31-40 DELB up to (BMAX $\cdot$ ALMAX). DELB and DELAL are
41-50 ALMAX step values. If BMAX is reached before
51-60 DELAL ALMAX, BMAX is kept unchanged until ALMAX is reached, and vice versa.
Each area (B(n) $\cdot$ AL(n)) is treated as a separate run with full output; that is "Data input reprint, Tables required and Summary of results".
If no area variation is wanted, these four fields should be left blank.



7. LINE (SHEAR STRENGTH INPUT DATA) ZZZ, SU, SUD, SUP, SU2.

Maximum 50 lines, FORMAT 5F10.0

- 01-10 ZZZ - The depth from mudline at which the shear strength values are given. The first line should always be given for ZZZ = 0.0.
- 11-20 SU - Soil shear strength determined by undrained triaxial compression tests, or just ordinary undrained shear strength in the case of a simple total stress analysis.
- 21-30 SUD - Soil shear strength determined by undrained simple shear tests. In the case of a simple total stress analysis, this field should be left blank. If values of SUD are to be given, the value at mudline must be > 0.0, and values have to be given for the whole table.
- 31-40 SUP - Soil shear strength determined by undrained triaxial extension tests. Further comments and limitations as for SUD.
- 41-50 SU2 - Minimum undrained shear strength only to be used on the horizontal part of the slip surface, (in the case of thin, weak horizontal layers or failure up in between skirts). If values of SU2 are to be given, the value at mudline must be > 0.0.
If values of SU2 are not given, values of SUD are copied into the field (or values of SU if SUD is not given).
It is not necessary to give values of SU2 below ZSF.

8. LINE (DIVIDING DATA SET)

- 01-04 **** Between different data sets in the same run, a line with **** in columns 1-4 has to be inserted. A "data set" consists of lines 1 to 6 plus NSU line 7.

9. LINE (END OF LAST DATA SET)

At the end of the last data set a blank line has
to be inserted.

PROGRAM "FLOW CHART" FOR CAP

MAIN

INPUT2

CALC2

ARRAY

RPRINT

SUBS

FACT

SHEAR2

SUBS

SAF2

SHEAR2

SUBS

FORCE

FACT

FORCC2

HHEAD

SHEAR2

SUBS

FORCE

FACT

HEAD

HANSEN

MEYER

FORCE2

HHEAD

SHEAR2

SUBS

FORCE

FACT

HEAD

OUTPUT

OUTPUT

SUBROUTINES

The stability program CAP consists of the following subroutines in alphabetic order:

- ARRAY** - Generates the necessary number of each variable for which calculations should be carried out.
INPUT: XF, XL, XS
OUTPUT: X, N
- CALC2** - Organizes all the calculations to be carried out.
INPUT: IPAGE, ITEXT(30), BB, AL, ZSS, ZSF, ZSSTEP, GAMS, GAMW, PVS, PVF, PVSTEP, NLOAD, ALF(M), ZMAX, ZSTEP, REDBS, REDSS, CRACK, PASTYP, NSU, NPR1, NPR2, NPR3, ZZZ(J), SU(J), SUD(J), SUP(J), SU2(J), PH0, AM0, DELPH0, DELPT0, NSPEC, ZSPEC(K), NUM, ALL(L), BBB(L), PS, NPS, IPROG, IUNIT(N), NPR5
OUTPUT: NPV, NZS, IFAIL(L,J,K)
The output parameters NPV and NZS are generated in subroutine ARRAY
INPUT/OUTPUT: IPAGE
This is updated in subroutines FORCC2 and FORCE2
EXTERNAL CALLS: ARRAY
RPRINT
SUBS
FACT
SHEAR2
SAF2
FORCC2
FORCE2
OUTPUT
- FACT** - Generates several factors to be used in the calculation of forces.

```

INPUT:  ALFA, REDBS, REDSS, CRACK, PASTYP
OUTPUT: F(I)

FORCE  - Calculates all the A- and B-values in the general
        equations for the forces.
INPUT:  E, ALPHA, PV, Z, ZS, BB, AL, PH, DELPH,
        DELPT, GAMS, GAMW, REDBS, REDSS, CRACK, PASTYP
OUTPUT: A(I), B(I)
EXTERNAL CALL:  FACT

FORCC2 - Calculates safety factor or horizontal force, for
        SF = 1.0, forces and mobilized shear stresses for the
        critical slip surface.
INPUT:  E(L), PV, ZS, ALF, NZ, Z(I), SFMIN(M), I2,
        SUMB, IPAGE, ITEXT(J), BB, AL, ZSS, ZSF,
        ZSSTEP, GAMS, GAMW, PVS, PVF, PVSTEP, PH,
        AM, DELPH, DELPT, NLOAD, ZZZ(K), SU(K),
        SUD(K), SUP(K), SU2(K), REDBS, REDSS, CRACK,
        PASTYP, NSU, NPR1, NPR2, NUM, NPR5,
        IFAIL(L,J,K)
INPUT/OUTPUT:  IPAGE (updated in this subroutine)
OUTPUT ON TAPE 7 (temporary work file):  ALF, PV, ZS,
        SF, Z(I), A2, CSS, CB, CM, I22, I33
EXTERNAL CALLS:  HHEAD
                  SHEAR2
                  FORCE
                  HEAD
                  HANSEN
                  MEYER

FORCE2 - Calculates safety factor or horizontal force for
        SF = 1.0, as well as forces and mobilized shear
        stresses for the specified slip surfaces.
INPUT:  E(I), ALPHA(J), PV, ZS, ALF, NZ, Z(K),
        SEMIN(L), I2, SUMB, IPAGE, ITEXT(M), BB, AL,
        ZSSTEP, PH, DELPH, DELPT, NPR1, PVS, PVF,
        B0, ZSF, ZSS, NLOAD, SU(N), SUD(N), SUP(N),
        SU2(N), GAMS, GAMW, REDBS, REDSS, CRACK,
        PASTYP, NSPEC, ZZZ(N), NSU, NUM, ZSPEC(I),

```



NPR5, IFAIL(L,J,K)

INPUT/OUTPUT: IPAGE, NPAGE (updated in this subroutine)

EXTERNAL CALLS: HHEAD, SHEAR2, FORCE, HEAD

HANSEN - Calculates mobilized shear stress according to Brinch Hansen's bearing capacity formula modified for offshore structures (see Appendices I and VII).

INPUT: B, AL, D, GAM, GAMW, PH, AMM, DPH, DPT, PV

OUTPUT: C3, I2

HEAD - Provides the writing of a general heading.

INPUT: NPAGE, IPAGE, ITEXT, ALF, PV, ZS, NPR4

HHEAD - Provides the writing of a general heading.

INPUT: NPAGE, IPAGE, ITEXT, ALF, PV, ZS, NPR5

INPUT2 - Provides the writing of "Input data reprint" and makes all data ready for processing in various subroutines.

INPUT/OUTPUT: ITEXT(I), IUNIT(J), B, AL, ZSS, ZSF, ZSSTEP, GAMS, GAMW, PS, PVS, PVF, PVSTEP, ZMAX, ZSTEP, REDBS, REDSS, CRACK, PASTYP, ZZZ(K), SU(K), SUP(K), SU2(K)

The above parameters are read from the input file.

IPAGE, NSU, NPR1, NPR2, NPR3, NSPEC, ZSPEC(I), ALF(J), NLOAD, PH0, AM0, DELPH0, DELPT0, NUM, ALL(K), BB(L), IPROG, NPR5

The above parameters are evaluated in this subroutine.

MEYER - Calculates mobilized shear stress according to Meyerhof's bearing capacity formula modified for offshore structures (see Appendices I and VIII).

INPUT: B, AL, D, GAM, GAMW, PH, AMM, DPH, DPT, PV

OUTPUT: C2, I2

OUTPUT - Provides the writing of "Summary of results" table.

INPUT: IPAGE, ITEXT(I), ZSS, ZSF, ZSSTEP, PVS, PVSTEP, NLOAD, NPR1, NPR2, NPV, NZS, NUM, PS, NPS, NPR5, IFAIL(L,J,K)



INPUT/OUTPUT: IPAGE

being updated during this routine

RPRINT

- Provides the writing of all the variables in use at the current time.

INPUT: B, AL, ZSS, ZSF, ZSSTEP, GAMS, GAMW, PVS,
PVSTEP, PH, AM, DELPH, DELPT, FAC, ZMAX,
ZSTEP, REDBS, REDSS, IUNIT(I), ZZZ(J),
SU(J), SUD(J), SUP(J), SU2(J), NSU, NLOAD,
NPR1, NPR2, NUM, IPROG, IPAGE, ITEXT(K), CRACK

OUTPUT: IPAGE (updated in this subroutine).

SAF2

- Calculates the safety factor and horizontal force for $SF = 1.0$ for a given data set.

INPUT: E, ALFA, PV, Z, ZS, BB, AL, PH, DELPH,
DELPT, NSU, ZZZ(I), SU(I), SUD(I), SUP(I),
SU2(I), GAMS, GAMW, REDBS, REDSS, CRACK,
PASTYP

OUTPUT: SF, CSS, SUMB, PH2

EXTERNAL CALLS: SHEAR2, FORCE

SHEAR2

- Calculates all the average shear strength values to be used in the general equations for the forces.

INPUT: ALFA, ZS, Z, NSU, ZZZ(I), SU(I), SUD(I),
SUP(I), SU2(I), PASTYP

OUTPUT: SA(J)

EXTERNAL CALL: SUBS

SUBS

- Calculates actual shear strength value at given depths by interpolation.

INPUT: ZZZ(I), SU(I), NSU, Z

OUTPUT: SS

ALPHABETIC LIST OF SYMBOLS USED IN CAP

A(20)	= factor in general force expression related to force (I)
AL	= base length normal to wave direction
ALF(2)	= load factor
ALFA	= angle in inclined section
ALL(50)	= AL
ALPHA	= ALFA
AM	= overturning moment at sea floor
AMM	= AM
AMO	= start value of AM
AS	= ALFA
B(20)	= factor in general force expression related to force (I)
BB	= base width in wave direction
BBB(50)	= BB, array of B-values
B0	= effective base width
CB	= mobilized shear stress according to Brinch Hansen's formula
CM	= mobilized shear stress according to Meyerhof's formula
CRACK	= option for crack on active side
CSS	= mobilized shear stress according to CAP
C2	= CM
C3	= CB
D	= depth of embedment
DELPH	= excess wave pressure at sea floor outside heel
DELPT	= excess wave pressure at sea floor outside toe
DELPH0	= start value for DELPH
DELPT0	= start value for DELPT
DPH	= DELPH
DPT	= DELPT

E = eccentricity

F(16) = factors used for calculating forces

FAC = load factor

GAM = total unit weight of soil

GAMS = total unit weight of soil

GAMW = unit weight of water

IFAIL(2,50,20)= control parameter array to check $B_0 > 0$ for every
slip surface (IFAIL = 0 : $B_0 > 0$; IFAIL = 1 : $B_0 \leq 0$).

IPAGE = page number

ITEXT(30) = array for storing of text from 1st line of input

IUNIT(16) = array for storing of text from 2nd line of input

N = number of values

NLOAD = number of load factors

NPAGE = number of pages in output

NPR1 = print option 1

NPR2 = print option 2

NPR3 = print option 3 (dummy)

NPR4 = print option 4 (dummy)

NPR5 = option 5

NPS = 15 = number of cases in parameter study

NPV = number of vertical loads

NSPEC = number of specified depths

NSU = number of shear strength values

NUM = number of loops for each variable

NZ = number of slip surfaces

NZS = number of depths of embedment

PASTYP = option for passive failure zone

PH = horizontal load at sea floor

PH0 = start value of PH

PS = selection of parameter study

PV(20) = vertical load at sea floor

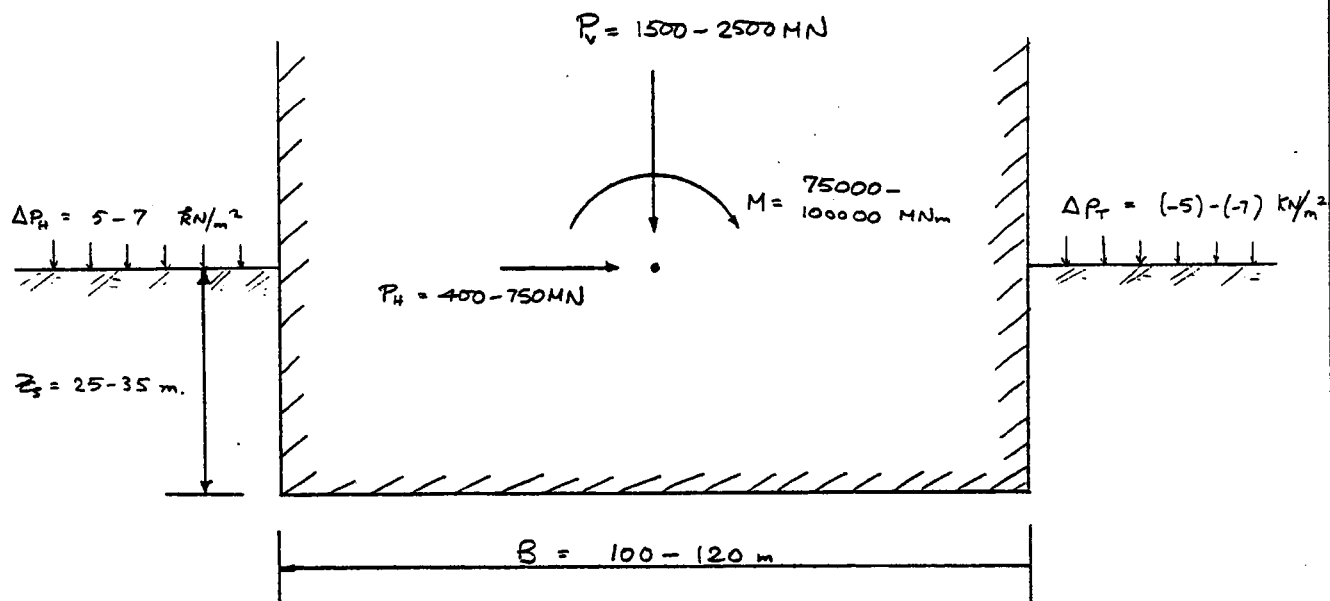
PVF	= vertical load at sea floor, final value
PVS	= vertical load at sea floor, start value
PVSTEP	= vertical load at sea floor, step value
REDBS	= reduction factor base/soil
REDSS	= reduction factor soil/soil
SA(20)	= average shear strength related to force (I)
SF	= safety factor
SFMIN(200)	= minimum safety factor if NPR5 = 0, or minimum horizontal force if NPR5 = 1
SS	= value of shear strength at specified depth
SU(50)	= undrained shear strength, by triaxial compression test
SUD(50)	= undrained shear strength, by direct simple shear test
SUP(50)	= undrained shear strength, by triaxial extension test
SU2(50)	= undrained shear strength, minimum
SUMB(200)	= the sum of all B(I)-values for each slip surface
SUMBE(200)	= SUMB - B(1) for each slip surface
Z(200)	= depth below sea floor
ZMAX	= maximum depth to which safety factors are evaluated
ZS(100)	= depth of embedment
ZSF	= depth of embedment, final value
ZSS	= depth of embedment, start value
ZSSTEP	= depth of embedment, step value
ZSPEC(3)	= depth of specified slip surface(s)
ZSTEP	= step value during the search of SFMIN
ZZZ(50)	= Z-value corresponding to the shear strength values
X	= variable
XF	= variable, first value
XL	= variable, last value
XS	= variable, step value

	JOB TITLE:	CONTRACT NO. 52406 - 46	PAGE II - 0
	SUBJECT: The Computer Program CAP	SIGN. RAL	DATE 20/9-85
		CONTR.	DATE

APPENDIX II : EXAMPLES

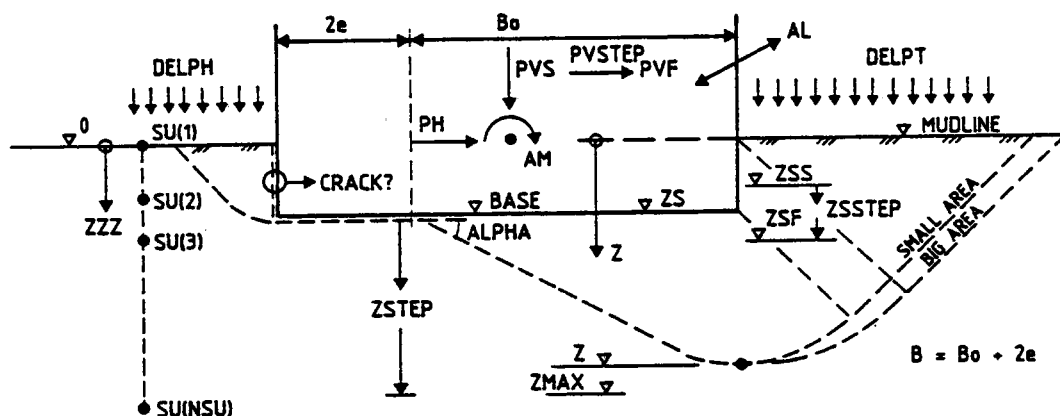
	JOB TITLE:	CONTRACT NO. 52406-46	PAGE II - 1
	SUBJECT:	SIGN. RAL / PT.	DATE 20/9/85
	DATA FOR CAP-FAMILY TEST EXAMPLES	CONTR.	DATE

Summary of Data Variations :-



	OPPDAG:	CAP-FAMILY EXAMPLES	OPPDAGSNR.	52406-46	SIDE	II-2
	FORSØK · BEREGNING		SIGN.	RAH	DATO	24/5-85
	FUNCTION CONTROL		KONTR.		DATO	

Function to be checked in RUN NO :			1	2	3	4	5	6	7
1.	Z STEP	, calculation for z_s -values		x	x				
2.	GAMS	, calculation independant of γ -value or not ?				x			
3.	PS	, parameter study		x					
4.	PVSTEP	, calculation for P_v -values					x		
5.	FAC	, 1.0 only $\gamma_f = 1.0$ 1.3 both $\gamma_f = 1.3$ and 1.0 - 1.3 only $\gamma_f = 1.3$		x			x	x	x
6.	REDBS	, reduction base / soil		.5	.5	.5	0	.5	1
7.	REDSS	, " soil / soil		.4	.4	.4	0	.4	1
8.	CRACK	, 0.0 open crack 1.0 closed crack						x	
9.	PASTYP	, 0.0 or 1.0 , difference ?		1	1	0	0	1	1
10.	ZSPEC 1 & 2	, Specified failure surface, $K(N)$ table							x
11.	NPR 1	, 0, 1 Table SF/z		0	1	0	1	1	0
12.	NPR 2	, 0, 1 Table $K(N)$ for SF_{min}		0	0	0	1	1	1
13.	NPR 4	, 0, 1 Full table SF/z		0	0	0	0	1	0
14.	NPR 5	, 0, 1 P_H for $SF = 1.0$		0	0	0	0	0	0
15.	BMAX/DELB	, more areas in one run							x
16.	RUH/RUH	, earth pressure coefficients		.5	.5	.5	0	.5	1
17.	SU	, shear strength options				x			
18.	SU/SU2	, - " -					x		
19.	SU, SUD, SUP	, - " -		x					
20.	SU, SUD, SUP, SU2,	, - " -							x
21.	More datasets	, in one run.					x		

REVISÉD INPUT
DATA FORM
FOR CAP

PROJECT									
EXAMPLE 1 *** CAP ***									

UNITS USED									
MN / M									

1	10	11	20	21	30	31	40	41	50	51	60	61	70	71	80
B		AL		ZSS		ZSF		ZSSTEP		GAMS 1)		GAMW 1)		PS 4)	
100.0		100.0		30.0		30.0		0.0		0.02		0.01		0.1	

1	10	11	20	21	30	31	40	41	50	51	60	61	70	71	80
PVS		PVF		PVSTEP		PH		AM		DELPH		DELPT		FAC 2)	
1500.0		1500.0		0.0		400.0		75000.0		0.005		-0.005		1.0	

1	10	11	20	21	30	31	40	41	50	51	60	61	70	71	80
ZMAX		ZSTEP		REDBS 3)		REDSS 4)		CRACK 9)		PASTYP 10)		ZSPEC 1 4)		ZSPEC 2 4)	
70.0		2.0		0.5		0.4		1.0		1.					

1	5	8	11	14	17	20	21	30	31	40	41	50	51	60	61	70	71	80	
NSU		NPR1 8)		NPR2 8)		NPR3 8)		NPR4 8)		NPR5 8)		BMAX 4)		DELB 4)		ALMAX 4)		DELAL 4)	
2		0		0		0		0		0									

1	10	11	20	21	30	31	40	41	50
ZZZ		SU 5)		SUD 6)		SUP 7)		SU2 11)	
0.0		0.001		0.001		0.001			
100.0		0.225		0.200		0.180			

1) TOTAL UNIT WEIGHT OF SOIL AND WATER

2) LOAD FACTOR ON PH, M, APH AND ΔPT = 1.2, LOAD FACTORS 1.0 AND 1.2 = -1.2, ONLY LOAD FACTOR 1.2

3) REDUCTION FACTOR FOR SIDE SHEAR, BASE/SOIL. NORMALLY 0.5

4) REDUCTION FACTOR FOR SIDE SHEAR, SOIL/SOIL. NORMALLY 0.4

5) TRIAXIAL COMPRESSION STRENGTH, OR SU

6) SIMPLE SHEAR STRENGTH

7) TRIAXIAL EXTENSION STRENGTH

8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
NPR1 = 0 GIVES TABLE OF SF-VALUES
NPR2 = 0 GIVES ALL K (N) FOR SFMIN
NPR3 = 0 GIVES ALL K (N) FOR ZSPEC
NPR4 = 0 OMITTED IN THIS MODE
NPR5 = 0 GIVES PH FOR SF = 1.0

9) CRACK ON HEEL SIDE: = 0. OPEN
CRACK = 1.0 CLOSED CRACK.

10) TYPES OF PASSIVE FAILURE: = 0. SMALL AREA. = 1.0 BIG AREA

11) MINIMUM HORIZONTAL STRENGTH

11) MINIMUM HORIZONTAL STRENGTH

*1) MAY BE LEFT BLANK.

////// INPUT DATA OMITTED IN THIS FAILURE MODE.

LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.

BLANK LINE ENDING ALL DATA SETS.

C A P

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406 DATED 01 OCT 1985

PROJECT : EKSEMPEL 1 ***CAP***

INPUT DATA REPRINT

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION

B = 100.000

LENGTH OF FOUNDATION

L = 100.000

DEPTH OF EMBEDMENT , START VALUE

ZSS = 30.000

DEPTH OF EMBEDMENT , FINAL VALUE

ZSF = 30.000

DEPTH OF EMBEDMENT , STEP VALUE

ZSSTEP = 0.000

SOIL TOTAL UNIT WEIGHT

GAMS = 0.020

WATER UNIT WEIGHT

GAMW = 0.010

PARAMETERSTUDY

PS = 0.100

SUBMERGED WEIGHT OF STRUCTURE , START VALUE

PVS = 1500.000

SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE

PVF = 1500.000

SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE

PVSTEP = 0.000

HORIZONTAL FORCE AT MUDLINE

PH = 400.000

OVERTURNING MOMENT AT MUDLINE

M = 75000.000

EXCESS WATER PRESSURE OUTSIDE HEEL

DELPH = 0.005

EXCESS WATER PRESSURE OUTSIDE TOE

DELPT = -0.005

LOAD FACTORS TO BE USED

LF = 1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE

ZMAX = 70.000

STEP VALUE FOR SLIP SURFACE DEPTH

ZSTEP = 2.000

REDUCTION FACTOR SIDE, BASE/SOIL

REDBS = 0.500

REDUCTION FACTOR SIDE, SOIL/SOIL

REDSS = 0.400

ACTIVE SIDE CRACK ASSUMPTION : CLOSED CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAxIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAxIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.200

PROJECT : EKSEMPEL 1 ***CAP***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITION THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$

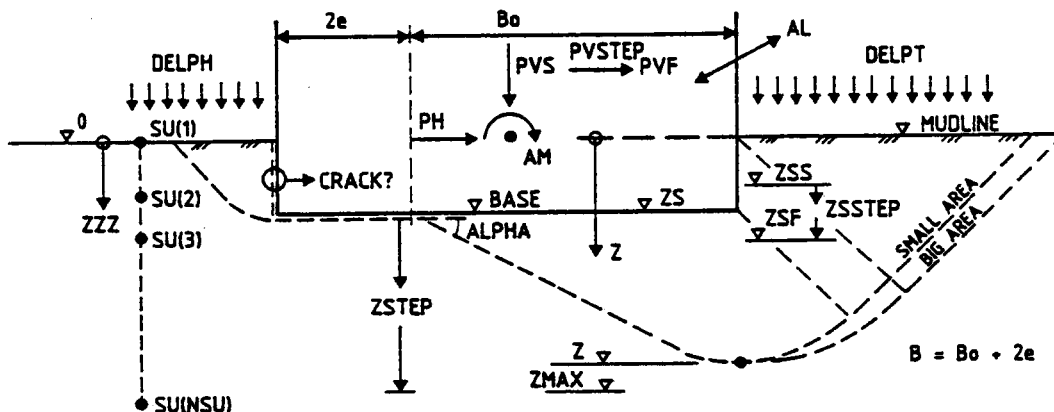
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)

MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR. HA	MOB SHEAR M' HOF	
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM	
1.00	1500.0	30.0	1.63	60.0	27.8	0.0396	0.0556	0.0573	BASIS
1.00	1500.0	30.0	1.18	60.0	37.0	0.0539	0.0747	0.0779	-0.10 B & L
1.00	1500.0	30.0	2.11	58.0	20.9	0.0300	0.0428	0.0439	0.10 B & L
1.00	1500.0	27.0	1.43	55.0	26.9	0.0412	0.0582	0.0601	-0.10 ZS
1.00	1500.0	33.0	1.82	65.0	28.8	0.0383	0.0531	0.0548	0.10 ZS
1.00	1350.0	30.0	1.65	60.0	28.4	0.0389	0.0540	0.0557	-0.10 PV
1.00	1650.0	30.0	1.60	60.0	27.3	0.0404	0.0571	0.0590	0.10 PV
1.00	1500.0	30.0	1.68	60.0	27.6	0.0383	0.0547	0.0566	-0.10 PH
1.00	1500.0	30.0	1.57	58.0	26.0	0.0399	0.0565	0.0582	0.10 PH
1.00	1500.0	30.0	1.73	58.0	24.5	0.0364	0.0516	0.0532	-0.10 M
1.00	1500.0	30.0	1.52	60.0	29.4	0.0422	0.0596	0.0618	0.10 M
1.00	1500.0	30.0	1.63	60.0	27.8	0.0395	0.0555	0.0572	-0.10 DELPH & DELPT
1.00	1500.0	30.0	1.62	60.0	27.8	0.0397	0.0557	0.0574	0.10 DELPH & DELPT
0.90	1500.0	30.0	1.80	60.0	26.2	0.0359	0.0507	0.0523	-0.10 LF
1.10	1500.0	30.0	1.47	60.0	29.6	0.0436	0.0607	0.0628	0.10 LF

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

REVISID INPUT
DATA FORM
FOR CAP[illegible]

* MAY BE LEFT BLANK.

////// INPUT DATA OMITTED IN THIS FAILURE MODE.

LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.

BLANK LINE ENDING ALL DATA SETS.

C A P

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406 DATED 01 JUL 1985

PROJECT : EKSEMPEL 2 ***CAP***

INPUT DATA REPRINT

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	25.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	35.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	5.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	400.000
OVERTURNING MOMENT AT MUDLINE	M =	75000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.005
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.005
LOAD FACTORS TO BE USED	LF =	1.3

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.500
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	0.400

ACTIVE SIDE CRACK ASSUMPTION : CLOSED CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

PROJECT : EKSEMPEL 2 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 25.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A*SA)/(-S(B)) S(A*SA)	S(B)
25.000	0.000	2.420	-0.1029E+04	0.4250E+03
27.000	1.943	2.309	-0.1182E+04	0.5116E+03
29.000	3.876	2.073	-0.1239E+04	0.5974E+03
31.000	5.803	1.903	-0.1299E+04	0.6824E+03
33.000	7.729	1.778	-0.1363E+04	0.7669E+03
35.000	9.657	1.683	-0.1432E+04	0.8510E+03
37.000	11.593	1.613	-0.1508E+04	0.9348E+03
39.000	13.540	1.562	-0.1591E+04	0.1019E+04
41.000	15.503	1.523	-0.1679E+04	0.1102E+04
43.000	17.484	1.495	-0.1773E+04	0.1186E+04
45.000	19.487	1.475	-0.1873E+04	0.1269E+04
47.000	21.527	1.462	-0.1979E+04	0.1354E+04
49.000	23.585	1.455	-0.2091E+04	0.1437E+04
51.000	25.679	1.453	-0.2210E+04	0.1521E+04
53.000	27.813	1.454	-0.2335E+04	0.1605E+04
55.000	29.994	1.460	-0.2467E+04	0.1690E+04
57.000	32.227	1.468	-0.2605E+04	0.1774E+04
59.000	34.521	1.480	-0.2751E+04	0.1859E+04
61.000	36.884	1.493	-0.2903E+04	0.1944E+04
63.000	39.349	1.508	-0.3062E+04	0.2030E+04
65.000	41.889	1.526	-0.3228E+04	0.2116E+04
67.000	44.539	1.545	-0.3402E+04	0.2202E+04

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 1.453
Z = 51.000
ALPHA = 25.679

PROJECT : EKSEMPEL 2 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A#SA)/(-S(B)) S(A#SA)	S(B)
30.000	0.000	3.122	-0.1342E+04	0.4300E+03
32.000	1.822	3.015	-0.1550E+04	0.5141E+03
34.000	3.634	2.708	-0.1617E+04	0.5972E+03
36.000	5.440	2.483	-0.1688E+04	0.6796E+03
38.000	7.243	2.314	-0.1762E+04	0.7613E+03
40.000	9.047	2.184	-0.1840E+04	0.8426E+03
42.000	10.855	2.082	-0.1923E+04	0.9234E+03
44.000	12.671	2.007	-0.2015E+04	0.1004E+04
46.000	14.499	1.949	-0.2114E+04	0.1085E+04
48.000	16.340	1.904	-0.2219E+04	0.1165E+04
50.000	18.200	1.871	-0.2330E+04	0.1245E+04
52.000	20.079	1.845	-0.2447E+04	0.1326E+04
54.000	21.983	1.828	-0.2571E+04	0.1407E+04
56.000	23.923	1.815	-0.2701E+04	0.1488E+04
58.000	25.882	1.809	-0.2838E+04	0.1568E+04
60.000	27.877	1.808	-0.2981E+04	0.1649E+04
62.000	29.913	1.810	-0.3132E+04	0.1730E+04
64.000	31.995	1.816	-0.3289E+04	0.1811E+04
66.000	34.128	1.825	-0.3454E+04	0.1893E+04
68.000	36.320	1.836	-0.3625E+04	0.1974E+04
70.000	38.579	1.850	-0.3804E+04	0.2056E+04

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 1.808
Z = 60.000
ALPHA = 27.877

PROJECT : EKSEMPEL 2 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 35.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A*SA)/(-S(B)) S(A*SA)	S(B)
35.000	0.000	3.895	-0.1694E+04	0.4350E+03
37.000	1.736	3.798	-0.1965E+04	0.5174E+03
39.000	3.461	3.413	-0.2044E+04	0.5989E+03
41.000	5.180	3.127	-0.2125E+04	0.6795E+03
43.000	6.896	2.909	-0.2209E+04	0.7593E+03
45.000	8.611	2.739	-0.2297E+04	0.8386E+03
47.000	10.329	2.604	-0.2389E+04	0.9175E+03
49.000	12.051	2.497	-0.2487E+04	0.9959E+03
51.000	13.783	2.416	-0.2595E+04	0.1074E+04
53.000	15.526	2.352	-0.2711E+04	0.1152E+04
55.000	17.284	2.303	-0.2833E+04	0.1231E+04
57.000	19.058	2.264	-0.2962E+04	0.1309E+04
59.000	20.851	2.234	-0.3098E+04	0.1387E+04
61.000	22.668	2.211	-0.3240E+04	0.1465E+04
63.000	24.519	2.195	-0.3388E+04	0.1544E+04
65.000	26.387	2.185	-0.3544E+04	0.1622E+04
67.000	28.288	2.180	-0.3707E+04	0.1701E+04
69.000	30.226	2.179	-0.3877E+04	0.1779E+04
70.000	31.211	2.180	-0.3965E+04	0.1818E+04

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 2.179
Z = 69.000
ALPHA = 30.226

PROJECT : EKSEMPEL 2 ***CAP***

LOAD FACTOR : 1.30 SUBMERGED WEIGHT : 1500.0 BASE EMB : 25.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A*SA)/(-S(B)) S(A*SA)	S(B)
25.000	0.000	1.862	-0.1029E+04	0.5525E+03
27.000	2.454	1.746	-0.1179E+04	0.6751E+03
29.000	4.882	1.550	-0.1233E+04	0.7955E+03
31.000	7.296	1.412	-0.1291E+04	0.9144E+03
33.000	9.705	1.311	-0.1353E+04	0.1032E+04
35.000	12.117	1.236	-0.1420E+04	0.1149E+04
37.000	14.544	1.182	-0.1495E+04	0.1265E+04
39.000	16.993	1.142	-0.1577E+04	0.1381E+04
41.000	19.473	1.113	-0.1665E+04	0.1496E+04
43.000	21.991	1.091	-0.1759E+04	0.1612E+04
45.000	24.556	1.076	-0.1859E+04	0.1728E+04
47.000	27.202	1.065	-0.1966E+04	0.1845E+04
49.000	29.884	1.060	-0.2080E+04	0.1961E+04
51.000	32.650	1.059	-0.2200E+04	0.2077E+04
53.000	35.511	1.060	-0.2327E+04	0.2194E+04
55.000	38.486	1.065	-0.2460E+04	0.2311E+04
57.000	41.600	1.071	-0.2601E+04	0.2429E+04
59.000	44.881	1.079	-0.2749E+04	0.2547E+04

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 1.059
Z = 51.000
ALPHA = 32.650

PROJECT : EKSEMPEL 2 ***CAP***

LOAD FACTOR : 1.30 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A*SA)/(-S(B)) S(A*SA)	S(B)
30.000	0.000	2.401	-0.1342E+04	0.5590E+03
32.000	2.213	2.295	-0.1547E+04	0.6741E+03
34.000	4.404	2.048	-0.1612E+04	0.7872E+03
36.000	6.582	1.870	-0.1680E+04	0.8987E+03
38.000	8.753	1.737	-0.1752E+04	0.1009E+04
40.000	10.924	1.635	-0.1829E+04	0.1118E+04
42.000	13.102	1.557	-0.1910E+04	0.1227E+04
44.000	15.293	1.499	-0.2002E+04	0.1335E+04
46.000	17.505	1.455	-0.2100E+04	0.1443E+04
48.000	19.742	1.422	-0.2205E+04	0.1551E+04
50.000	22.012	1.396	-0.2316E+04	0.1659E+04
52.000	24.319	1.378	-0.2434E+04	0.1766E+04
54.000	26.672	1.365	-0.2559E+04	0.1875E+04
56.000	29.095	1.356	-0.2690E+04	0.1984E+04
58.000	31.555	1.352	-0.2828E+04	0.2092E+04
60.000	34.090	1.351	-0.2973E+04	0.2201E+04
62.000	36.709	1.353	-0.3126E+04	0.2310E+04
64.000	39.427	1.358	-0.3285E+04	0.2419E+04
66.000	42.263	1.365	-0.3452E+04	0.2529E+04

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 1.351
Z = 60.000
ALPHA = 34.090

PROJECT : EKSEMPEL 2 ***CAP***

LOAD FACTOR : 1.30 SUBMERGED WEIGHT : 1500.0 BASE EMB : 35.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A*SA)/(-S(B)) S(A*SA)	S(B)
35.000	0.000	2.996	-0.1694E+04	0.5655E+03
37.000	2.052	2.904	-0.1962E+04	0.6758E+03
39.000	4.085	2.601	-0.2039E+04	0.7842E+03
41.000	6.104	2.377	-0.2118E+04	0.8911E+03
43.000	8.116	2.207	-0.2200E+04	0.9966E+03
45.000	10.126	2.077	-0.2287E+04	0.1101E+04
47.000	12.139	1.974	-0.2378E+04	0.1205E+04
49.000	14.160	1.892	-0.2474E+04	0.1308E+04
51.000	16.195	1.830	-0.2582E+04	0.1410E+04
53.000	18.250	1.783	-0.2698E+04	0.1513E+04
55.000	20.327	1.746	-0.2820E+04	0.1615E+04
57.000	22.433	1.717	-0.2949E+04	0.1718E+04
59.000	24.573	1.695	-0.3085E+04	0.1820E+04
61.000	26.751	1.679	-0.3228E+04	0.1923E+04
63.000	28.993	1.667	-0.3378E+04	0.2027E+04
65.000	31.262	1.660	-0.3536E+04	0.2130E+04
67.000	33.593	1.657	-0.3700E+04	0.2233E+04
69.000	35.994	1.657	-0.3872E+04	0.2336E+04
70.000	37.225	1.658	-0.3960E+04	0.2388E+04

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 1.657
Z = 67.000
ALPHA = 33.593

PROJECT : EKSEMPEL 2 ***CAP***
 SUMMARY OF RESULTS OF STABILITY CALCULATIONS

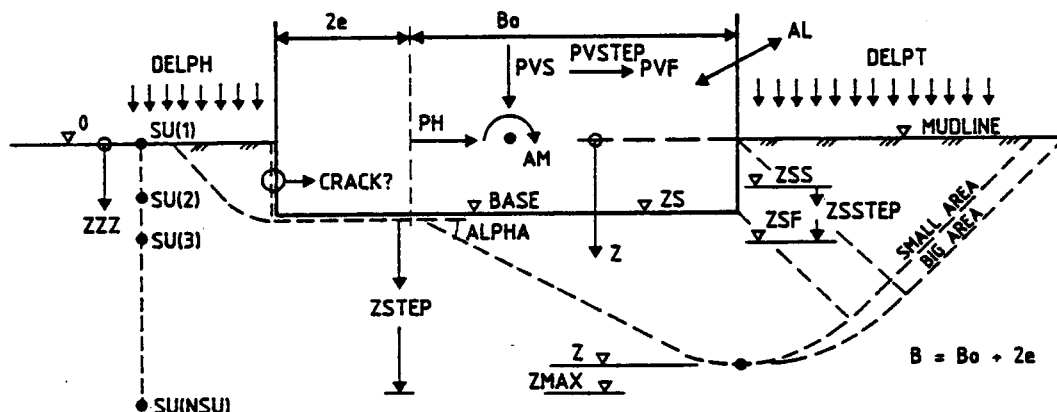
IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M' HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	1500.0	25.0	1.45	51.00	25.7	0.0421	0.0601	0.0621
1.00	1500.0	30.0	1.81	60.00	27.9	0.0397	0.0556	0.0573
1.00	1500.0	35.0	2.18	69.00	30.2	0.0378	0.0515	0.0533
1.30	1500.0	25.0	1.06	51.00	32.6	0.0571	0.0799	0.0832
1.30	1500.0	30.0	1.35	60.00	34.1	0.0528	0.0723	0.0751
1.30	1500.0	35.0	1.66	67.00	33.6	0.0484	0.0661	0.0687

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

REVISÉ INPUT
DATA FORM
FOR CAP[illegible]

7) MAY BE LEFT BLANK.
 //// INPUT DATA OMITTED IN THIS FAILURE MODE.
 LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.
 BLANK LINE ENDING ALL DATA SETS.

C A P

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406 DATED 01 JUL 1985

PROJECT : EKSEMPEL 3 ***CAP***

INPUT DATA REPRINT

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION

B = 100.000

LENGTH OF FOUNDATION

L = 100.000

DEPTH OF EMBEDMENT , START VALUE

ZSS = 25.000

DEPTH OF EMBEDMENT , FINAL VALUE

ZSF = 30.000

DEPTH OF EMBEDMENT , STEP VALUE

ZSSTEP = 5.000

SOIL TOTAL UNIT WEIGHT

GAMS = 0.015

WATER UNIT WEIGHT

GAMW = 0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE

PVS = 1500.000

SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE

PVF = 2500.000

SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE

PVSTEP = 500.000

HORIZONTAL FORCE AT MUDLINE

PH = 400.000

OVERTURNING MOMENT AT MUDLINE

M = 75000.000

EXCESS WATER PRESSURE OUTSIDE HEEL

DELPH = 0.005

EXCESS WATER PRESSURE OUTSIDE TOE

DELPT = -0.005

LOAD FACTORS TO BE USED

LF = -1.3

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE

ZMAX = 70.000

STEP VALUE FOR SLIP SURFACE DEPTH

ZSTEP = 2.000

REDUCTION FACTOR SIDE, BASE/SOIL

REDBS = 0.500

REDUCTION FACTOR SIDE, SOIL/SOIL

REDSS = 0.400

ACTIVE SIDE CRACK ASSUMPTION : CLOSED CRACK

PASSIVE SIDE AREA ASSUMPTION : SMALL AREA

SHEAR STRENGTH USED :

DEPTH BELOW
MUDLINE

UNDRAINED
SHEAR STRENGTH

MINIMUM
SHEAR STRENGTH

0.000

0.001

0.001

100.000

0.225

0.050

PROJECT : EKSEMPEL 3 ***CAP***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$

MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)

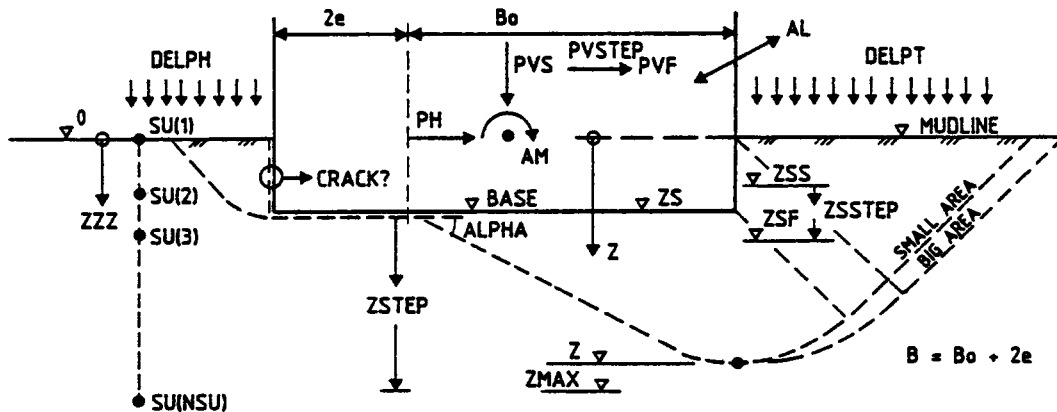
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.30	1500.0	25.0	0.54	41.00	41.3	0.0715	0.1017	0.1072
1.30	1500.0	30.0	0.75	50.00	42.2	0.0650	0.0888	0.0934
1.30	2000.0	25.0	0.65	45.00	34.6	0.0676	0.0957	0.1008
1.30	2000.0	30.0	0.84	54.00	37.7	0.0639	0.0871	0.0915
1.30	2500.0	25.0	0.71	45.00	27.3	0.0640	0.0954	0.0999
1.30	2500.0	30.0	0.89	54.00	30.9	0.0620	0.0887	0.0928

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

BEARING CAPACITY CALCULATED BY SLIP SURFACES

REVISED INPUT
DATA FORM
FOR CAP

1										57															
PROJECT																									
EXAMPLE 4 *** CAP ***																									
1										30															
UNITS USED																									
MN/M																									
1 10 11 20 21 30 31 40 41 50 51 60 61 70 71 80																									
B		AL		ZSS		ZSF		ZSSTEP		GAMS 1)		GAMW 1)		PS 2)											
100.0		100.0		30.0		30.0		0.0		0.020		0.010		1.0											
1 10 11 20 21 30 31 40 41 50 51 60 61 70 71 80																									
PVS		PVF		PVSTEP		PH		AM		DELPH		DELPT		FAC 2)											
1500.0		1500.0		0.0		1.0		75000.0		0.005		-0.005		1.0											
1 10 11 20 21 30 31 40 41 50 51 60 61 70 71 80																									
ZMAX		ZSTEP		REDBS 3)		REDSS 4)		CRACK 9)		PASTYP 10)		ZSPEC 1 11)		ZSPEC 2 11)											
70.0		2.0		0.0		0.0		0.0		0.0		0.0		0.0											
1 5 8 11 14 17 20 21 30 31 40 41 50 51 60 61 70 71 80																									
NSU		NPR1		NPR2		NPR3		NPR4		NPR5		BMAX 4)		DELB 4)		ALMAX 4)		DELAL 4)		RUV		RUH		BHMIN	
2		1		1		1		1		1															
1 10 11 20 21 30 31 40 41 50																									
ZZZ		SU 5)		SUD 6)		SUP 7)		SU2 11)																	
0.0		0.001																							
100.0		0.225																							
100.0		0.150																							
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15																									

1) TOTAL UNIT WEIGHT OF SOIL AND WATER
2) LOAD FACTOR ON PH, M, ΔPH AND ΔPT = 1.2, LOAD FACTORS 1.0 AND 1.2 = -1.2, ONLY LOAD FACTOR 1.2
3) REDUCTION FACTOR FOR SIDE SHEAR, BASE/SOIL NORMALLY 0.5
4) REDUCTION FACTOR FOR SIDE SHEAR, SOIL/SOIL NORMALLY 0.4
5) TRIAXIAL COMPRESSION STRENGTH, OR SU
6) SIMPLE SHEAR STRENGTH
7) TRIAXIAL EXTENSION STRENGTH
8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
NPR1 = 0 GIVES TABLE OF SF-VALUES
NPR2 = 0 GIVES ALL K (N) FOR SFMIN
NPR3 = 0 GIVES ALL K (N) FOR ZSPEC
NPR4 = 0 OMITTED IN THIS MODE
NPR5 = 0 GIVES PH FOR SF = 1.0
9) CRACK ON HEEL SIDE: = 0. OPEN
CRACK = 1.0 CLOSED CRACK
10) TYPES OF PASSIVE FAILURE: = 0. SMALL AREA. = 1.0 BIG AREA
11) MINIMUM HORIZONTAL STRENGTH

*) MAY BE LEFT BLANK.

//// INPUT DATA OMITTED IN THIS FAILURE MODE.

LINE WITH ***** IN COLUMN 1-4 BETWEEN DATA SETS.

BLANK LINE ENDING ALL DATA SETS.

C A P

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406 DATED 01 JUL 1985

PROJECT : EKSEMPEL 4 ***CAP***

INPUT DATA REPRINT

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION

B = 100.000

LENGTH OF FOUNDATION

L = 100.000

DEPTH OF EMBEDMENT , START VALUE

ZSS = 30.000

DEPTH OF EMBEDMENT , FINAL VALUE

ZSF = 30.000

DEPTH OF EMBEDMENT , STEP VALUE

ZSSTEP = 0.000

SOIL TOTAL UNIT WEIGHT

GAMS = 0.020

WATER UNIT WEIGHT

GAMW = 0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE

PVS = 1500.000

SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE

PVF = 1500.000

SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE

PVSTEP = 0.000

HORIZONTAL FORCE AT MUDLINE

PH = 400.000

OVERTURNING MOMENT AT MUDLINE

M = 75000.000

EXCESS WATER PRESSURE OUTSIDE HEEL

DELPH = 0.005

EXCESS WATER PRESSURE OUTSIDE TOE

DELPT = -0.005

LOAD FACTORS TO BE USED

LF = 1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE

ZMAX = 70.000

STEP VALUE FOR SLIP SURFACE DEPTH

ZSTEP = 2.000

REDUCTION FACTOR SIDE, BASE/SOIL

REDBS = 0.000

REDUCTION FACTOR SIDE, SOIL/SOIL

REDSS = 0.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : SMALL AREA

SHEAR STRENGTH USED :

DEPTH BELOW
MUDLINE

UNDRAINED
SHEAR STRENGTH

MINIMUM
SHEAR STRENGTH

0.000

0.001

0.001

100.000

0.225

0.225

SF SF2 -44.480 -44.480

1/ 4

PROJECT : EKSEMPEL 4 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= S(A*SA)/(-S(B))	
			S(A*SA)	S(B)
30.000	0.000	-44.480	-0.8896E+03	-0.2000E+02
32.000	1.814	14.837	-0.9409E+03	0.6341E+02
34.000	3.625	6.802	-0.9963E+03	0.1465E+03
36.000	5.436	4.607	-0.1056E+04	0.2293E+03
38.000	7.250	3.593	-0.1121E+04	0.3119E+03
40.000	9.069	3.017	-0.1190E+04	0.3943E+03
42.000	10.895	2.651	-0.1263E+04	0.4765E+03
44.000	12.732	2.402	-0.1342E+04	0.5587E+03
46.000	14.581	2.225	-0.1426E+04	0.6409E+03
48.000	16.445	2.096	-0.1515E+04	0.7229E+03
50.000	18.327	1.999	-0.1609E+04	0.8050E+03
52.000	20.230	1.926	-0.1709E+04	0.8871E+03
54.000	22.157	1.872	-0.1814E+04	0.9692E+03
56.000	24.111	1.831	-0.1925E+04	0.1051E+04
58.000	26.095	1.801	-0.2041E+04	0.1133E+04
60.000	28.115	1.779	-0.2163E+04	0.1216E+04
62.000	30.176	1.764	-0.2290E+04	0.1298E+04
64.000	32.278	1.756	-0.2423E+04	0.1380E+04
66.000	34.432	1.752	-0.2562E+04	0.1462E+04
68.000	36.643	1.752	-0.2707E+04	0.1545E+04
70.000	38.920	1.756	-0.2857E+04	0.1627E+04

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 1.752
Z = 66.000
ALPHA = 34.432

PROJECT : EKSEMPEL 4 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 1.752
 FOUND FOR DEPTH Z = 66.000
 AND SLIP SURFACE ANGLE ALPHA = 34.432

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.4000E+03	0.0000E+00	0.000	0.000	0.4000E+03
2	PA	0.1500E+02	0.0000E+00	0.035	0.020	0.1500E+02
3	-RH	-0.1414E+03	-0.3633E+04	0.068	0.039	0.0000E+00
4	HA	0.2603E+04	-0.9076E+04	0.114	0.065	0.3192E+04
5	-PP	-0.5535E+03	-0.6000E+04	0.035	0.020	-0.4350E+03
6	-HP	-0.2323E+04	-0.9255E+04	0.116	0.066	-0.1710E+04
7	-RS1	0.0000E+00	0.0000E+00	0.035	0.020	0.0000E+00
8	-RS2	0.0000E+00	0.0000E+00	0.099	0.057	0.0000E+00
9	-RS3	0.0000E+00	0.0000E+00	0.035	0.020	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.013	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.024	0.013	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.024	0.013	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.024	0.013	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.020	0.0000E+00

PROJECT : EKSEMPEL 4 ***CAP***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	1500.0	30.0	1.75	66.00	34.4	0.0523	0.0556	0.0573

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

C A P

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406 DATED 01 JUL 1985

PROJECT : EKSEMPEL 4 ***CARL***

INPUT DATA REPRINT

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	2000.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	2000.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	750.000
OVERTURNING MOMENT AT MUDLINE	M =	100000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.007
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.007
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	0.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : SMALL AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.150	0.150
100.000	0.150	0.150

PROJECT : EKSEMPEL 4 ***CARL***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 2000.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= $S(A*SA)/(-S(B))$ S(A*SA)	S(B)
30.000	0.000	7.018	-0.2400E+04	0.3420E+03
32.000	2.139	5.273	-0.2479E+04	0.4701E+03
34.000	4.263	4.293	-0.2561E+04	0.5965E+03
36.000	6.376	3.668	-0.2646E+04	0.7215E+03
38.000	8.484	3.235	-0.2735E+04	0.8453E+03
40.000	10.592	2.920	-0.2826E+04	0.9681E+03
42.000	12.703	2.680	-0.2921E+04	0.1090E+04
44.000	14.824	2.493	-0.3019E+04	0.1211E+04
46.000	16.957	2.342	-0.3119E+04	0.1331E+04
48.000	19.108	2.220	-0.3222E+04	0.1451E+04
50.000	21.281	2.118	-0.3327E+04	0.1571E+04
52.000	23.483	2.032	-0.3434E+04	0.1690E+04
54.000	25.717	1.960	-0.3544E+04	0.1808E+04
56.000	27.990	1.897	-0.3656E+04	0.1927E+04
58.000	30.308	1.843	-0.3769E+04	0.2045E+04
60.000	32.680	1.796	-0.3884E+04	0.2163E+04
62.000	35.114	1.754	-0.4000E+04	0.2280E+04
64.000	37.621	1.717	-0.4117E+04	0.2398E+04
66.000	40.213	1.684	-0.4236E+04	0.2515E+04
68.000	42.906	1.654	-0.4355E+04	0.2633E+04

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 1.654
Z = 68.000
ALPHA = 42.906

PROJECT : EKSEMPEL 4 ***CARL***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 2000.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 1.654
 FOUND FOR DEPTH Z = 68.000
 AND SLIP SURFACE ANGLE ALPHA = 42.906

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.7500E+03	0.0000E+00	0.000	0.000	0.7500E+03
2	PA	0.2100E+02	0.0000E+00	0.150	0.091	0.2100E+02
3	-RH	-0.4006E+03	-0.4418E+04	0.150	0.091	0.0000E+00
4	HA	0.3324E+04	-0.8847E+04	0.150	0.091	0.4126E+04
5	-PP	-0.9730E+03	-0.6000E+04	0.150	0.091	-0.4290E+03
6	-HP	-0.2721E+04	-0.9769E+04	0.150	0.091	-0.1835E+04
7	-RS1	0.0000E+00	0.0000E+00	0.150	0.091	0.0000E+00
8	-RS2	0.0000E+00	0.0000E+00	0.150	0.090	0.0000E+00
9	-RS3	0.0000E+00	0.0000E+00	0.150	0.091	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.150	0.091	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.150	0.091	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.150	0.091	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.150	0.091	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.150	0.091	0.0000E+00

PROJECT : EKSEMPEL 4 ***CARL***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$

MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)

MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOFF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	2000.0	30.0	1.65	68.00	42.9	0.0907	0.0840	0.0867

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

C A P

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406 DATED 01 OCT 1985

PROJECT : EXAMPLE 5 ***CAP***

INPUT DATA REPRINT

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION

B = 100.000

LENGTH OF FOUNDATION

L = 100.000

DEPTH OF EMBEDMENT , START VALUE

ZSS = 30.000

DEPTH OF EMBEDMENT , FINAL VALUE

ZSF = 30.000

DEPTH OF EMBEDMENT , STEP VALUE

ZSSTEP = 0.000

SOIL TOTAL UNIT WEIGHT

GAMS = 0.020

WATER UNIT WEIGHT

GAMW = 0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE

PVS = 1500.000

SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE

PVF = 1500.000

SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE

PVSTEP = 0.000

HORIZONTAL FORCE AT MUDLINE

PH = 400.000

OVERTURNING MOMENT AT MUDLINE

M = 75000.000

EXCESS WATER PRESSURE OUTSIDE HEEL

DELPH = 0.005

EXCESS WATER PRESSURE OUTSIDE TOE

DELPT = -0.005

LOAD FACTORS TO BE USED

LF = 1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE

ZMAX = 70.000

STEP VALUE FOR SLIP SURFACE DEPTH

ZSTEP = 2.000

REDUCTION FACTOR SIDE, BASE/SOIL

REDBS = 0.400

REDUCTION FACTOR SIDE, SOIL/SOIL

REDSS = 0.500

ACTIVE SIDE CRACK ASSUMPTION : CLOSED CRACK

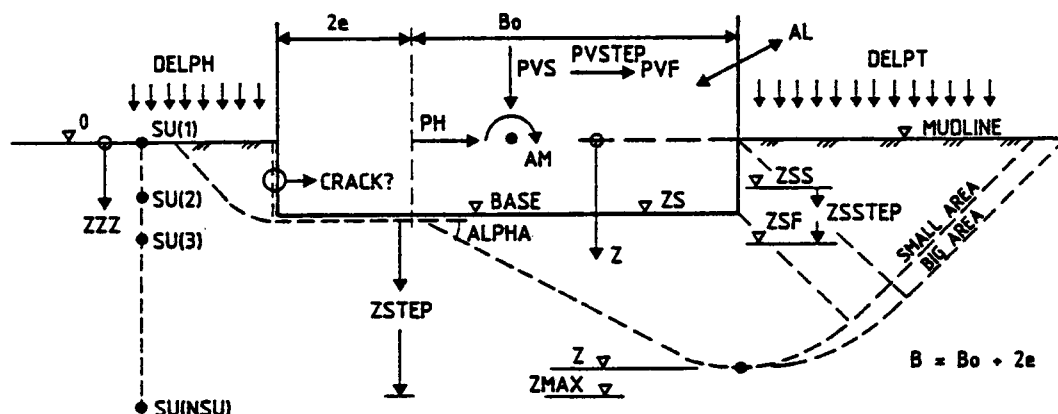
PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

1/ 4

BEARING CAPACITY CALCULATED BY SLIP SURFACES

REVISED INPUT
DATA FORM
FOR CAP

1 PROJECT										57																																																																															
EXAMPLE 5 *** CAP ***																																																																																									
1 UNITS USED										30																																																																															
MN/M																																																																																									
1 B										10 11										20 21										30 31										40 41										50 51										60 61										70 71										80									
100.0										100.0										30.0										30.0										0.0										0.020										0.010										PS 1)																			
1 PVS										10 11										20 21										30 31										40 41										50 51										60 61										70 71										80									
1500.0										1500.0										0.0										100.0										75000.0										0.005										-0.005										FAC 2)																			
1 ZMAX										10 11										20 21										30 31										40 41										50 51										60 61										70 71										80									
70.0										2.0										0.4										0.5										1.0										1.0										45.0										75.0																			
1 NSU										10 11										20 21										30 31										40 41										50 51										60 61										70 71										80									
2										1										1										1										1										0																																							
1 ZSS										10 11										20 21										30 31										40 41										50 51										60 61										70 71										80									
0.0										0.001																																																																															
1 SU 5)										10 11										20 21										30 31										40 41										50 51										60 61										70 71										80									
100.0										0.225																																																																															
1 SUD 6)										10 11										20 21										30 31										40 41										50 51										60 61										70 71										80									
1 SUP 7)										10 11										20 21										30 31										40 41										50 51										60 61										70 71										80									
1 SU2 11)										10 11										20 21										30 31										40 41										50 51										60 61										70 71										80									
1																																																																																									
2																																																																																									
3																																																																																									
4																																																																																									
5																																																																																									
6																																																																																									
7																																																																																									
8																																																																																									
9																																																																																									
10																																																																																									
11																																																																																									
12																																																																																									
13																																																																																									
14																																																																																									
15																																																																																									

- 1) TOTAL UNIT WEIGHT OF SOIL AND WATER
- 2) LOAD FACTOR ON PH, M, ΔPH AND ΔPT = 1.2, LOAD FACTORS 1.0 AND 1.2 = -1.2, ONLY LOAD FACTOR 1.2
- 3) REDUCTION FACTOR FOR SIDE SHEAR, BASE/SOIL. NORMALLY 0.5
- 4) REDUCTION FACTOR FOR SIDE SHEAR, SOIL/SOIL. NORMALLY 0.4
- 5) TRIAXIAL COMPRESSION STRENGTH, OR SU
- 6) SIMPLE SHEAR STRENGTH
- 7) TRIAXIAL EXTENSION STRENGTH
- 8) PRINT OPTIONS FOR VARIOUS PRINT-OUTS
NPR1 = 0 GIVES TABLE OF SF-VALUES
NPR2 = 0 GIVES ALL K (N) FOR SFMIN
NPR3 = 0 GIVES ALL K (N) FOR ZSPEC
NPR4 = 0 OMITTED IN THIS MODE
NPR5 = 0 GIVES PH FOR SF = 1.0
- 9) CRACK ON HEEL SIDE: = 0. OPEN CRACK. = 1.0 CLOSED CRACK.
- 10) TYPES OF PASSIVE FAILURE: = 0. SMALL AREA. = 1.0 BIG AREA
- 11) MINIMUM HORIZONTAL STRENGTH

*) MAY BE LEFT BLANK.

//// INPUT DATA OMITTED IN THIS FAILURE MODE.

LINE WITH * * * * IN COLUMN 1-4 BETWEEN DATA SETS.

BLANK LINE ENDING ALL DATA SETS.

PROJECT : EXAMPLE 5 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

SAFETY FACTOR VERSUS DEPTH

DEPTH OF SLIP SURFACE BELOW MUDLINE Z	BASE/SLIP SURFACE ANGLE ALPHA	SAFETY FACTOR SF	SF= $S(A*SA)/(-S(B))$ S(A*SA)	S(B)
30.000	0.000	3.087	-0.1327E+04	0.4300E+03
32.000	1.827	2.993	-0.1540E+04	0.5145E+03
34.000	3.645	2.692	-0.1610E+04	0.5981E+03
36.000	5.457	2.470	-0.1682E+04	0.6809E+03
38.000	7.267	2.304	-0.1758E+04	0.7633E+03
40.000	9.079	2.175	-0.1838E+04	0.8451E+03
42.000	10.896	2.075	-0.1923E+04	0.9266E+03
44.000	12.721	2.002	-0.2018E+04	0.1008E+04
46.000	14.558	1.947	-0.2121E+04	0.1089E+04
48.000	16.410	1.905	-0.2230E+04	0.1170E+04
50.000	18.280	1.874	-0.2345E+04	0.1251E+04
52.000	20.170	1.851	-0.2467E+04	0.1333E+04
54.000	22.094	1.835	-0.2595E+04	0.1415E+04
56.000	24.034	1.826	-0.2731E+04	0.1496E+04
58.000	26.006	1.822	-0.2873E+04	0.1577E+04
60.000	28.014	1.822	-0.3023E+04	0.1659E+04
62.000	30.062	1.827	-0.3179E+04	0.1740E+04
64.000	32.157	1.835	-0.3343E+04	0.1822E+04
66.000	34.305	1.846	-0.3515E+04	0.1904E+04
68.000	36.512	1.860	-0.3694E+04	0.1986E+04
70.000	38.788	1.876	-0.3881E+04	0.2069E+04

MINIMUM SAFETY FACTOR
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

SF = 1.822
Z = 58.000
ALPHA = 26.006

PROJECT : EXAMPLE 5 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 1.822
 FOUND FOR DEPTH Z = 58.000
 AND SLIP SURFACE ANGLE ALPHA = 26.006

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.4000E+03	0.0000E+00	0.000	0.000	0.4000E+03
2	PA	0.3185E+03	-0.7712E+04	0.035	0.019	0.4650E+03
3	-RH	-0.1353E+03	-0.3614E+04	0.068	0.037	0.0000E+00
4	HA	0.1910E+04	-0.8101E+04	0.102	0.056	0.2365E+04
5	-PP	-0.5815E+03	-0.7712E+04	0.035	0.019	-0.4350E+03
6	-HP	-0.1711E+04	-0.7198E+04	0.125	0.069	-0.1218E+04
7	-RS1	-0.4558E+02	-0.2400E+04	0.035	0.019	0.0000E+00
8	-RS2	-0.1169E+03	-0.1974E+04	0.108	0.059	0.0000E+00
9	-RS3	-0.3819E+02	-0.2011E+04	0.035	0.019	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	-0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.013	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.024	0.013	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.024	0.013	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.024	0.013	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.019	0.0000E+00

PROJECT : EXAMPLE 5 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

SPECIFIED SURFACE CALCULATION

MINIMUM SAFETY FACTOR SF = 1.981
 FOUND FOR DEPTH Z = 45.000
 AND SLIP SURFACE ANGLE ALPHA = 13.587

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.4000E+03	0.0000E+00	0.000	0.000	0.4000E+03
2	PA	0.3303E+03	-0.7712E+04	0.035	0.017	0.4650E+03
3	-RH	-0.1244E+03	-0.3614E+04	0.068	0.034	0.0000E+00
4	HA	0.8717E+03	-0.6903E+04	0.085	0.043	0.1169E+04
5	-PA	-0.5697E+03	-0.7712E+04	0.035	0.017	-0.4350E+03
6	-HP	-0.7992E+03	-0.3856E+04	0.125	0.063	-0.5550E+03
7	-RS1	-0.4192E+02	-0.2400E+04	0.035	0.017	0.0000E+00
8	-RS2	-0.4127E+02	-0.8407E+03	0.097	0.049	0.0000E+00
9	-RS3	-0.2549E+02	-0.1459E+04	0.035	0.017	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.012	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.024	0.012	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.024	0.012	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.024	0.012	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.017	0.0000E+00

PROJECT : EXAMPLE 5 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

SPECIFIED SURFACE CALCULATION

MINIMUM SAFETY FACTOR SF = 1.927
 FOUND FOR DEPTH Z = 75.000
 AND SLIP SURFACE ANGLE ALPHA = 44.809

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.4000E+03	0.0000E+00	0.000	0.000	0.4000E+03
2	PA	0.3265E+03	-0.7712E+04	0.035	0.018	0.4650E+03
3	-RH	-0.1279E+03	-0.3614E+04	0.068	0.035	0.0000E+00
4	HA	0.3497E+04	-0.1034E+05	0.128	0.066	0.4183E+04
5	-PA	-0.5735E+03	-0.7712E+04	0.035	0.018	-0.4350E+03
6	-HP	-0.3170E+04	-0.1157E+05	0.138	0.072	-0.2340E+04
7	-RS1	-0.4309E+02	-0.2400E+04	0.035	0.018	0.0000E+00
8	-RS2	-0.2605E+03	-0.4101E+04	0.122	0.064	0.0000E+00
9	-RS3	-0.4905E+02	-0.2732E+04	0.035	0.018	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.012	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.024	0.012	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.024	0.012	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.024	0.012	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.018	0.0000E+00

PROJECT : EXAMPLE 5 ***CAP***
 SUMMARY OF RESULTS OF STABILITY CALCULATIONS

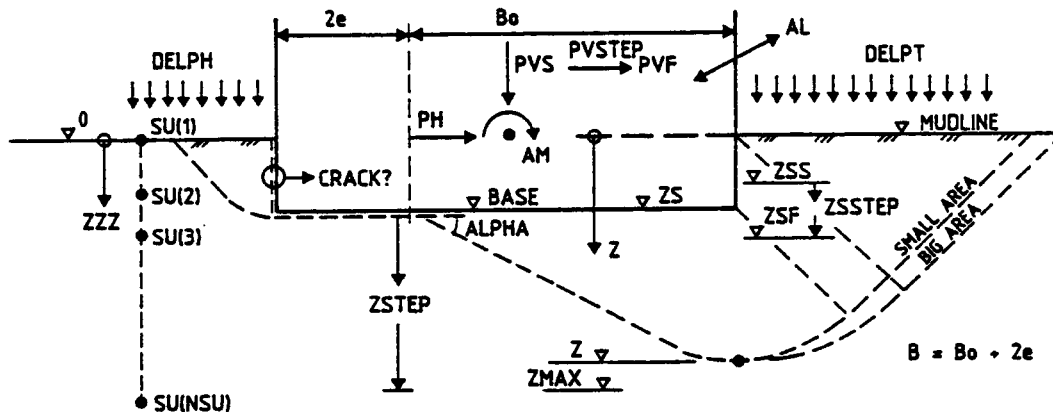
IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOFF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	1500.0	30.0	1.82	58.00	26.0	0.0387	0.0556	0.0573

*)
 THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

BEARING CAPACITY CALCULATED BY SLIP SURFACES

REVISD INPUT
DATA FORM
FOR CAP[illegible]

C A P

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406 DATED 01 JUL 1985

PROJECT : EKSEMPEL 6 ***CAP***

INPUT DATA REPRINT

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	400.000
OVERTURNING MOMENT AT MUDLINE	M =	75000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.005
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.005
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	1.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : CLOSED CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

PROJECT : EKSEMPEL 6 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 2.135
 FOUND FOR DEPTH Z = 56.000
 AND SLIP SURFACE ANGLE ALPHA = 23.515

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.4000E+03	0.0000E+00	0.000	0.000	0.4000E+03
2	PA	0.3400E+03	-0.7712E+04	0.035	0.016	0.4650E+03
3	-RH	-0.1113E+03	-0.3484E+04	0.068	0.032	0.0000E+00
4	HA	0.1762E+04	-0.7984E+04	0.099	0.047	0.2133E+04
5	-PP	-0.5600E+03	-0.7712E+04	0.035	0.016	-0.4350E+03
6	-HP	-0.1493E+04	-0.6684E+04	0.124	0.058	-0.1105E+04
7	-RS1	-0.9724E+02	-0.6000E+04	0.035	0.016	0.0000E+00
8	-RS2	-0.1778E+03	-0.3578E+04	0.106	0.050	0.0000E+00
9	-RS3	-0.6242E+02	-0.3852E+04	0.035	0.016	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.011	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.024	0.011	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.024	0.011	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.024	0.011	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.016	0.0000E+00

PROJECT : EKSEMPEL 6 ***CAP***

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$

MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)

MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	1500.0	30.0	2.13	56.00	23.5	0.0310	0.0556	0.0573

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406 DATED 01 JUL 1985

PROJECT : EKSEMPEL 6 ***CAP***

THIS RUN WAS EXECUTED AT 0/ 0 0

INPUT DATA REPRINT

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION	B =	110.000
LENGTH OF FOUNDATION	L =	90.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	400.000
OVERTURNING MOMENT AT MUDLINE	M =	75000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.005
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.005
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	1.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

PROJECT : EKSEMPEL 6 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 2.304
 FOUND FOR DEPTH Z = 56.000
 AND SLIP SURFACE ANGLE ALPHA = 20.405

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.4000E+03	0.0000E+00	0.000	0.000	0.4000E+03
2	PA	0.3142E+03	-0.6941E+04	0.035	0.015	0.4185E+03
3	-RH	-0.9440E+02	-0.3188E+04	0.068	0.030	0.0000E+00
4	HA	0.1524E+04	-0.7882E+04	0.099	0.043	0.1863E+04
5	-PP	-0.4958E+03	-0.6941E+04	0.035	0.015	-0.3915E+03
6	-HP	-0.1318E+04	-0.6016E+04	0.124	0.054	-0.9945E+03
7	-RS1	-0.9914E+02	-0.6600E+04	0.035	0.015	0.0000E+00
8	-RS2	-0.1730E+03	-0.3805E+04	0.105	0.045	0.0000E+00
9	-RS3	-0.5786E+02	-0.3852E+04	0.035	0.015	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.010	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.024	0.010	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.024	0.010	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.024	0.010	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.015	0.0000E+00

PROJECT : EKSEMPEL 6 ***CAP***
 SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOFF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	1500.0	30.0	2.30	56.00	20.4	0.0286	0.0524	0.0538

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406 DATED 01 JUL 1985

PROJECT : EKSEMPEL 6 ***CAP***

THIS RUN WAS EXECUTED AT 0/ 0 0

INPUT DATA REPRINT

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION	B =	120.000
LENGTH OF FOUNDATION	L =	80.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	1500.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	400.000
OVERTURNING MOMENT AT MUDLINE	M =	75000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.005
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.005
LOAD FACTORS TO BE USED	LF =	1.0

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	1.000
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	1.000

ACTIVE SIDE CRACK ASSUMPTION : OPEN CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	UNDRAINED SHEAR STRENGTH	MINIMUM SHEAR STRENGTH
0.000	0.001	0.001
100.000	0.225	0.225

PROJECT : EKSEMPEL 6 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 1500.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM SAFETY FACTOR SF = 2.418
 FOUND FOR DEPTH Z = 54.000
 AND SLIP SURFACE ANGLE ALPHA = 16.693

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.4000E+03	0.0000E+00	0.000	0.000	0.4000E+03
2	PA	0.2837E+03	-0.6170E+04	0.035	0.014	0.3720E+03
3	-RH	-0.8223E+02	-0.2916E+04	0.068	0.028	0.0000E+00
4	HA	0.1191E+04	-0.7487E+04	0.096	0.040	0.1489E+04
5	-PP	-0.4363E+03	-0.6170E+04	0.035	0.014	-0.3480E+03
6	-HP	-0.1048E+04	-0.4936E+04	0.123	0.051	-0.7968E+03
7	-RS1	-0.1030E+03	-0.7200E+04	0.035	0.014	0.0000E+00
8	-RS2	-0.1522E+03	-0.3613E+04	0.102	0.042	0.0000E+00
9	-RS3	-0.5268E+02	-0.3682E+04	0.035	0.014	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.010	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.024	0.010	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.024	0.010	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.024	0.010	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.014	0.0000E+00

PROJECT : EKSEMPEL 6 ***CAP***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS

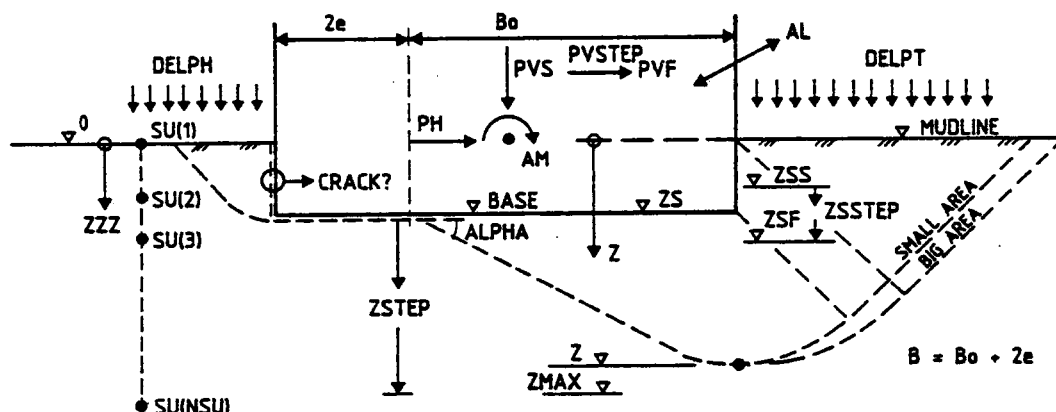
IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	1500.0	30.0	2.42	54.00	16.7	0.0265	0.0504	0.0515

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

REVISÉ INPUT
DATA FORM
FOR CAP[illegible]

*) MAY BE LEFT BLANK.
 //// INPUT DATA OMITTED IN THIS FAILURE MODE.
 LINE WITH * * * IN COLUMN 1-4 BETWEEN DATA SETS.
 BLANK LINE ENDING ALL DATA SETS.

C A P

----- NGI 1985 -----

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 1

METHOD DESCRIBED IN NGI INTERNAL REPORT 52406 DATED 01 JUL 1985

PROJECT : EKSEMPEL 7 ***CAP***

INPUT DATA REPRINT

UNITS USED : SI-ENHETER DVS MN & M

WIDTH OF FOUNDATION	B =	100.000
LENGTH OF FOUNDATION	L =	100.000
DEPTH OF EMBEDMENT , START VALUE	ZSS =	30.000
DEPTH OF EMBEDMENT , FINAL VALUE	ZSF =	30.000
DEPTH OF EMBEDMENT , STEP VALUE	ZSSTEP =	0.000
SOIL TOTAL UNIT WEIGHT	GAMS =	0.020
WATER UNIT WEIGHT	GAMW =	0.010

SUBMERGED WEIGHT OF STRUCTURE , START VALUE	PVS =	2000.000
SUBMERGED WEIGHT OF STRUCTURE , FINAL VALUE	PVF =	2000.000
SUBMERGED WEIGHT OF STRUCTURE , STEP VALUE	PVSTEP =	0.000
HORIZONTAL FORCE AT MUDLINE	PH =	750.000
OVERTURNING MOMENT AT MUDLINE	M =	100000.000
EXCESS WATER PRESSURE OUTSIDE HEEL	DELPH =	0.070
EXCESS WATER PRESSURE OUTSIDE TOE	DELPT =	-0.070
LOAD FACTORS TO BE USED	LF =	1.3

MAXIMUM DEPTH OF SLIP SURFACE BELOW BASE	ZMAX =	70.000
STEP VALUE FOR SLIP SURFACE DEPTH	ZSTEP =	2.000
REDUCTION FACTOR SIDE, BASE/SOIL	REDBS =	0.500
REDUCTION FACTOR SIDE, SOIL/SOIL	REDSS =	0.400

ACTIVE SIDE CRACK ASSUMPTION : CLOSED CRACK

PASSIVE SIDE AREA ASSUMPTION : BIG AREA

SHEAR STRENGTH USED :

DEPTH BELOW MUDLINE	TRIAXIAL COMPRESSION STRENGTH	SIMPLE SHEAR STRENGTH	TRIAXIAL EXTENSION STRENGTH	MINIMUM HORIZONTAL STRENGTH
0.000	0.001	0.001	0.001	0.001
100.000	0.225	0.200	0.180	0.050

PROJECT : EKSEMPEL 7 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 2000.0 BASE EMB : 30.00

HORIZONTAL FORCE VERSUS DEPTH

DEPTH OF SLIP	BASE/SLIP	HORIZ.	PH= S(A*SA)+S(B)-B(1)	
SURFACE BELOW MUDLINE	SURFACE ANGLE	FORCE	S(A*SA)	S(B)
Z	ALPHA	PH		
30.000	0.000	331.5	-0.7515E+03	0.7515E+03
32.000	1.975	646.3	-0.1193E+04	0.1193E+04
34.000	4.228	537.2	-0.1234E+04	0.1234E+04
36.000	6.198	480.2	-0.1302E+04	0.1302E+04
38.000	8.173	424.1	-0.1371E+04	0.1371E+04
40.000	10.113	376.2	-0.1444E+04	0.1444E+04
42.000	12.038	334.2	-0.1521E+04	0.1521E+04
44.000	13.956	302.4	-0.1606E+04	0.1606E+04
46.000	15.890	276.4	-0.1697E+04	0.1697E+04
48.000	17.839	255.9	-0.1794E+04	0.1794E+04
50.000	19.810	240.6	-0.1895E+04	0.1895E+04
52.000	21.813	230.1	-0.2002E+04	0.2002E+04
54.000	23.857	224.3	-0.2115E+04	0.2115E+04
56.000	25.949	222.7	-0.2233E+04	0.2233E+04
58.000	28.100	225.3	-0.2356E+04	0.2356E+04
60.000	30.321	231.7	-0.2486E+04	0.2486E+04
62.000	32.624	241.7	-0.2621E+04	0.2621E+04
64.000	35.022	255.0	-0.2762E+04	0.2762E+04
66.000	37.533	271.5	-0.2910E+04	0.2910E+04
68.000	40.175	290.9	-0.3063E+04	0.3063E+04
70.000	42.975	313.0	-0.3221E+04	0.3221E+04

MINIMUM HORIZONTAL FORCE
FOUND FOR DEPTH
AND SLIP SURFACE ANGLE

PH = 222.7
Z = 56.000
ALPHA = 25.949

PROJECT : EKSEMPEL 7 ***CAP***

LOAD FACTOR : 1.00 SUBMERGED WEIGHT : 2000.0 BASE EMB : 30.00

FORCES CALCULATED FOR CRITICAL SURFACE

MINIMUM HORIZONTAL FORCE PH = 222.7
 FOUND FOR DEPTH Z = 56.000
 AND SLIP SURFACE ANGLE ALPHA = 25.949

$$K(N) = A(N) * SA(N)/SF + B(N)$$

N	CODE	K(N)	A(N)	SA(N)	SA(N)/SF	B(N)
1	PH	0.2227E+03	0.0000E+00	0.000	0.000	0.2227E+03
2	PA	0.3932E+03	-0.7712E+04	0.035	0.035	0.6600E+03
3	-RH	-0.6371E+02	-0.4058E+04	0.016	0.016	0.0000E+00
4	HA	0.1799E+04	-0.7531E+04	0.097	0.097	0.2526E+04
5	-PP	-0.4548E+03	-0.7712E+04	0.028	0.028	-0.2400E+03
6	-HP	-0.1625E+04	-0.6684E+04	0.103	0.103	-0.9360E+03
7	-RS1	-0.9255E+02	-0.3000E+04	0.031	0.031	0.0000E+00
8	-RS2	-0.1311E+03	-0.1377E+04	0.095	0.095	0.0000E+00
9	-RS3	-0.4753E+02	-0.1541E+04	0.031	0.031	0.0000E+00
11	DHA01	0.0000E+00	0.0000E+00	0.000	0.000	0.0000E+00
12	DHA02	0.0000E+00	0.0000E+00	0.024	0.024	0.0000E+00
15	DHA05	0.0000E+00	0.0000E+00	0.019	0.019	0.0000E+00
17	DHA07	0.0000E+00	0.0000E+00	0.021	0.021	0.0000E+00
19	DHA09	0.0000E+00	0.0000E+00	0.021	0.021	0.0000E+00
20	DHA10	0.0000E+00	0.0000E+00	0.035	0.035	0.0000E+00

PROJECT : EKSEMPEL 7 ***CAP***
SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	MINIMUM HORIZON FORCE	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR.HA	MOB SHEAR M'HOFF
LF	PV	ZS	PH	Z	ALPHA	CSS	CB	CM
1.00	2000.0	30.0	223.	56.00	25.9	0.0564	0.0837	0.0873

*)

THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT

	JOB TITLE:	CONTRACT NO. 52406 - 46	PAGE IV - 0
	SUBJECT: The Computer Program CAP	SIGN. RAL	DATE 20/9 - 85
		CONTR.	DATE

APPENDIX IV : PARAMETER STUDY

	JOB TITLE:	CONTRACT NO.	PAGE
	SUBJECT:	52406-46	IV - 1
		RAL	DATE 20/9-85
CAP	CONTR.	DATE	

PARAMETER STUDY

The CAP results are compared with the results of Brinch Hansen's and Meyerhof's formulas.

Base case parameters are as follows:

$B = 100 \text{ m}$	$\Delta P_H = 50 \text{ kPa}$
$L = 100 \text{ m}$	$\Delta P_T = -50 \text{ kPa}$
$Z_s = 5 \text{ m}$	$L_F = 1.0$
$\gamma = 20 \text{ kN/m}^3$	$RED_{BS} = 0.5$
$\gamma_w = 10 \text{ kN/m}^3$	$RED_{SS} = 0.4$
$P_v = 2000 \text{ MN}$	$CRACK = 0$
$P_H = 600 \text{ MN}$	$PASTYP = 1$
$M = 30000 \text{ MNm}$	$S_u = 10 \text{ kPa}$

In the parameter study one parameter is varied at the time, the others are kept constant.



JOB TITLE:

SUBJECT:

CAP

CONTRACT NO.

52406-46

PAGE

IV - 2

SIGN.

RAL

DATE

29/9-85

CONTR.

DATE

RESULTS PARAMETER STUDY 1, REDSS

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 2 / 2

PROJECT : PARAMETER STUDY

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITION THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

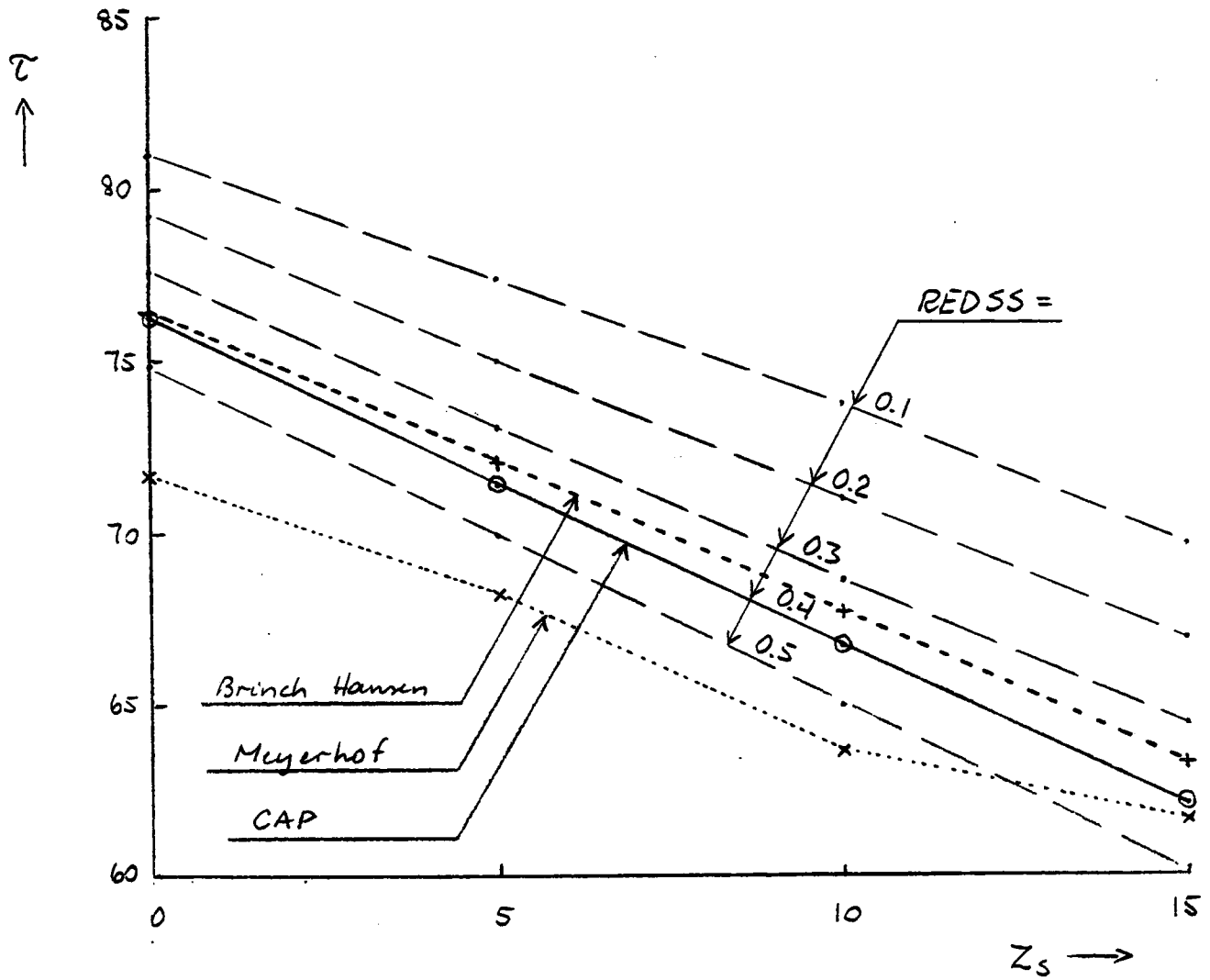
AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR. HA	MOB SHEAR M' HOF
REDSS	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
0.1	2000.0	0.0	1.14	46.00	41.1	0.0810	0.0764	0.0717
	2000.0	5.0	1.22	57.00	44.2	0.0773	0.0720	0.0681
	2000.0	10.0	1.30	65.00	44.8	0.0736	0.0676	0.0636
	2000.0	15.0	1.39	72.00	45.0	0.0695	0.0632	0.0616
0.2	2000.0	0.0	1.18	39.00	33.9	0.0792	0.0764	0.0717
	2000.0	5.0	1.26	55.00	42.1	0.0749	0.0720	0.0681
	2000.0	10.0	1.35	65.00	44.9	0.0709	0.0676	0.0636
	2000.0	15.0	1.45	72.00	45.0	0.0668	0.0632	0.0616
0.3	2000.0	0.0	1.21	35.00	30.0	0.0776	0.0764	0.0717
	2000.0	5.0	1.30	48.00	35.2	0.0730	0.0720	0.0681
	2000.0	10.0	1.40	63.00	42.8	0.0685	0.0676	0.0636
	2000.0	15.0	1.51	71.00	44.0	0.0643	0.0632	0.0616
0.4	2000.0	0.0	1.23	32.00	27.2	0.0762	0.0764	0.0717
	2000.0	5.0	1.34	44.00	31.5	0.0713	0.0720	0.0681
	2000.0	10.0	1.45	57.00	37.1	0.0666	0.0676	0.0636
	2000.0	15.0	1.57	71.00	44.1	0.0620	0.0632	0.0616
0.5	2000.0	0.0	1.26	29.00	24.5	0.0749	0.0764	0.0717
	2000.0	5.0	1.37	41.00	28.9	0.0698	0.0720	0.0681
	2000.0	10.0	1.50	53.00	33.5	0.0649	0.0676	0.0636
	2000.0	15.0	1.63	66.00	39.3	0.0601	0.0632	0.0616



JOB TITLE:	CONTRACT NO. S2406-46	PAGE IV - 3
SUBJECT:	SIGN. RAL	DATE 20/9-85
CAP	CONTR.	DATE

PARAMETER REDSS



	JOB TITLE:	CONTRACT NO. 52406-46	PAGE IV - 4
	SUBJECT: CAP	SIGN. RAL	DATE 20/6-85
		CONTR.	DATE

RESULTS PARAMETER STUDY 2 , B/L

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 2 / 2

PROJECT : PARAMETER STUDY

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITON THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$

MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)

MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR. HA	MOB SHEAR M' HOF
B/L	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
0	2000000.0	0.0	1.11	49.00	44.4	0.0835	0.0811	0.0804
0	2000000.0	5.0	1.15	57.00	44.2	0.0815	0.0788	0.0771
0.5	4000.0	0.0	1.18	39.00	33.9	0.0792	0.0786	0.0758
0.5	4000.0	5.0	1.25	53.00	40.1	0.0756	0.0752	0.0728
1.0	2000.0	0.0	1.23	32.00	27.2	0.0762	0.0764	0.0717
1.0	2000.0	5.0	1.34	44.00	31.5	0.0713	0.0720	0.0681
2.0	1000.0	0.0	1.32	24.00	20.1	0.0717	0.0730	0.0646
2.0	1000.0	5.0	1.49	35.00	23.7	0.0647	0.0665	0.0605



JOB TITLE:

SUBJECT:

CAP

CONTRACT NO.

S2406-46

PAGE

IV - 5

SIGN.

RAL

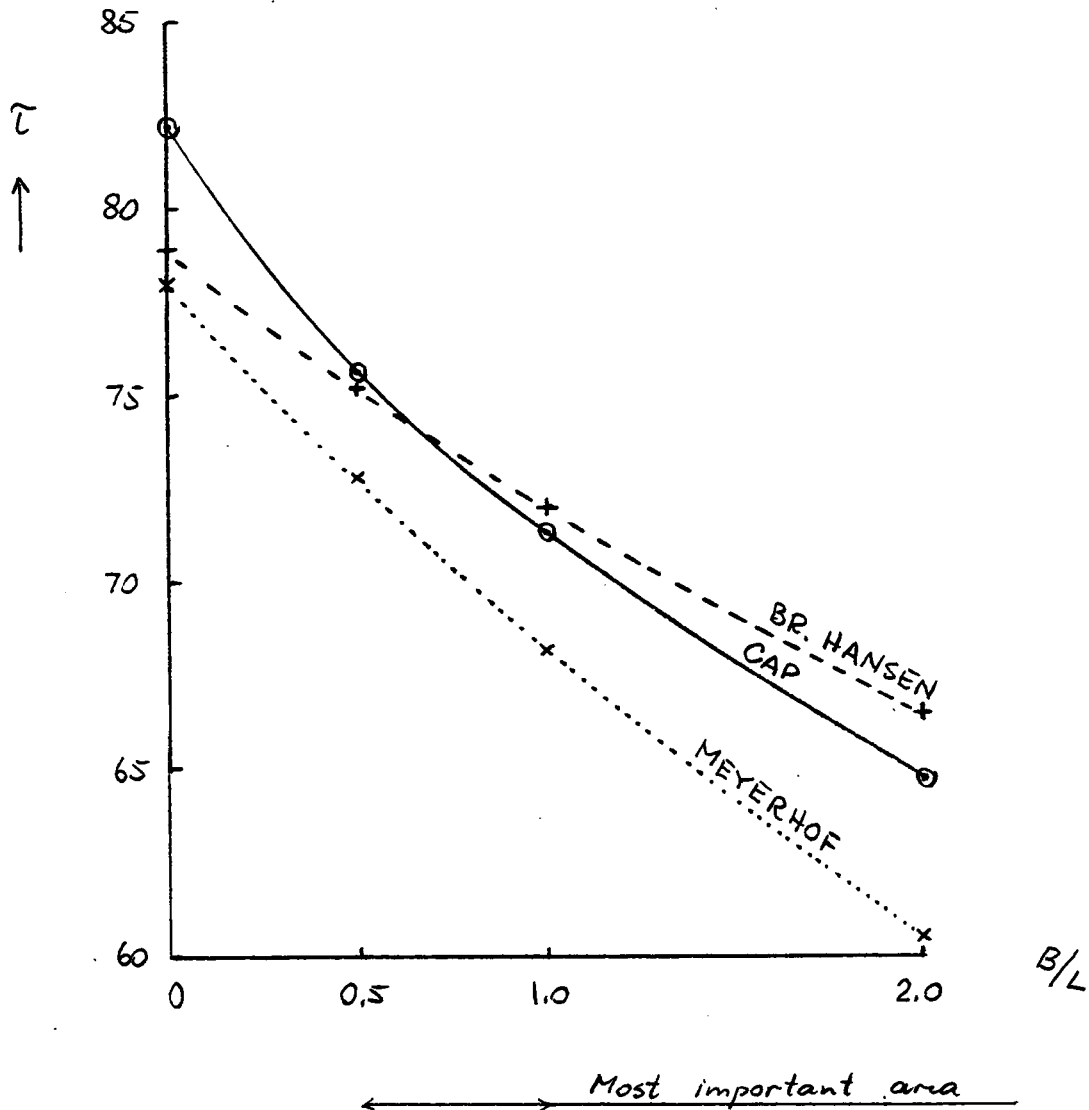
DATE

20/9-85

CONTR.

DATE

PARAMETER B/L





JOB TITLE:

SUBJECT:

CAP

CONTRACT NO.

52406-46

PAGE

IV - 6

SIGN.

RAL

DATE

20/9 - 85

CONTR.

DATE

RESULTS PARAMETER STUDY 3, Z_s & P_v

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 2 / 2

PROJECT : PARAMETER STUDY

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITION THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR. HA	MOB SHEAR M' HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
1.00	1000.0	0.0	1.36	23.00	35.1	0.0698	0.0695	0.0676
1.00	1000.0	5.0	1.57	32.00	28.1	0.0614	0.0622	0.0587
1.00	1000.0	10.0	1.76	44.00	30.9	0.0556	0.0570	0.0538
1.00	1000.0	15.0	1.94	57.00	35.8	0.0507	0.0523	0.0499
1.00	1500.0	0.0	1.33	27.00	26.7	0.0710	0.0713	0.0672
1.00	1500.0	5.0	1.46	38.00	29.0	0.0656	0.0664	0.0625
1.00	1500.0	10.0	1.60	51.00	34.1	0.0606	0.0619	0.0589
1.00	1500.0	15.0	1.75	65.00	40.6	0.0560	0.0575	0.0556
1.00	2000.0	0.0	1.23	32.00	27.2	0.0762	0.0764	0.0717
1.00	2000.0	5.0	1.34	44.00	31.5	0.0713	0.0720	0.0681
1.00	2000.0	10.0	1.45	57.00	37.1	0.0666	0.0676	0.0636
1.00	2000.0	15.0	1.57	71.00	44.1	0.0620	0.0632	0.0616
1.00	2500.0	0.0	1.13	37.00	29.1	0.0826	0.0827	0.0776
1.00	2500.0	5.0	1.22	49.00	33.9	0.0778	0.0782	0.0745
1.00	2500.0	10.0	1.32	62.00	39.8	0.0730	0.0737	0.0703
1.00	2500.0	15.0	1.42	73.00	44.2	0.0684	0.0693	0.0680
1.00	3000.0	0.0	1.04	41.00	30.8	0.0898	0.0895	0.0844
1.00	3000.0	5.0	1.12	53.00	35.8	0.0848	0.0849	0.0812
1.00	3000.0	10.0	1.20	67.00	42.9	0.0798	0.0802	0.0780
1.00	3000.0	15.0	1.30	75.00	44.6	0.0750	0.0756	0.0746

*)

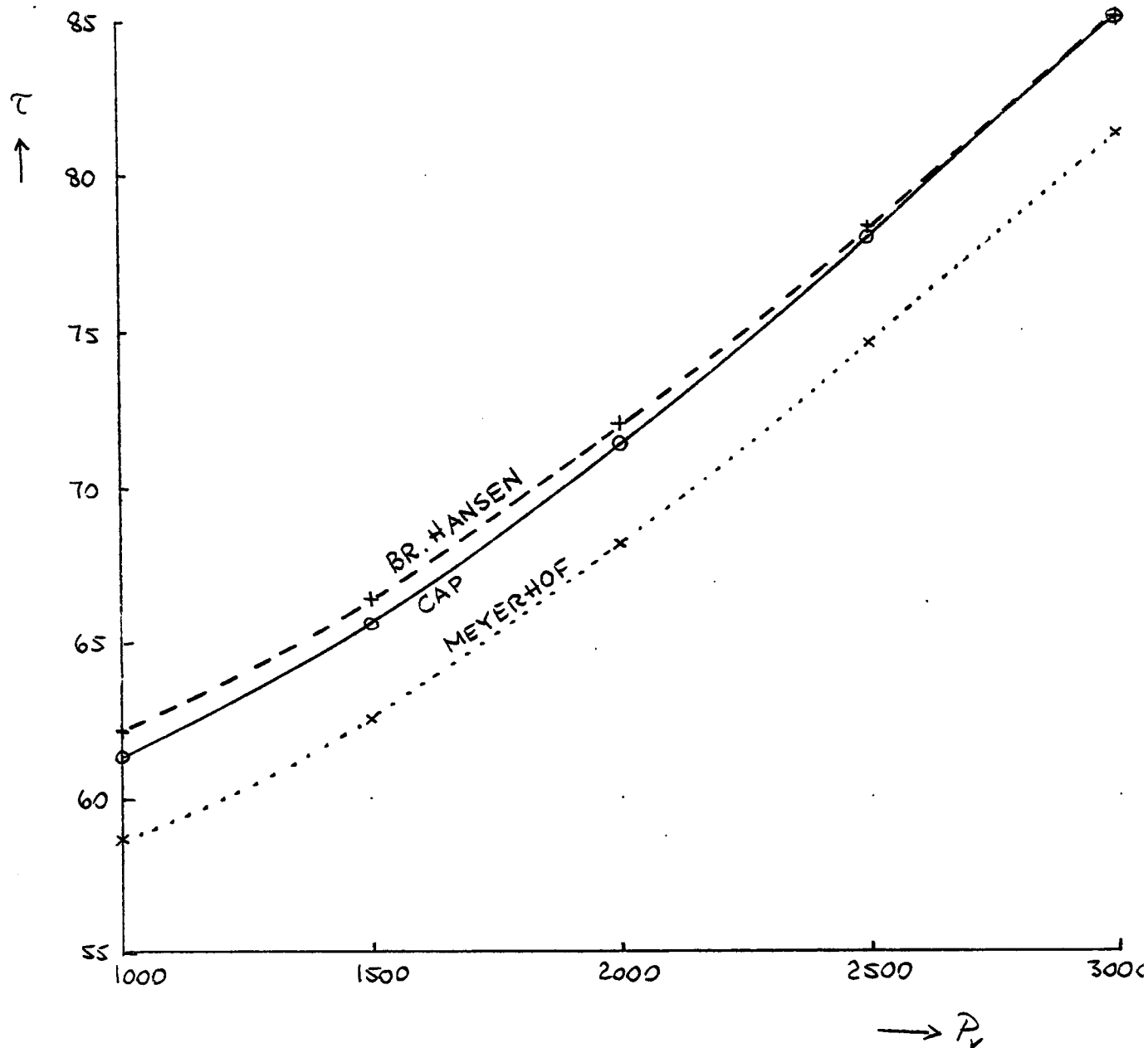
THE TWO MENTIONED FORMULAS ARE BASED ON THE EFFECTIVE AREA CONCEPT



JOB TITLE:	
SUBJECT:	CAP

CONTRACT NO.	52406-46	PAGE	JV - 7
SIGN.	RAL	DATE	20/9-85
CONTR.		DATE	

PARAMETER P_v



	JOB TITLE:	CONTRACT NO. 52406-46	PAGE IV - 8
	SUBJECT: CAP	SIGN. <i>RAL</i>	DATE 20/9 - 85
		CONTR.	DATE

RESULTS PARAMETER STUDY 4, P_H

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 2 / 2

PROJECT : PARAMETER STUDY

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITION THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

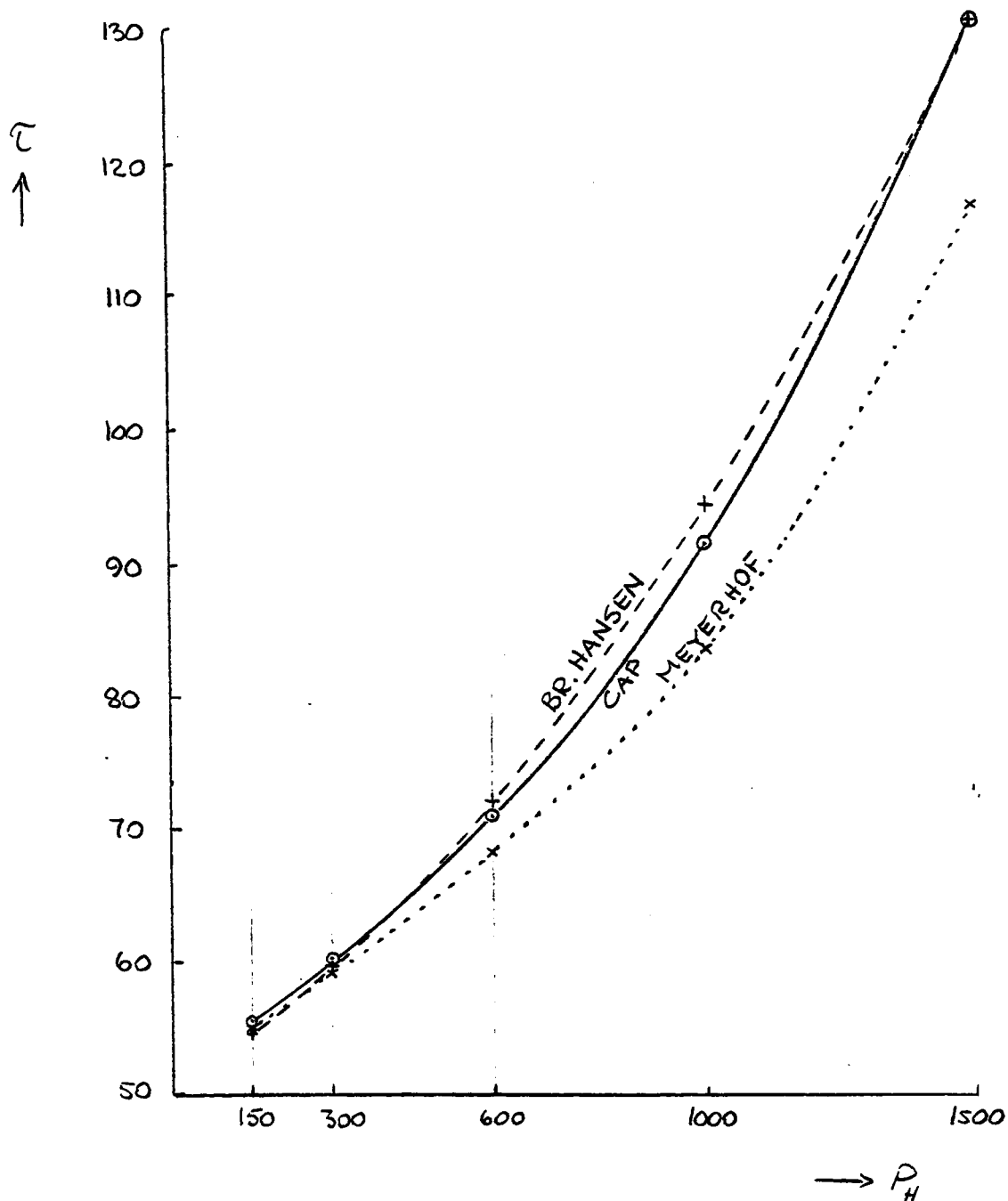
AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR. HA	MOB SHEAR M' HOF
P_H	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
150.0	2000.0	0.0	1.57	49.00	44.4	0.0592	0.0567	0.0569
150.0	2000.0	5.0	1.69	58.00	44.1	0.0559	0.0547	0.0550
300.0	2000.0	0.0	1.45	49.00	44.4	0.0641	0.0622	0.0611
300.0	2000.0	5.0	1.56	58.00	44.5	0.0606	0.0593	0.0586
600.0	2000.0	0.0	1.23	32.00	27.2	0.0762	0.0764	0.0717
600.0	2000.0	5.0	1.34	44.00	31.5	0.0713	0.0720	0.0681
1000.0	2000.0	0.0	0.95	13.00	10.7	0.1012	0.1031	0.0881
1000.0	2000.0	5.0	1.06	25.00	15.9	0.0916	0.0940	0.0833
1500.0	2000.0	0.0	0.67	0.00	0.0	0.1500	0.100000.	0.1176
1500.0	2000.0	5.0	0.77	5.00	0.0	0.1303	0.1305	0.1065



JOB TITLE:	CONTRACT NO. 52406-46	PAGE IV - 9
SUBJECT: CAP	SIGN. RAL	DATE 20/9-85
	CONTR.	DATE

PARAMETER P_H





JOB TITLE:

CONTRACT NO.

52406-46

PAGE

IV - 10

SUBJECT:

CAP

SIGN.

RAL

DATE

20/5-85

CONTR.

DATE

RESULTS PARAMETER STUDY 5 , M

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 2/ 2

PROJECT : PARAMETER STUDY

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITION THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$

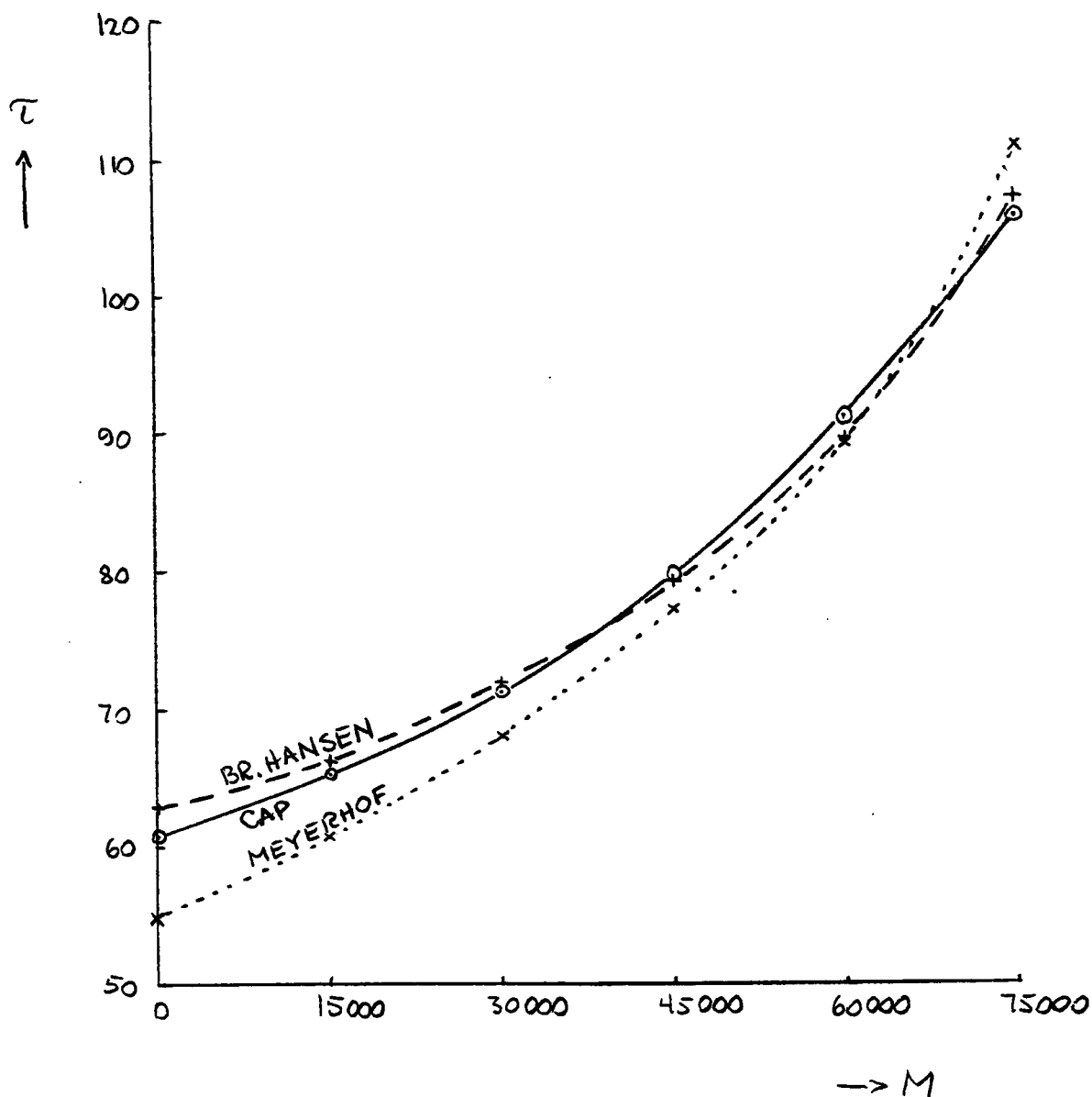
MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)

MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR. HA	MOB SHEAR M' HOF
M	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
0.0	2000.0	0.0	1.45	24.00	13.9	0.0652	0.0670	0.0570
0.0	2000.0	5.0	1.58	36.00	18.4	0.0610	0.0629	0.0549
15000.0	2000.0	0.0	1.35	28.00	19.2	0.0696	0.0708	0.0634
15000.0	2000.0	5.0	1.47	40.00	23.9	0.0654	0.0668	0.0609
30000.0	2000.0	0.0	1.23	32.00	27.2	0.0762	0.0764	0.0717
30000.0	2000.0	5.0	1.34	44.00	31.5	0.0713	0.0720	0.0681
45000.0	2000.0	0.0	1.08	37.00	42.3	0.0864	0.0850	0.0828
45000.0	2000.0	5.0	1.20	49.00	44.6	0.0796	0.0791	0.0771
60000.0	2000.0	0.0	0.92	28.00	44.4	0.1030	0.0993	0.0993
60000.0	2000.0	5.0	1.05	40.00	43.5	0.0913	0.0892	0.0888

	JOB TITLE:	CONTRACT NO. S2406-46	PAGE IV - 11
	SUBJECT: CAP	SIGN. RAL	DATE 20/9-85
		CONTR.	DATE

PARAMETER M





JOB TITLE:

CONTRACT NO.

52406-46

PAGE

IV - 12

SUBJECT:

CAP

SIGN.

RAL

DATE

20/9-85

CONTR.

DATE

RESULTS PARAMETER STUDY 6 , Δp

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 2 / 2

PROJECT : PARAMETER STUDY

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

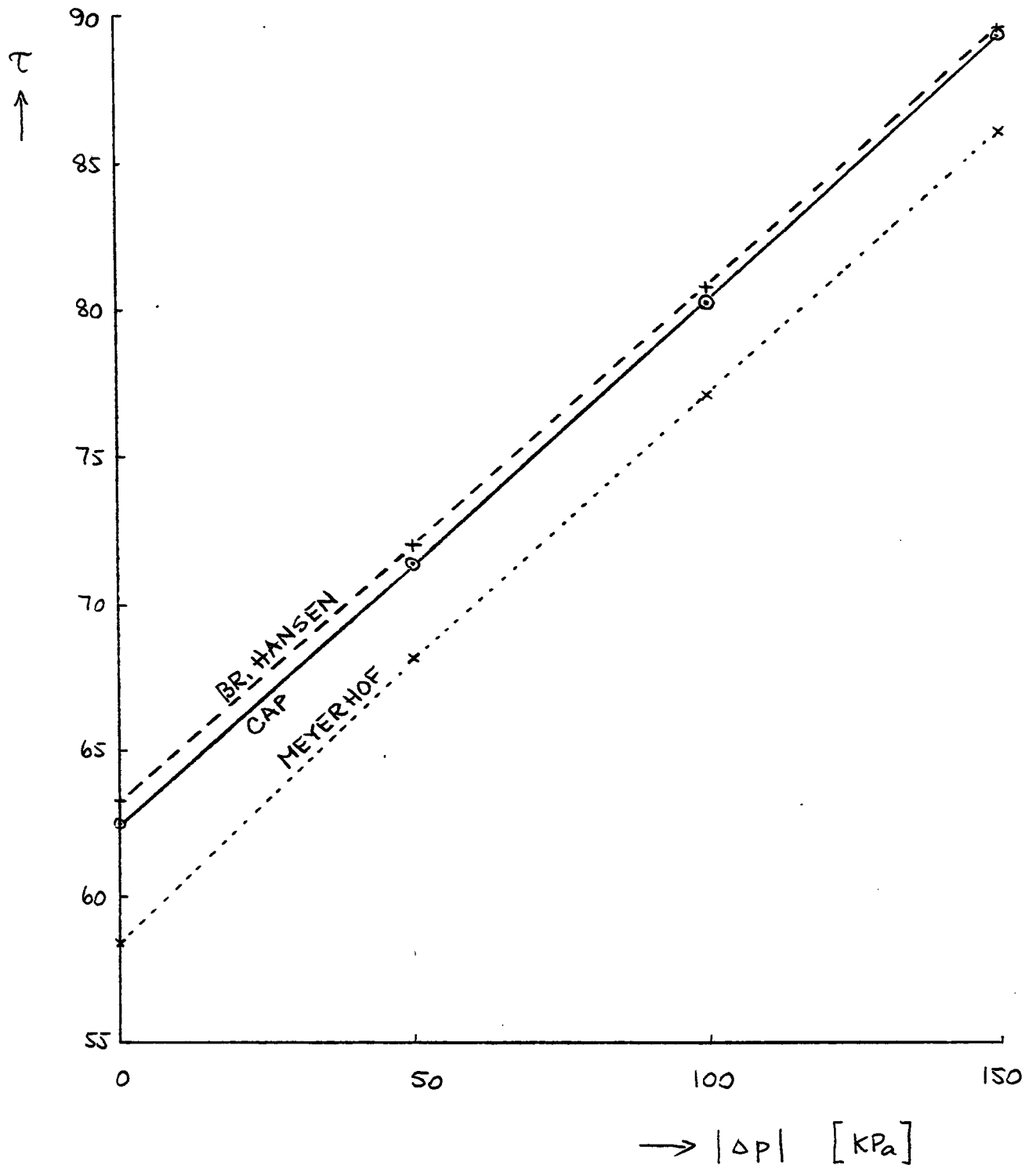
IN ADDITION THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$
 MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = CB *)
 MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = CM *)

	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR. HA	MOB SHEAR M' HOF
Δp	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
0	2000.0	0.0	1.36	26.00	21.8	0.0695	0.0702	0.0641
	2000.0	5.0	1.53	41.00	28.9	0.0625	0.0633	0.0584
± 0.05	2000.0	0.0	1.23	32.00	27.2	0.0762	0.0764	0.0717
	2000.0	5.0	1.34	44.00	31.5	0.0713	0.0720	0.0681
± 0.10	2000.0	0.0	1.12	36.00	30.9	0.0834	0.0831	0.0792
	2000.0	5.0	1.19	46.00	33.3	0.0802	0.0807	0.0770
± 0.15	2000.0	0.0	1.03	40.00	34.8	0.0909	0.0900	0.0868
	2000.0	5.0	1.07	48.00	35.2	0.0891	0.0894	0.0859

	JOB TITLE:	CONTRACT NO.	PAGE
	SUBJECT:	52406-46	IV - 13
		SIGN.	DATE
	CAP	RAL	20/9-85
		CONTR.	DATE

PARAMETER Δp





JOB TITLE:

CONTRACT NO.

52406-46

PAGE

IV - 14

SUBJECT:

CAP

SIGN.

RAL

DATE

20/9-85

CONTR.

DATE

RESULTS PARAMETER STUDY 7 , LF

BEARING CAPACITY CALCULATED BY SLIP SURFACES

PAGE 2/ 2

PROJECT : PARAMETER STUDY

SUMMARY OF RESULTS OF STABILITY CALCULATIONS

IN ADDITION THE FOLLOWING RESULTS HAVE BEEN INCLUDED :

AVERAGE MOBILIZED SHEAR STRESS ALONG SLIP SURFACE $CSS = -S(B)/S(A)$ MOBILIZED SHEAR STRESS ACCORDING TO BRINCH HANSEN FORMULA = $CB *$ MOBILIZED SHEAR STRESS ACCORDING TO MEYERHOF FORMULA = $CM *$

LOAD FACT USED	SUBMERGED PLATFORM WEIGHT	DEPTH BASE EMBED	SAFETY FACTOR	DEPTH SLIP SURFA	SLIP SURFA ANGLE	MOB SHEAR NGI	MOB SHEAR BR. HA	MOB SHEAR M' HOF
LF	PV	ZS	SF	Z	ALPHA	CSS	CB	CM
0.50	2000.0	0.0	1.77	47.00	33.6	0.0526	0.0520	0.0495
0.50	2000.0	5.0	1.91	60.00	38.9	0.0495	0.0492	0.0475
1.00	2000.0	0.0	1.23	32.00	27.2	0.0762	0.0764	0.0717
1.00	2000.0	5.0	1.34	44.00	31.5	0.0713	0.0720	0.0681
1.50	2000.0	0.0	0.90	25.00	27.0	0.1048	0.1051	0.0996
1.50	2000.0	5.0	0.99	36.00	30.2	0.0970	0.0982	0.0932
2.00	2000.0	0.0	0.68	23.00	35.1	0.1397	0.1390	0.1352
2.00	2000.0	5.0	0.76	33.00	35.0	0.1270	0.1279	0.1237
2.50	2000.0	0.0	0.51	17.00	42.8	0.1891	0.1844	0.1825
2.50	2000.0	5.0	0.59	30.00	43.8	0.1644	0.1633	0.1599



JOB TITLE:

SUBJECT:

CAP

CONTRACT NO.

52406-46

PAGE

IV - 15

SIGN.

PAL

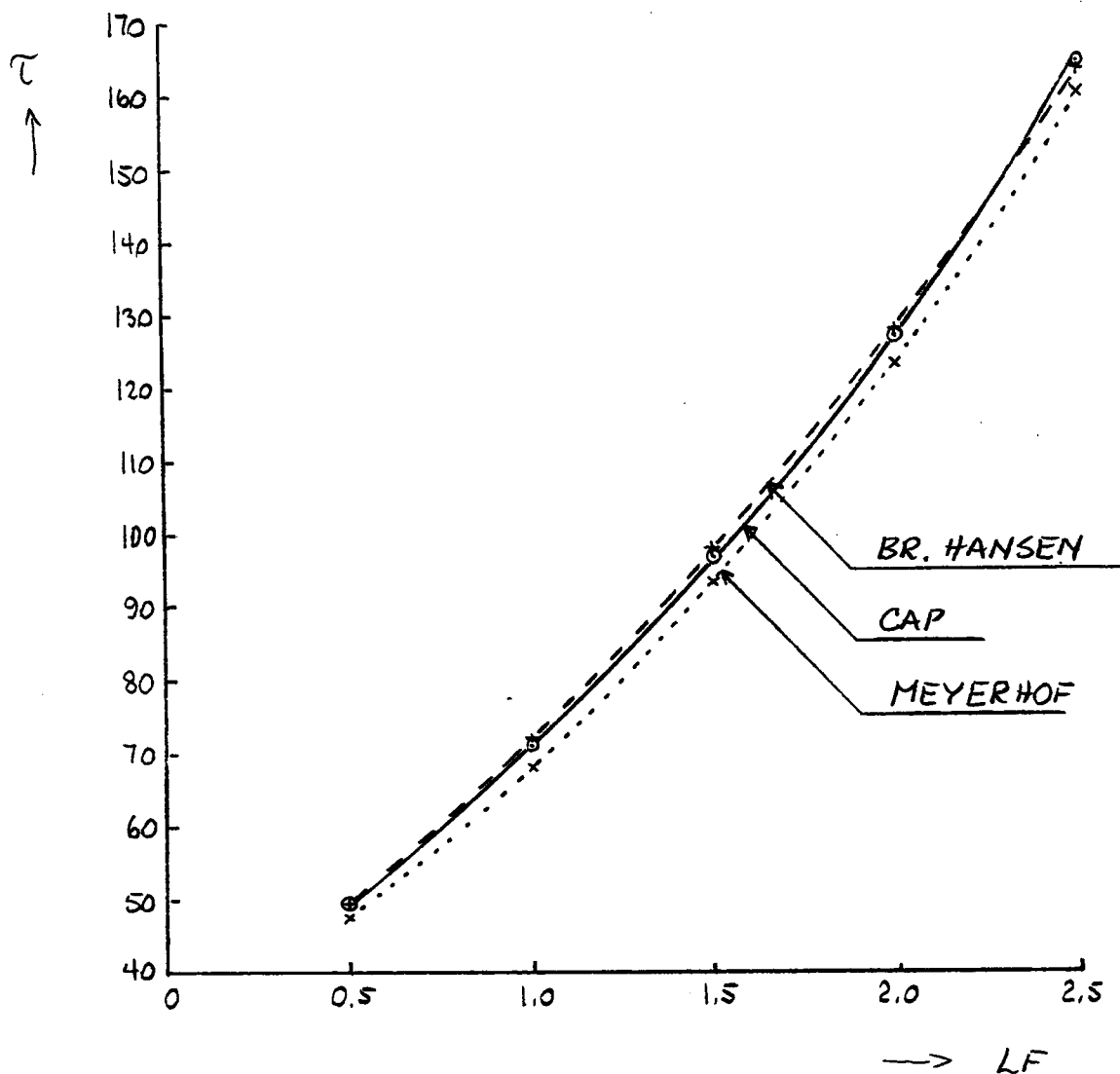
DATE

20/9-85

CONTR.

DATE

PARAMETER LF



	JOB TITLE:	CONTRACT NO.	PAGE
	SUBJECT: CAP SUMMARY PARAMETER STUDY	SIGN. P4L	IV - 16
		CONTR.	DATE 20/9-85

PARAMETER	DEPTH	MOB SHEAR CAP	MOB SHEAR BR. HANSEN	DIFFERENCE IN %	AVERAGE DIFF. %
$B/L = 0$	0	83.5	81.1	3.0	0.2
	5	81.5	78.8	3.4	
	0	79.2	78.6	0.8	
	5	75.6	75.2	0.5	
	0	76.2	76.4	-0.3	
	5	71.3	72.0	-1.0	
0.5	0	71.7	73.0	-1.8	-1.1
	5	64.7	66.5	-2.7	
	0	69.8	69.5	0.4	
	5	61.4	62.2	-1.3	
	10	55.6	57.0	-2.5	
	15	50.7	52.3	-3.1	
1.0	0	71.0	71.3	-0.4	-1.1
	5	65.6	66.4	-1.2	
	10	60.6	61.9	-2.1	
	15	56.0	57.5	-2.6	
	0	76.2	76.4	-0.3	
	5	71.3	72.0	-1.0	
2.0	0	66.6	67.6	-1.5	-1.1
	5	62.0	63.2	-1.9	
	10	82.6	82.7	-0.1	
	5	77.8	78.2	-0.5	
	10	73.0	73.7	-0.9	
	15	68.4	69.3	-1.3	
2500	0	89.8	89.5	0.3	0.7
	5	84.8	84.9	-0.1	
	10	79.8	80.2	-0.5	
	15	75.0	75.6	-0.8	
	0	59.2	56.7	4.4	
	5	55.9	54.7	2.2	
3000	0	64.1	62.2	3.1	0.7
	5	60.6	59.3	2.2	
	0	76.2	76.4	-0.3	
	5	71.3	72.0	-1.0	
	0	101.2	103.1	-1.8	
	5	91.6	94.0	-2.6	
1500	0	150.0	-	-	0.7
	5	130.3	130.5	-0.2	
M = 0	0	65.2	67.0	-2.7	0.7
	5	61.0	62.9	-3.0	
	0	69.6	70.8	-1.7	
	5	65.4	66.8	-2.1	
	0	65.2	67.0	-2.7	
	5	61.0	62.9	-3.0	

	JOB TITLE:	CONTRACT NO. 52406-46	PAGE IV - 17
	SUBJECT: CAP	SIGN. PAL	DATE 20/9-85
		CONTR.	DATE

PARAMETER	DEPTH	MOB SHEAR CAP	MOB SHEAR CAP	DIFFERENCE IN %	
M = 30 000	0	76.2	76.4	-0.3	-0.8
	5	71.3	72.0	-1.0	
	0	86.4	85.0	1.6	
	5	79.6	79.1	0.6	
	0	103.0	99.3	3.7	
	5	91.3	89.2	2.4	
45 000	0	125.9	133.4	-5.6	
	5	106.0	108.3	-2.1	
60 000	0	69.5	70.2	-1.0	-0.4
	5	62.5	63.3	-1.3	
75 000	0	76.2	76.4	-0.3	
	5	71.3	72.0	-1.0	
$\Delta p = 0$	0	83.4	83.1	0.4	
	5	80.2	80.7	-0.6	
± 0.050	0	90.9	90.0	1.0	
	5	89.1	89.4	-0.3	
± 0.100	0	52.6	52.0	1.2	
	5	49.5	49.2	0.6	
± 0.150	0	76.2	76.4	-0.3	
	5	71.3	72.0	-1.0	
LF = 0.5	0	104.8	105.1	-0.3	0.2
	5	97.0	98.2	-1.2	
1.0	0	139.7	139.0	0.5	
	5	127.0	127.9	-0.7	
1.5	0	189.1	184.4	2.5	
	5	164.4	163.3	0.7	
2.0	0				
	5				
2.5	0				
	5				

Average difference for all cases -0.4 %

Maximum difference +4.4 %

Minimum difference -5.6 %

It should be noted that the iteration process in CAP stops when the difference between two SF are less than 1 %.

	JOB TITLE:	CONTRACT NO.	PAGE
	SUBJECT:	52406-46	IV - 17
		SIGN.	DATE
	CAP	CONTR.	DATE

PARAMETER	DEPTH	MOB SHEAR CAP	MOB SHEAR CAP	DIFFERENCE IN %	
M = 30 000	0	76.2	76.4	-0.3	-0.8
	5	71.3	72.0	-1.0	
45 000	0	86.4	85.0	1.6	
	5	79.6	79.1	0.6	
60 000	0	103.0	99.3	3.7	
	5	91.3	89.2	2.4	
75 000	0	125.9	133.4	-5.6	-0.4
	5	106.0	108.3	-2.1	
$\Delta p = 0$	0	69.5	70.2	-1.0	
	5	62.5	63.3	-1.3	
± 0.050	0	76.2	76.4	-0.3	
	5	71.3	72.0	-1.0	
± 0.100	0	83.4	83.1	0.4	0.2
	5	80.2	80.7	-0.6	
± 0.150	0	90.9	90.0	1.0	
	5	89.1	89.4	-0.3	
LF = 0.5	0	52.6	52.0	1.2	
	5	49.5	49.2	0.6	
1.0	0	76.2	76.4	-0.3	0.2
	5	71.3	72.0	-1.0	
1.5	0	104.8	105.1	-0.3	
	5	97.0	98.2	-1.2	
2.0	0	139.7	139.0	0.5	
	5	127.0	127.9	-0.7	
2.5	0	189.1	184.4	2.5	0.7
	5	164.4	163.3	0.7	

Average difference for all cases -0.4 %

Maximum difference +4.4 %

Minimum difference -5.6 %

It should be noted that the iteration process in CAP stops when the difference between two SF are less than 1 %.

	JOB TITLE:	CONTRACT NO. 52406-46	PAGE V-0
	SUBJECT: The Computer Program CAP	SIGN. RAL	DATE 23/9-75
		CONTR.	DATE

APPENDIX V

Comparison between CAP and the Bearing
Capacity Formula's of : Brinch Hansen
Meyerhof
Janbu

	JOB TITLE:	CONTRACT NO. 52406-46	PAGE V-1
	SUBJECT: CAP	SIGN. RAL	DATE 23/9-75
		CONTR.	DATE

COMPARISON

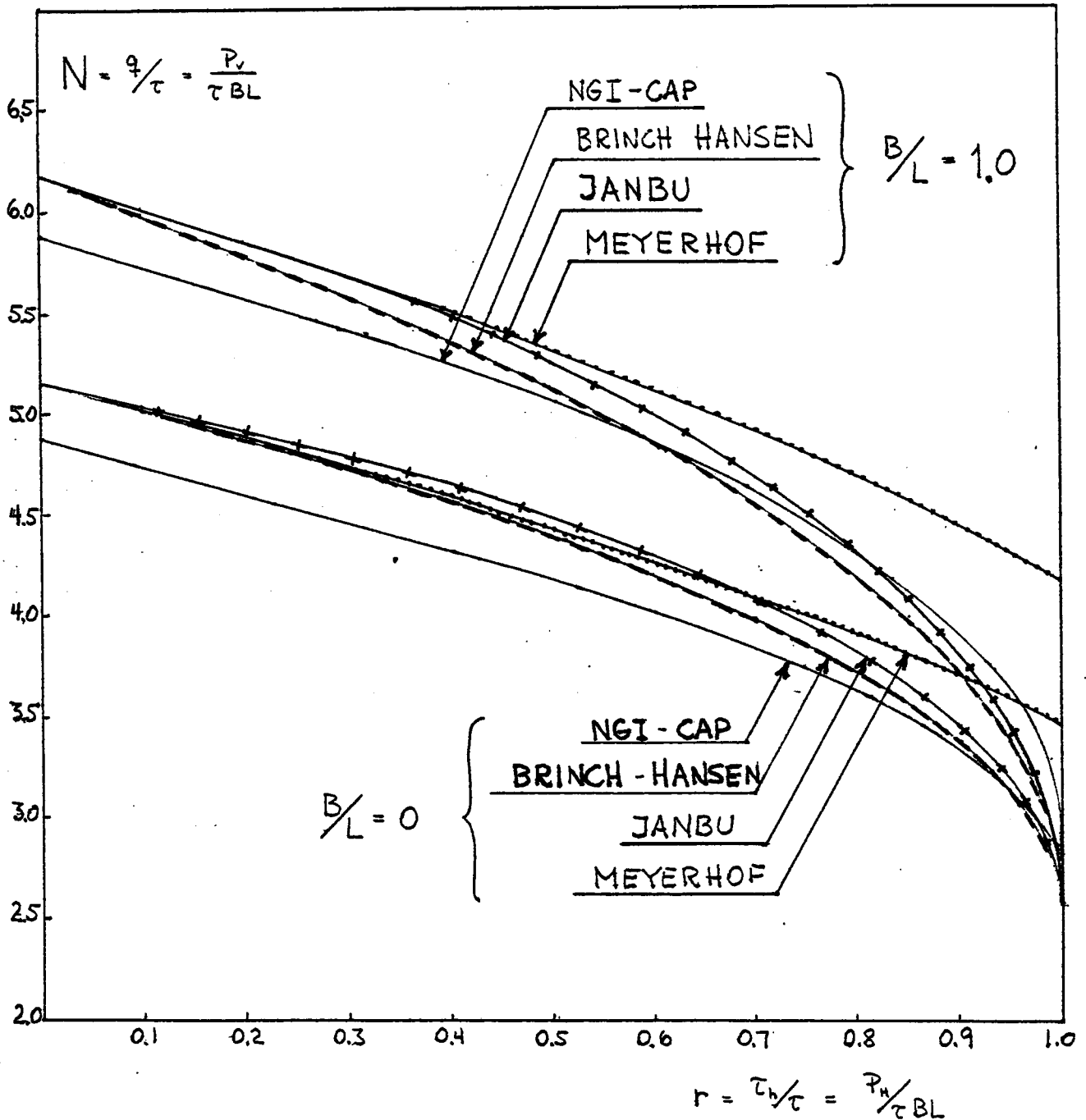
For the special case : Surface load $z_s = 0$
 No moment $M = 0$
 No surcharge $q_t = 0$

Cap results are back-calculated to give figures comparable to : Bearing capacity factor N
 Inclination factor i_c
 Shape factor s_c

Results are compared to the bearing capacity formula's of : Brinch Hansen
 Meyerhof
 Janbu

	JOB TITLE:	CONTRACT NO. 52406-46	PAGE I-2
	SUBJECT:	SIGN. PAL	DATE 23/9-75
	CAP	CONTR.	DATE

BEARING CAPACITY : $N = \frac{q}{\tau}$ versus $r = \frac{\tau_h}{\tau}$
 for surface loading, $D=0$, $q_t=0$, $M=0$



Note that : $\frac{P_h}{P_v} = \frac{r}{N}$



JOB TITLE:

CONTRACT NO.

52406-46

PAGE

V-3

SUBJECT:

CAP

SIGN.

DAL

DATE

23/9-75

CONTR.

DATE

BRINCH HANSEN

$$\frac{P_v}{B L} = c N_c (1 + s_c^a + d_c^a - i_c^a) + q_t$$

$$s_c^a = 0,2(1 - 2i_c^a) B/L$$

$$d_c^a = 0,4 \arctan D/B$$

$$i_c^a = 0,5 - 0,5 \sqrt{1 - r}$$

$$r = P_v / c B L, \quad N_c = 2 + \pi$$

r	i_c^a	s_c^a	N, B/L = 0	N, B/L = 1.0
0	0	0,2	5,14	6,17
0,1	0,026	0,190	5,01	5,98
0,2	0,0528	0,179	4,87	5,79
0,3	0,0817	0,167	4,72	5,58
0,4	0,1127	0,155	4,56	5,36
0,5	0,146	0,141	4,39	5,12
0,6	0,184	0,126	4,20	4,84
0,7	0,226	0,110	3,98	4,55
0,8	0,276	0,0894	3,72	4,18
0,9	0,342	0,0632	3,38	3,71
0,95	0,388	0,0447	3,15	3,38
1,0	0,5	0	2,57	2,57

	JOB TITLE:	CONTRACT NO. 52406-46	PAGE V-4
	SUBJECT: CAP	SIGN. DAL	DATE 23/9 75
		CONTR.	DATE

MEYERHOF

$$\frac{P_v}{BL} = c N_c \cdot s_c d_c i_c + q_t i_q$$

$$s_c = 1 + 0,2 B/L$$

$$d_c = 1 + 0,2 D/B$$

$$i_c = \left(1 - \arctan \left(\frac{P_H}{P_v} \right) / 90 \right)^2 = i_q$$

$$r = P_H / cBL, \quad N_c = 2 + \pi$$

r	i _c	s _c	N, B/L = 0	N, B/L = 1.0
0	1,0	1,2	5,14	6,17
0,1				
0,2	0,949	1,2	4,88	5,85
0,3				
0,4	0,892	1,2	4,59	5,51
0,5				
0,6	0,830	1,2	4,27	5,12
0,7				
0,8	0,759	1,2	3,90	4,68
0,9	0,719	1,2	3,70	4,44
1,0	0,675	1,2	3,47	4,16



JOB TITLE:

SUBJECT:

CAP

CONTRACT NO.

52406-46

PAGE

V-5

SIGN.

RAL

DATE

23/9-75

CONTR.

DATE

JANBU

$$\frac{P_v}{BL} = c N_c (1 + s_c - i_c) + q_t$$

$$i_c = [1 + \arcsin r - \cos(\arcsin r)] / (2 + \pi)$$

$$s_c = 0,2 (1 - 2 i_c) \frac{B}{L} \quad [\text{Brinch Hansen}]$$

$$"N" = N_c (1 + s_c - i_c)$$

$$r = \frac{P_v}{cBL}, \quad N_c = 2 + \pi$$

r	i_c	s_c	"N", $\frac{B}{L} = 0$	"N", $\frac{B}{L} = 1.0$
0	0	0,2	5,14	6,17
0,1	0,0204	0,1918	5,04	6,02
0,2	0,0431	0,1828	4,92	5,86
0,3	0,0682	0,1727	4,79	5,68
0,4	0,0963	0,1615	4,65	5,48
0,5	0,128	0,149	4,48	5,25
0,6	0,164	0,134	4,30	4,99
0,7	0,206	0,117	4,08	4,68
0,8	0,258	0,0967	3,82	4,31
0,9	0,3275	0,0690	3,46	3,81
0,95	0,3775	0,0490	3,20	3,45
1,0	0,5	0	2,57	2,57

(Janbu's solution with Brinch Hansen's shape factor.)



JOB TITLE:

SUBJECT:

CAP

CONTRACT NO.

52406 - 46

PAGE

V - 6

SIGN.

PAL

DATE

23/9 - 75

CONTR.

DATE

NGI - CAP

$$N = \frac{\frac{2 \cos \alpha - \cos 2\alpha}{\tan \alpha} + \ln \left[\tan \left(\frac{\alpha}{2} + 45^\circ \right) \right] + \frac{2 + \pi}{2} + \left[1 + \frac{\pi}{4} + \frac{\alpha \pi}{180} + \frac{1}{\tan \alpha} \right] \sin \alpha}{1 + \frac{P_u}{P_v} \frac{1}{\sin \alpha}} f_8$$

$N_{\min} = f(\alpha)$ has to be found for each $\frac{P_u}{P_v}$

$$"N" = N_{(r=0, B/L=0)} (1 + "s" - "i")$$

$$"r" = "N" \cdot \frac{P_u}{P_v}$$

"r"	"i"	"s"	"N", B/L = 0	"N", B/L = 1.0
0	0	0,21	4,87	5,88
0,1	0,0287	0,20	4,73	5,73
0,2	0,0575	0,20	4,59	5,57
0,3	0,0842	0,195	4,46	5,41
0,4	0,113	0,189	4,32	5,24
0,5	0,143	0,181	4,18	5,06
0,6	0,175	0,169	4,02	4,84
0,7	0,211	0,158	3,84	4,61
0,8	0,255	0,138	3,63	4,30
0,9	0,314	0,115	3,34	3,90
0,95	0,355	0,100	3,14	3,63
1.0	0,446 ?	0,031 ?	~ 2,7 ?	~ 2,85 ?

	JOB TITLE:	CONTRACT NO. 52406-46	PAGE VI - 0
	SUBJECT: The Computer Program CAP	SIGN. RAL	DATE 30/9-85
		CONTR.	DATE

APPENDIX VI

STABILITY CALCULATIONS FOR OFFSHORE GRAVITY STRUCTURES

OTC 1976

Stability Calculations for Offshore Gravity Structures

By

Rolf Lauritzen and Knut Schjetne, Norwegian Geotechnical Institute

THIS PAPER IS SUBJECT TO CORRECTION

©Copyright 1976

Offshore Technology Conference on behalf of the American Institute of Mining, Metallurgical, and Petroleum Engineers, Inc. (Society of Mining Engineers, The Metallurgical Society and Society of Petroleum Engineers), American Association of Petroleum Geologists, American Institute of Chemical Engineers, American Society of Civil Engineers, American Society of Mechanical Engineers, Institute of Electrical and Electronics Engineers, Marine Technology Society, Society of Exploration Geophysicists, and Society of Naval Architects and Marine Engineers.

This paper was prepared for presentation at the Eighth Annual Offshore Technology Conference, Houston, Tex., May 3-6, 1976. Permission to copy is restricted to an abstract of not more than 300 words. Illustrations may not be copied. Such use of an abstract should contain conspicuous acknowledgment of where and by whom the paper is presented.

ABSTRACT

The paper contains a general discussion of bearing capacity stability calculations for offshore gravity structures founded on clay deposits. An alternative to the present methods called the slip surface method is presented and a parameter study is described. Comparison is made with the methods suggested by Brinch Hansen, Janbu, Karal and Meyerhof. The applicability of the different methods is discussed and evaluated.

INTRODUCTION

As geotechnical consultants to oil companies and contractors involved in the North Sea oil development, the Norwegian Geotechnical Institute is engaged in the foundation design of offshore gravity structures. The seabed in the North Sea consists mainly of clay deposits. Therefore stability analysis is one of the most important aspects to be considered in feasibility studies of actual platform locations.

Because of the vital influence on the platform design, stability analyses are usually carried out before an extensive soil investigation

References and illustrations at end of paper.

of the site is completed. At that stage the only soil strength parameters available are undrained shear strength values evaluated from in situ cone penetration tests or preliminary laboratory tests on board the exploration vessel.

The critical design forces on the structure are short term environmental forces generating an undrained state of stress in the supporting soil mass. This makes it advantageous to use methods based upon undrained shear strength of the soil in the stability analysis. Several such methods are available today, but none of them can fully satisfy the special conditions for offshore structures. A method is needed that is efficient, reliable and applicable to inhomogeneous soil conditions.

In this paper two bearing capacity formulas applied to offshore gravity structures are reviewed and an alternative calculation procedure is presented called the slip surface method.

BASIC ASSUMPTIONS

As none of the present methods are applicable to offshore gravity structures in their existing form, certain assumptions must be made in order to adapt these methods to our purpose. Several assumptions common for all the methods will be briefly described in this section.

1. Shape. The most frequently used foundation shapes are circular, square, hexagonal or octagonal. In all cases the actual shape is transformed to a square, of side lengths B and L, with the same cross section area. By definition B is the dimension in the wave force direction and L the dimension perpendicular to the wave direction.

2. Base. The design loads are usually defined at sea floor level. As virtually all concrete gravity structures are equipped with some type of skirts, the level of the skirt tip will in this paper be defined as the "foundation base".

3. Loads at the base. The vertical load at the seafloor is increased by the submerged weight of the soil within the skirts. No vertical shear forces on the skirts are taken into account. In the horizontal load at the base are included the horizontal load at sea floor level, active and passive earth pressures and horizontal side shear resistance on skirts.

The active earth pressure is not allowed to be negative. In that case, an open crack is assumed to occur between skirt and soil. A shear strength reduction of 50% is applied for the horizontal side shear resistance due to assumed soil disturbance during penetration of the skirts.

The moment at the base is the sum of the moment at sea floor level, and the moment of all the forces acting on the structure between sea floor level and the base.

4. Effective area. Moment equilibrium is satisfied by applying the vertical load at the base concentrically on the effective area, calculated on the basis of the actual platform width reduced by twice the eccentricity.

5. Full contact. Normally the structure is assumed to be in contact with the subsoil under the whole platform base. From this assumption it follows that the platform cannot move horizontally unless a failure surface is developed under the whole base area.

6. Horizontal load on effective area. The part of the horizontal load to be applied simultaneously with the vertical load on the effective area, is found by subtracting the shear force taken by the area outside the effective area from the horizontal load at the base.

7. Constant safety factor. All the calculated forces that involve the shear strength of the clay are assumed to have the same degree of strength mobilization. This requires an iteration process to be carried out

since the loads at the base level are dependent upon the safety factor.

PRESENT CALCULATION METHODS

For structures on clay having constant shear strength with depth the bearing capacity formulas of Brinch Hansen, (1970) and Meyerhof (1963) may be used, of which the Brinch Hansen's method almost always gives the lowest safety factor and is therefore usually recommended. All the basic assumptions outlined in this paper are valid for this method.

The bearing capacity formulas are based on development of critical shear zones in the soil, but they give no description of the location of these failure zones. N. Janbu (1975) has, however, recently developed a bearing capacity calculation method in which the location of the critical failure zones can be found for an infinitely long foundation with inclined load. This method gives almost the same result as Brinch Hansen's formula, if both the shape factor and the depth factor are neglected. The geometry of the failure zone is shown in fig. 1. The angle alpha is given by:

$$\tan(\alpha) = \sqrt{\frac{1-r}{1+r}} \dots \dots \dots (1)$$

where r is the ratio of the actual horizontal stress to the mobilized critical shear stress:

$$r = \frac{\tau_h}{\tau} = \frac{H/B_0L}{\tau_0/SF} \dots \dots \dots (2)$$

In the case of inhomogeneous soil conditions the bearing capacity formulas are less useful. They are based on a fixed geometry of the critical shear zone for an infinitely long foundation and on empirical factors that take shape, depth etc. into account. They cannot, however, give the safety factor for a chosen failure surface. Methods that take into account a chosen failure surface have been published among others by Janbu (1973), Morgenstern and Price (1965) and Karal (1973). The problem with these methods is that they are both complicated to use and time consuming.

THE SLIP SURFACE METHOD

As already stated, it is desirable to have a method which gives quick and reliable answers to various configurations of soil properties, platform geometries and applied loads. This has been the aim in the development of the slip surface method. It is also desirable that the method should correlate well with Brinch Hansen's formula, modi-

fied as outlined in this paper, when applied to homogeneous soil conditions.

The work has been concentrated on a geometrical model of the sliding body as shown in fig. 1. The cross-section over the entire length L is constant, and the sliding body is cut off by vertical planes at both ends of the foundation. In order to compute the forces involved it is convenient to divide the sliding body into four sections, an active section, a flat section, an inclined section and a passive section.

The resistance to sliding for all surfaces of the sliding body are evaluated, and the resulting horizontal force in each section is calculated under force equilibrium conditions. The safety factor is found by overall horizontal force equilibrium. The depth z to the lowest point on the slip surface is varied in steps and the critical depth giving the minimum safety factor is established.

In the slip surface method the reduction factor for the forces on soil side areas is of great importance. If the chosen geometry of the sliding body were the correct one, a factor of 1.0 should have been applied, giving full contribution from the soil side areas on the overall stability. A factor of 0 means no benefit of the soil side areas. However, a reduction factor of 0.4 has proven to give results in good agreement with Brinch Hansen's formula for homogeneous soil conditions, and this value is therefore recommended.

PARAMETER STUDY

A comprehensive parameter study was carried out in order to compare the results of Brinch Hansen's formula, Meyerhof's formula and the slip surface method. The study was based on a basic set of parameters chosen within the range of platform data available at NGI.

The data representing nearly 40 different platform projects for the North Sea, is summarized in table 1 below.

	lowest value	average value	highest value
A m ²	5,000	8,460	12,000
D m	0.5	3.5	12.0
P _v t	60,000	185,000	300,000
P _H t	26,000	49,000	79,000
M tm	800,000	2200,000	4200,000

Table 1 : Summary of platform data.

Basic case:

B =	L =	100 m	M =	3 000 000 tm
D =	5 m	Δp =	± 5 t/m ²	
P _v =	200 000 t	γ =	2.0 t/m ³	
P _H =	60 000 t	τ _c =	15 t/m ²	

The resulting safety factors using the different methods were:

Brinch Hansen's formula	SF = 2.08
Meyerhof's formula	SF = 2.20
The slip surface method	SF = 2.09

The parameter study was carried out by varying one parameter at a time keeping all the others constant. In addition a study was made where load factors were applied to P_H, M and Δp simultaneously. The parameters were extended beyond the highest and lowest values in table 1. Two representative diagrams from the parameter study are shown in fig. 2 and 3.

The study proved that the results of the slip surface method correlated well with the results using Brinch Hansen's formula. Considering all cases the slip surface method gave at the maximum 4% lower (P_H = 15 000) and 2.4% higher (M = 0) safety factor than Brinch Hansen's formula. On the average the difference in safety factor as found by the two methods was less than ± 1%.

In order to study what influence the different parameters have on the safety factor, a new parameter called "safety factor response", SFR, is introduced, with the following definition: A change in a parameter of 1% gives a change in the safety factor of SFR%. Calculated values of SFR valid for the basic case is given in table 2 below:

1% change in parameter	Safety factor response, SFR, %			
	Brinch Hansen formula	Meyerhof's formula	Slip surface method	Average
- B/L	- 0.09	- 0.13	- 0.11	- 0.11
± D	± 0.06	± 0.05	± 0.06	± 0.06
± P _v	± 0.33	± 0.35	± 0.32	± 0.33
± P _H	± 0.39	± 0.30	± 0.35	± 0.35
± M	± 0.17	± 0.24	± 0.19	± 0.20
± Δp	± 0.12	± 0.13	± 0.13	± 0.13
± LF	± 0.66	± 0.59	± 0.65	± 0.63

Table 2: Summary of the parameter study.

For homogeneous soil conditions and a constant foundation area it is clearly demonstrated that P_H and P_V are the dominating parameters in a bearing capacity calculation. According to table 2 a change of 1% gives a 1/3% response in the safety factor. The corresponding response of 1% change in the moment is 1/5%. The other parameters are of minor interest in this connection. However, for inhomogeneous soil conditions the value of D may prove to be of great importance.

A non-linear relationship was observed between load factor and safety factor, see fig. 3. The SFR of approximately 2/3 given in the table above corresponds to a safety factor of 2.0. The product of the load factor and the safety (material) factor has occasionally been presented as the overall safety factor. This product is, however, not constant as can be seen from the numbers below, calculated by Brinch Hansen's formula:

LF	:	0.40	0.75	1.00	1.50	2.00
SF	:	3.33	2.50	2.08	1.53	1.17
LF x SF	:	1.33	1.88	2.08	2.30	2.34

A second interesting observation was that the maximum depth of the critical slip surface calculated by the slip surface method was 39 m in the basic case, whereas the maximum depth of the critical shear zone calculated by Janbu's procedure for Brinch Hansen's Formula was 37 m.

Also interesting was the observation that small variations in the maximum depth of the slip surface had very little influence on the safety factor. The lowest point on the slip surface could be moved 9 m upwards and 14 m downwards before the calculated safety factor changed by more than 1% for the basic case.

DISCUSSION

Below is discussed some aspects of the stability calculations that are found to have influence on the safety factor.

Transfer of Horizontal Load

In the parameter study it was noticed that the safety factor is strongly influenced by the horizontal force. Since the actual bearing capacity calculation is related to the effective base area, all the horizontal resistance that can be mobilized elsewhere on the structure is of great importance. This is demonstrated in the following by calculating the safety factor for the basic case with different assumptions concerning the transfer of the horizontal force to the soil.

1. The horizontal force at sea floor level, is applied on the effective area, no forces on skirts or side areas are taken into account. SF1 = 1.53
2. The horizontal force is uniformly distributed over the total area, no forces on skirts or side areas are taken into account. SF2 = 1.65
3. The area outside the effective area is assumed to belong to the critical slip surface, with constant safety factor along the whole slip surface, no forces on skirts or side areas are taken into account. SF3 = 1.72
4. The same assumption as 3. In addition active and passive force, but no forces on side areas, later referred to as "plane strain". SF4 = 1.84
5. The same assumption as 4, in addition side forces on skirts, but no forces on side areas below the foundation base. SF5 = 1.88
6. The same assumption as 5, in addition side forces soil/soil below the foundation base. SF6 = 2.09

Assumption 6 is applied in the slip surface method.

Calculation of Average Shear Strength

In a layered clay profile with shear strength varying with depth, there are two different ways of calculating the average shear strength. The conventional way in a slip surface method is to calculate the average shear strength along the critical shear surface. In this case the shear strength of a layer contributes to the average shear strength in proportion to the intersection length of the shear surface through the actual layer. This is called the "surface method". Brinch Hansen's formula is, however, evaluated by composing critical shear stress zones with uniform shear stress mobilisation within the zones. In this case it should be more correct to calculate the average shear strength based on volume instead of shear surface. This is called the "volume method".

To compare the influence of these two methods of determining the shear strength, the slip surface method was used to calculate the basic case with 7 different soil profiles. The first had linearly increasing shear strength with depth, the others were typical North Sea profiles. All profiles are shown in fig. 4.

Profile No.	Surface method		Volume method	
	SF	critical depth in meter	SF	critical depth in meter
1	1.86	18	1.79	21
2	1.81	23	1.84	26
3	2.20	24	2.27	27
4	1.86	25	2.00	33
5	2.03	25	2.16	33
6	2.29	25	2.42	33
7	2.38	25	2.53	33

Table 3. Summary of stability analysis for seven soil profiles.

As can be seen from table 3 the "volume method" gave the lowest safety factor when the shear strength increased with depth. For all the other profiles tested the surface method gave on the average a 5% lower calculated safety factor. By using a bearing capacity formula with fixed geometry of the shear zones, the differences would have been larger.

From the authors point of view the "volume method" should be used in bearing capacity formulas, whereas in the slip surface method the "surface method" seems to be more correct.

Spacing of skirts

The slip surface method is based on the assumption that the level of the tip of the skirts is the foundation level and that a failure in the soil must occur at or below that level. This assumption is related to the spacing of the skirts and the soil properties. For widely spaced skirts the failure surface may tend to go up in between the skirts, if the resistance to sliding is lower at higher levels. The problem is discussed by Young et al (1975)

With no regard to the reasons, be it weak soil layers or poor platform base - sea floor contact that causes lower resistance to sliding, the problem may be treated in the following manner. The resistance to horizontal sliding between the skirts is evaluated and compared to the resistance against sliding at the tip of the skirts. The ratio of the two resistances is then used as a reduction factor for the shear strength at the foundation base.

Cyclic loading

A study of the behaviour of clay subjected to cyclic loading has recently been carried

out by a group of four European geotechnical laboratories, sponsored by thirteen oil companies and institutions involved in the North Sea oil development. (Andersen 1975) It was found that the reduction in undrained shear strength due to cyclic loading can be established as a function of the load history and platform geometry. With the reduced shear strength profile the safety against foundation failure may be evaluated from the slip surface method.

NUMERICAL EXAMPLE

The stability of an actual offshore gravity structure has been analysed by different calculation methods, see fig. 5.

Geometry:	B = L =	88.3 m
	D =	3.5 m
Forces at sea floor :	P_v =	187 000 t
	P_H =	49 100 t
	M =	2240 000 tm
	Δp =	$\pm$ 3.5 t/m ²
Soil properties:	ϕ =	2.0 t/m ³
	design shear strength profile as given in fig. 5.	

The following calculation methods are used:

1. The Slip Surface Method as described in this paper.
2. Janbu's General Procedure of Slices, (1973)
3. Energy Method by K. Karal, (1973)
4. Brinch Hansen's Formula, (1970) modified
5. Meyerhof's Formula, (1963) modified.

Geometrical shape of the various critical failure mechanisms are shown in fig. 5. The corresponding calculated safety factors are given in table 4 below.

Method	Calculated safety factors	
	Plane strain B/L = 0	Actual case B/L = 1.0
1	2.00	2.15
2	1.96	-
3	2.03	-
4	2.15	2.35
5	2.17	2.49

Table 4. Summary of calculated safety factors.

The applicability of the slip surface method is demonstrated by a "design chart" for the actual platform site and foundation geometry, see fig. 6. For a required safety factor of 2.0, corresponding values of vertical loads, necessary penetration depths and

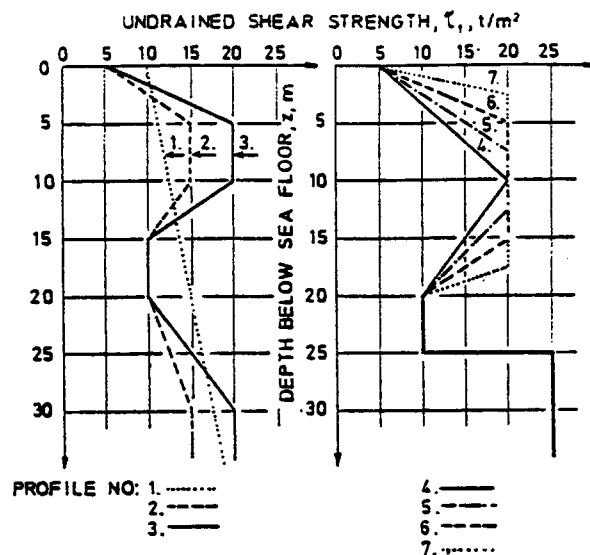


Fig. 4 - Shear strength profiles.

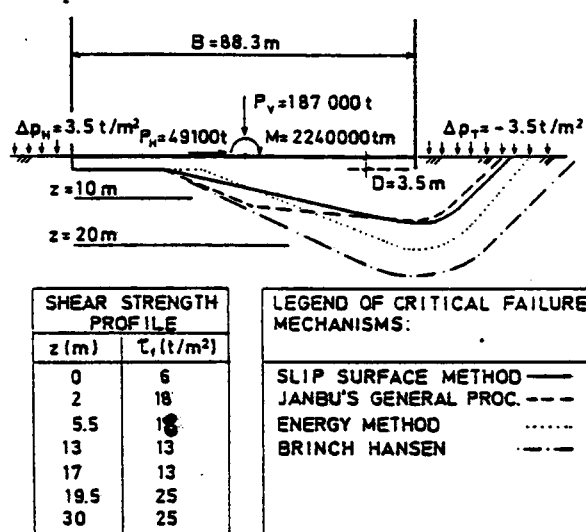


Fig. 5 - Example, actual offshore gravity structure.

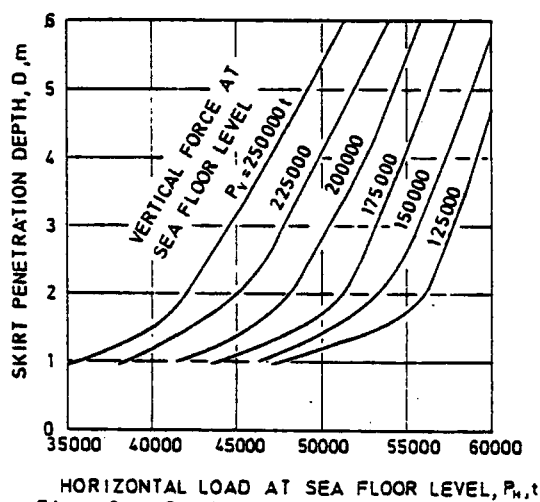


Fig. 6 - Design chart, safety factor=2.0 corresponding values of P_v , P_h , and D .

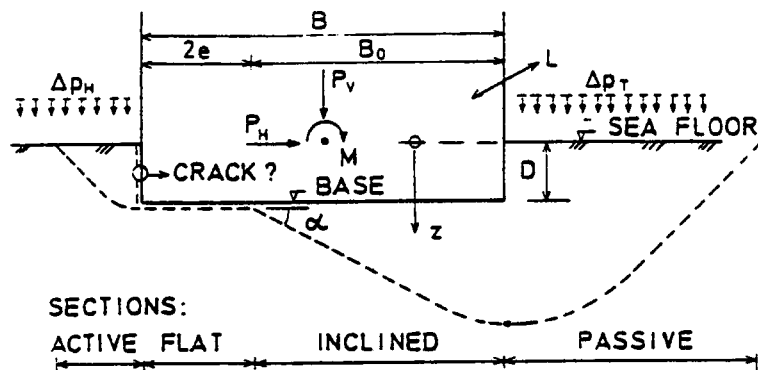


Fig. 1 - Geometry of failure surface.

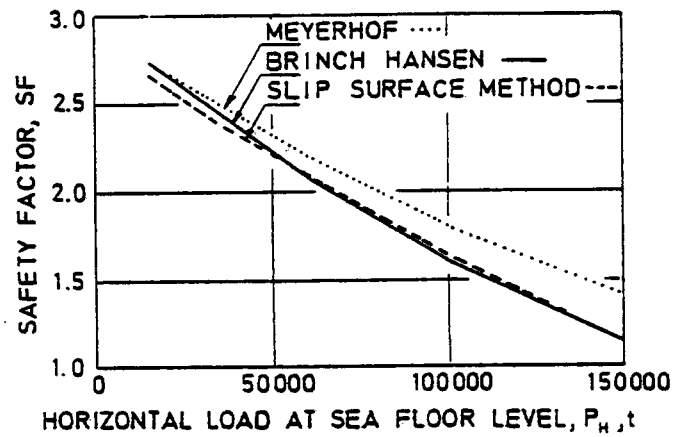


Fig. 2 - Parameter study, horizontal load.

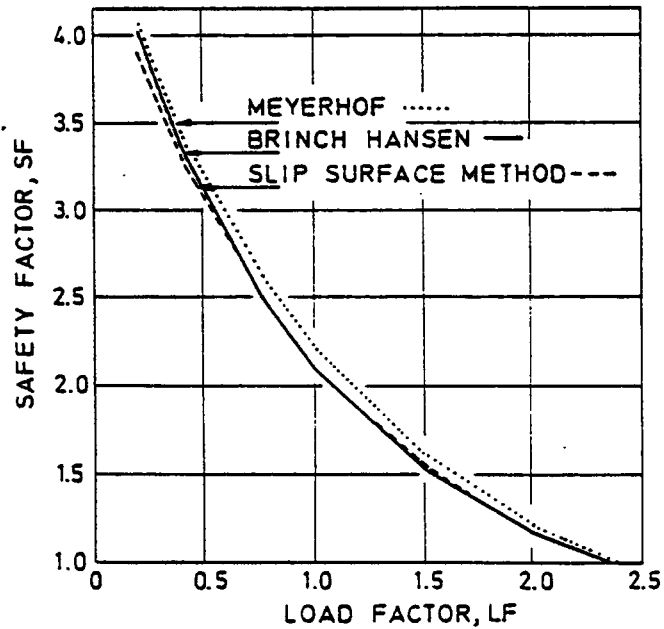


Fig. 3 - Parameter study, load factor.

horizontal loads are calculated. The other wave forces, M and Δp , are assumed to change proportionally to the horizontal load.

CONCLUSIONS

Bearing capacity stability calculations for offshore gravity structures founded on clay deposits have to be based upon undrained shear strength values at certain stages in the design. The formula of Brinch Hansen (1970) may be used for homogeneous soil conditions with some modifications as recommended in this paper.

For inhomogeneous soil conditions an alternative calculation procedure is outlined, called the slip surface method. This method correlate well with the above mentioned formula when applied to homogeneous soil conditions. Calculated differences in corresponding safety factors do not exceed $\pm 5\%$, average differences being approximately $\pm 1\%$. The method is simple and quick and is thus convenient for studying the effect of varying the different design parameters on the foundation stability.

An example of the stability calculation for an actual platform in the North Sea is included, where the slip surface method shows remarkably good correlation to the methods of Janbu (1973) and Karal (1973).

NOMENCLATURE

Geometry:

A	: structure base area	m^2
B	: base width in wave direction	m
L	: base length	m
D	: skirt penetration depth	m
z	: depth below sea floor	m
α	: angle in inclined section	degree
e	= M_B / V_B : eccentricity	m
B_o	= $B - 2e$: effective base width	m

Loads:

P_v	: vertical load at sea floor	t
P_H	: horizontal load at sea floor	t
M	: moment at sea floor	tm
Δp	: excess wave pressure at sea floor	t/m^2
V_B, H_B, M_B	: loads at the base	
H	: horizontal load on the effective area	t

Soil parameters:

γ	: total unit weight of soil	t/m^3
----------	-----------------------------	---------

$\bar{\tau}_u$	: undrained shear strength	t/m^2
τ	= $\bar{\tau}_u / SF$: mobilized shear stress	t/m^2

SF : safety factor

Unit used: meter (m)
metric tons (t)

ACKNOWLEDGEMENTS

The authors wish to express their appreciation for the assistance given during the preparation of this paper by Karel Karal, The River and Harbour Laboratory at the Technical University in Trondheim, Fritz Nowacki, Geateam at present NGI and the staff at the Norwegian Geotechnical Institute.

REFERENCES

- Andersen, K.H. (1975) Research project, repeated loading on clay. Summary and interpretation of test results. Norwegian Geotechnical Institute, 74037-9, 15. October 1975.
- Hansen, J.B. (1970) A revised and extended formula for bearing capacity. The Danish Geotechnical Institute, Copenhagen, Bulletin No.28, pp. 5 - 11.
- Janbu, N (1975) Betongkonstruksjoner på kontinentalsokkelen, kursdagene ved NTH 1975 (in norwegian)
- Janbu, N (1973) Slope stability computations Embankment-dam engineering. Casagrande volume. Ed.: R.C.Hirschfeld and S.J.Poulos. N.Y. Wiley. pp. 47-86
- Karal, K (1973) En energimetode for geotekniske stabilitetsanalyser (in norwegian). Thesis Ph.D. The Norwegian Institute of Technology.
- Meyerhof, G.G. (1963) Some Recent Research on the Bearing Capacity of Foundations. Canadian Geotechnical Journal, Vol.1, No.1, pp.16-26
- Morgenstern, N.R. and V.E.Price (1965) The analysis of the stability of general slip surfaces. Géotechnique, Vol.15, No.1, pp.79-93
- Young, A.G., L.M.Kraft and J.A.Focht (1975). Geotechnical considerations in Foundation Design of Offshore Gravity Structures. Offshore Technology Conference, Vol.III pp. 367-386.

	JOB TITLE:	CONTRACT NO. S2406-46	PAGE <u>VII</u> - 0
	SUBJECT: The Computer Program CAP	SIGN. RAL	DATE 30/9-85
		CONTR.	DATE

APPENDIX VII

A REVISED AND EXTENDED FORMULA FOR BEARING CAPACITY

J. BRINCH HANSEN 1970

A Revised and Extended Formula for Bearing Capacity

by J. Brinch Hansen

(Reprint of Lecture in Japan, October 1968)

1. Original Formula.

A simple formula for the bearing capacity of a shallow foundation was developed around 1943 by Buisman, Caquot and Terzaghi. With the latter's notations it reads:

$$Q/B = \frac{1}{2} \bar{\gamma} B N_{\gamma} + \bar{q} N_q + c N_c \quad (1)$$

This formula is developed for an infinitely long foundation of width B , placed upon the horizontal surface of soil with an effective unit weight $\bar{\gamma}$, a friction angle φ and a cohesion c . $\bar{q}$ is a unit surcharge acting upon the soil surface outside the foundation, and Q is the ultimate bearing capacity per unit length of this foundation, provided that it is loaded centrally and vertically.

2. Bearing Capacity Factors.

The exact formulas for N_q and N_c were indicated already by Prandtl:

$$N_q = e^{\pi \tan \varphi} \tan^2 (45^\circ + \varphi/2) \quad (2)$$

$$N_c = (N_q - 1) \cot \varphi \quad (3)$$

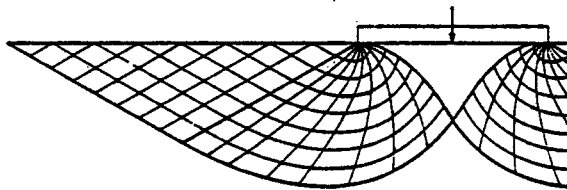


Fig. 1 Lundgren-Mortensen rupture figure for calculation of N_{γ} . Vertical load on heavy earth (no surface load).

The best available calculations of N_{γ} were made, first by Lundgren-Mortensen, and later by Odgaard and N. H. Christensen, using the rupture-figure shown in fig. 1. The results correspond closely to the empirical formula:

$$N_{\gamma} = 1.5 (N_q - 1) \tan \varphi \quad (4)$$

Curves for all 3 factors are shown in fig. 2.

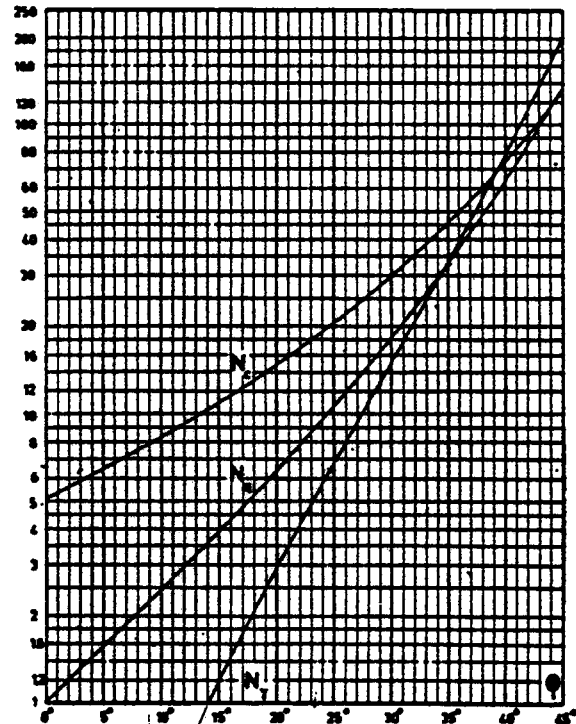


Fig. 2 Bearing capacity factors N_q , N_c , and N_{γ} as functions of φ .

Since N_q and N_c are calculated for one rupture-figure, and N_{γ} for another, the simple superposition implied by equation (1) must actually be an approximation. However, it is always on the safe side, and the error is usually less than 20 % (Lundgren-Mortensen).

3. Practical Cases.

Actual foundations deviate in several respects from the simple case considered above.

Thus, the load may be eccentric or inclined or both. The base of the foundation is usually placed at a depth D below the soil surface. The foundation has always a limited length L , and its shape may not even be rectangular. Finally, both the foundation base and the ground surface may be inclined.

Apart from the eccentricity, which is best taken

into account by considering the so-called effective foundation area, all the other influences can be expressed by means of suitable factors to the 3 terms in the original formula.

The different factors can be found by considering rather simple cases, in which only one complication occurs at a time. When these factors are then used together for more complicated cases this will, of course, be an approximation.

For the different new factors we shall use the following symbols:

- s shape factors.
- d depth factors.
- i inclination factors.
- b base inclination factors.
- g ground inclination factors.

4. Effective Foundation Area.

All loads acting from above upon the base of the foundation are combined into one resultant. It has a component V normal to the base, a component H in the base, and intersects the base in a point called the load centre.

Now, a so-called effective foundation area of rectangular shape is determined in such a way, that its geometric centre coincides with the load centre, and that it follows as closely as possible the nearest contour of the actual base area. A few examples are shown in fig. 3.

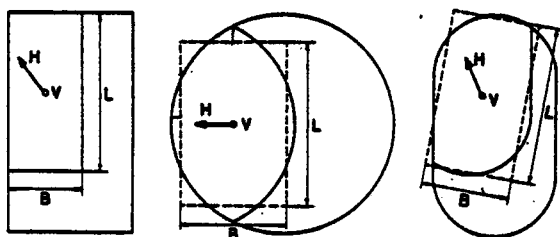


Fig. 3 Equivalent and effective foundation areas.

The short side of this equivalent rectangle is called B and the long one L . The effective foundation area is $A = BL$. For a strip foundation the effective width B will simply be twice the distance from the load centre to the nearest edge of the base.

Meyerhof, the writer and others have shown that the actual bearing capacity of an eccentrically loaded foundation will be very nearly equal to the bearing capacity of the centrally loaded effective foundation area. Consequently, in the following we shall only consider central loading, and the symbols B , L and A will always refer to the effective rectangle.

5. Extended Formulas.

Applying the 5 new kinds of factors to the original equation (1), we get the following formula (Brinch Hansen):

$$Q/A = \frac{1}{2} \bar{\gamma} B N_{\gamma} s_{\gamma} d_{\gamma} i_{\gamma} b_{\gamma} g_{\gamma} + \bar{q} N_q s_q d_q i_q b_q g_q + c N_c s_c d_c i_c b_c g_c \quad (5)$$

$\bar{q}$ is now to be understood as the effective overburden pressure at base level.

In the special case of a horizontal ground surface, the ground inclination factors g disappear, of course, and the equation can then be written somewhat simpler:

$$Q/A = \frac{1}{2} \bar{\gamma} B N_{\gamma} s_{\gamma} d_{\gamma} i_{\gamma} b_{\gamma} + (\bar{q} + c \cot \varphi) N_q s_q d_q i_q b_q - c \cot \varphi \quad (6)$$

In the other special case of $\varphi = 0$ (~ undrained failure in clay), it will theoretically be more correct to introduce additive constants instead of factors. Since the c -term is usually dominant, we may write:

$$Q/A = (\pi + 2) c_u (1 + s_c^a + d_c^a - i_c^a - b_c^a - g_c^a) \quad (7)$$

6. Load Inclination Factors.

An inclination of the load will always mean a reduced bearing capacity, and the reduction is often very considerable.

Exact formulas for i_q and i_c have been derived by several authors, using the rupture-figure shown in fig. 4. The rather complicated results can be approximated by means of simple empirical formulas, however (Brinch Hansen).

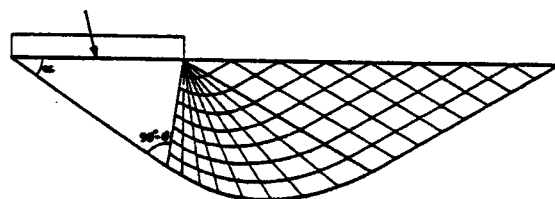


Fig. 4 Rupture figure for calculation of i_q and i_c . Inclined load on weightless earth (with vertical surface load).

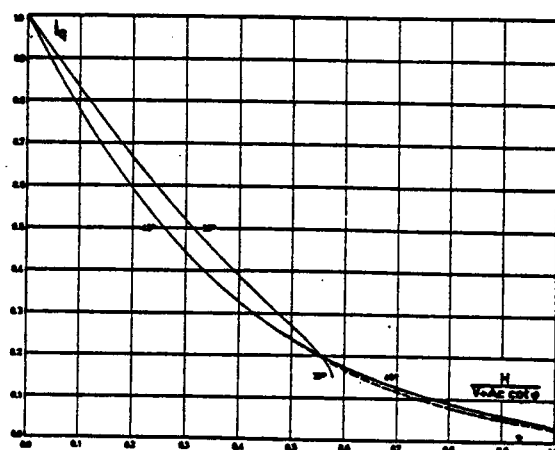


Fig. 5 Inclination factor i_q for q -term.

In the case of $\varphi = 0$ we get:

$$i_c^a = 0.5 - 0.5 \sqrt{1 - H/Ac_u} \quad (8)$$

For $\varphi = 30^\circ$ and 45° the results are shown in fig. 5. The dotted line corresponds to the formula:

$$i_q = [1 - 0.5 H : (V + Ac \cot \varphi)]^5 \quad (9)$$

Calculations of i_γ have been made by Odgaard and N. H. Christensen, using the rupture-figure shown in fig. 6. Their results for $\varphi = 30^\circ$ and 45° are shown in fig. 7. The dotted line corresponds to the formula:

$$i_\gamma = [1 - 0.7 H : (V + Ac \cot \varphi)]^5 \quad (10)$$

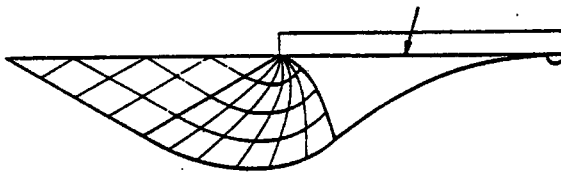


Fig. 6 Rupture figure for calculation of i_γ . Inclined load on heavy earth (no surface load).

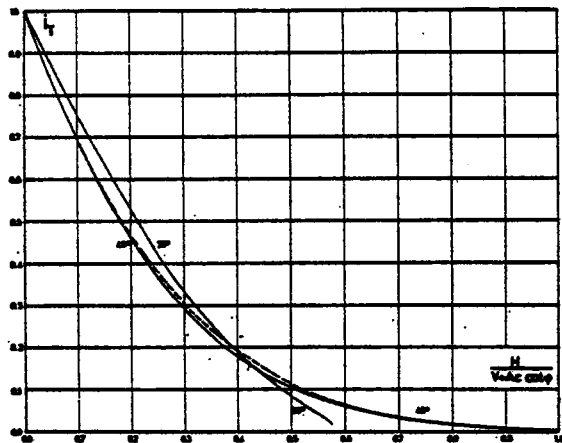


Fig. 7 Inclination factor i_γ for γ -term.

As we shall see later, i_γ must be modified if the foundation base is inclined.

To avoid misunderstandings it should be mentioned that (9) and (10) must not be used, if the quantity inside the bracket becomes negative. The bearing capacity will in such cases be negligible.

The above inclination factors are valid for a horizontal force $H = H_B$, acting parallel with the short sides B of the equivalent effective rectangle. If we substitute H by H_B in the equations (8-10), the corresponding factors may be termed i_{cB}^a , i_{qB} and $i_{\gamma B}$ respectively.

In the more general case, where there is also a horizontal force component H_L , acting parallel with the long sides L , we can find another set of factors i_{cL}^a , i_{qL} and $i_{\gamma L}$ by substituting H by H_L in (8-10).

The first set of factors (with second subscript B)

should be used for investigating the usual failure along the long sides L , occurring when H_B is dominant. The second set (with second subscript L) are used for investigating a possible failure along the short sides B , which may occur when H_L is dominant.

7. Base and Ground Inclination.

The general case is shown in fig. 8. The slope angle is called β , and since the ground will usually be sloping away from the foundation, β shall be defined as positive in this case. The foundation depth D is measured vertically.

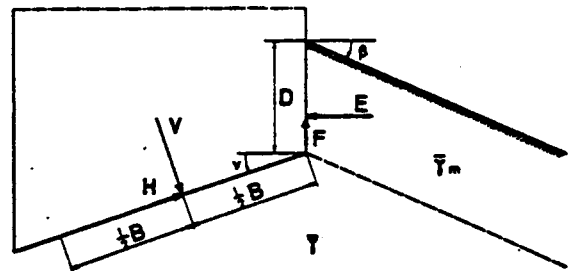


Fig. 8 General case of base and front of retaining wall with inclination of base and ground.

The effective width of the inclined base is called B , whereas V indicates the foundation load normal to the base and H the load in the base. ν is the angle between base and horizon.

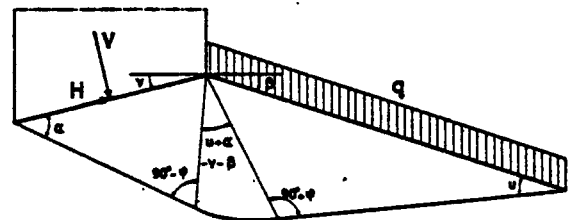


Fig. 9 Rupture figure for calculation of b_c , b_q , g_c and g_q . Frictionless earth or weightless earth with vertical surface load.

For the theoretical case of frictionless or weightless earth we get the rupture-figure shown in fig. 9. In the case of $\varphi = 0$ it gives the exact formulas:

$$b_c^a = \frac{2\nu}{\pi + 2} = \frac{\nu^\circ}{147^\circ} \quad (11)$$

$$g_c^a = \frac{2\beta}{\pi + 2} = \frac{\beta^\circ}{147^\circ} \quad (12)$$

For other friction angles we can find the following exact formula for b_q :

$$b_q = e^{-2\nu \tan \varphi} \quad (13)$$

For b_γ we can develop an empirical formula by utilizing the fact that, for a vertical wall, we must have $K_\gamma = K_p$ according to Coulomb. This gives:

$$b_\gamma = e^{-2.7 \nu \tan \varphi} \quad (14)$$

With regard to g_q , exact formulas can be developed. A closer inspection reveals that g_q is exactly the same function of $\tan \beta$, as i_q is of H : ($V + Ac \cot \varphi$). Hence we can write approximately:

$$g_q = [1 - 0.5 \tan \beta]^5 = g_\gamma \quad (15)$$

We have here assumed g_γ equal to g_q in accordance with the fact that, for a vertical wall and any ground inclination, we must have $K_\gamma = K_p$ according to Coulomb.

The above formulas should only be used for positive values of ν and β , the latter being smaller than φ . Also, $\nu + \beta$ must not exceed 90° .

In the case that the foundation base becomes a vertical wall ($\nu = 90^\circ$), we shall, according to Coulomb's earth pressure theory, have $K_\gamma = K_p$, independent of the roughness of the wall. Therefore it will, in the general case, be necessary to let i_γ depend upon ν in such a way, that for $\nu = 90^\circ$ we get $i_\gamma = i_q$. This is easily achieved by writing:

$$i_\gamma = [1 - (0.7 - \nu/450^\circ) H : (V + Ac \cot \varphi)]^5 \quad (16)$$

8. Shape Factors.

Theoretical values of the shape factors can hardly be indicated at present, since their calculation would require a 3-dimensional theory of plasticity. Thus, experimental evidence must be used.

Extensive and careful plate loading tests on sand have led de Beer to propose the empirical formulas:

$$s_\gamma = 1 - 0.4 B/L \quad (17)$$

$$s_q = 1 + \sin \varphi \cdot B/L \quad (18)$$

As a result of loading tests on undrained clay Skempton found the value:

$$s_c^a = 0.2 B/L \quad (19)$$

which can be shown to be a limiting case of (18).

The shape factors indicated above are actually only valid for vertical loading. In the other limiting case of the foundation sliding on its base, the shape can have little influence. For inclined loads we must, therefore, modify the formulas by introducing the inclination factors. And since failure can take place either along the long sides, or along the short sides, we shall need

two sets of inclination factors. The following formulas are proposed (Brinch Hansen):

$$s_{cB}^a = 0.2 i_{cB}^a B/L \quad (20)$$

$$s_{cL}^a = 0.2 i_{cL}^a L/B \quad (21)$$

$$s_{qB} = 1 + \sin \varphi \cdot Bi_{qB}/L \quad (22)$$

$$s_{qL} = 1 + \sin \varphi \cdot Li_{qL}/B \quad (23)$$

$$s_{\gamma B} = 1 - 0.4 (Bi_{\gamma B}) : (Li_{\gamma L}) \quad (24)$$

$$s_{\gamma L} = 1 - 0.4 (Li_{\gamma L}) : (Bi_{\gamma B}) \quad (25)$$

For the last two factors the special rule must be followed, that the value exceeding 0.6 should always be used.

9. Depth Effects.

Actual foundations are always placed at a certain depth D below the surface. One consequence of this is, that we must take into account the effective weight of the soil above base level. This is done by putting:

$$\bar{q} = \bar{\gamma}_m D \quad (26)$$

where $\bar{\gamma}_m$ is the average effective weight of the soil above base level.

Another effect is due to the fact that the soil above base level has a certain shearing strength. In the following this soil is assumed to be identical with the soil below base level. When it is inferior (or possesses no strength at all) the indicated depth effect will have to be reduced (or neglected completely).

At least one rupture-line will always pass through the soil layer above base level, and the effect will be an increase of the bearing capacity. This we express by means of a depth factor, which is used for foundations with mainly vertical loads.

If an upper Rankine zone is continued through the soil above the base without changing the rupture-figure below base level, the result will be an extra, horizontal force. Therefore, on foundations subjected to considerable horizontal loads (e.g. retaining walls), the simplest way to take the depth effect into consideration is to assume a passive earth pressure acting upon the side of the foundation.

10. Depth Factors.

The depth factor d_γ presents no problem, because we have always, according to definitions:

$$d_\gamma = 1 \quad (27)$$

For small values of D/B it is easy to calculate d_q or d_c , because we can just extend the known rupture-figure for $D = 0$ up to the actual surface. In this way the following approximate formulas have been found (Brinch Hansen):

$$d_c^a = 0.4 D/B \quad (28)$$

$$d_q = 1 + 2 \tan \varphi (1 - \sin \varphi)^2 D/B \quad (29)$$

These formulas may be used for $D \leq B$. For greater depth it is difficult to calculate the depth factors, but we know that they must ultimately approach an asymptotic value. I have, therefore, tentatively proposed the following formulas:

$$d_c^a = 0.4 \arctan D/B \quad (30)$$

$$d_q = 1 + 2 \tan \varphi (1 - \sin \varphi)^2 \arctan D/B \quad (31)$$

We can actually test these formulas by applying them, together with the formulas for the shape factors, to a square pile base at great depth ($D \rightarrow \infty$). This gives for $\varphi = 0$:

$$Q/A = (\pi + 2) c_u (1 + 0.2 + 0.4 \pi/2) = 9.4 c_u \quad (32)$$

which is a well-known result for pile point resistance in clay.

For other friction angles we get:

$$Q/A = \bar{q} N_q (1 + \sin \varphi) (1 + \pi \tan \varphi (1 - \sin \varphi)^2) \quad (33)$$

For values of φ between 30° and 40° this formula gives the result $2.2 \bar{q} N_q$ in very good agreement with Danish experience for pile point resistances in sand, provided that φ is taken as the friction angle in plane strain.

The above formulas are valid for the usual case of failure along the long sides L of the base, and formulas (30)–(31) give the corresponding depth factors d_{cB}^a and d_{qB}^a .

For the investigation of a possible failure along the short sides B we must use another set of depth factors:

$$d_{cL}^a = 0.4 \arctan D/L \quad (34)$$

$$d_{qL} = 1 + 2 \tan \varphi (1 - \sin \varphi)^2 \arctan D/L \quad (35)$$

11. General Formulas.

In the general case, where the horizontal force has both a component H_B parallel with the short sides B , and a component H_L parallel with the long sides L of the equivalent effective rectangle, we must use the following formulas:

$$Q/A \leq \frac{1}{2} \bar{\gamma} N_\gamma B s_{\gamma B} i_{\gamma B} b_\gamma + (\bar{q} + c \cot \varphi) N_q d_{qB} s_{qB} i_{qB} b_q - c \cot \varphi \quad (36)$$

$$\frac{1}{2} \bar{\gamma} N_\gamma L s_{\gamma L} i_{\gamma L} b_\gamma + (\bar{q} + c \cot \varphi) N_q d_{qL} s_{qL} i_{qL} b_q$$

This formula should be used in the following way. Of the two possibilities for the γ -term, the upper one should be used when $B i_{\gamma B} \leq L i_{\gamma L}$, whereas the lower one should be used when $B i_{\gamma B} \geq L i_{\gamma L}$. A check on the right choice is that $s_\gamma \geq 0.6$. Of the two possibilities for the q -term we must always choose the one giving the smallest numerical value.

In the special case of $\varphi = 0$ we must choose the smallest of the following two values:

$$Q/A \leq \frac{(\pi + 2) c_u (1 + s_{cB}^a + d_{cB}^a - i_{cB}^a - b_c^a - g_c^a)}{(\pi + 2) c_u (1 + s_{cL}^a + d_{cL}^a - i_{cL}^a - b_c^a - g_c^a)} \quad (37)$$

12. Passive Earth Pressure.

For foundations subjected to considerable horizontal forces it has always been a question, whether it is permissible to assume a passive earth pressure acting on one vertical side of the foundation. The fear has been expressed that this would require too great horizontal movements of the foundation.

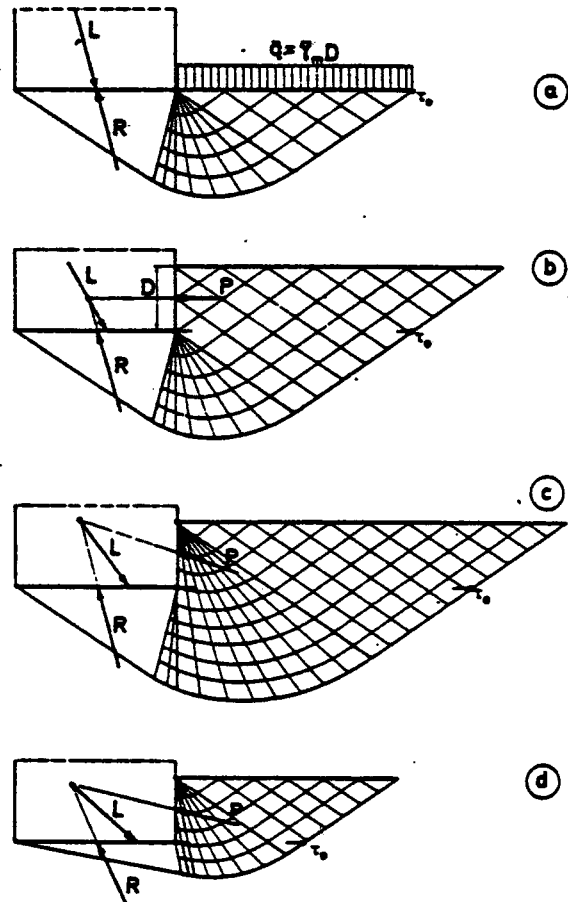


Fig. 10 Statically admissible rupture figures. Frictionless or weightless earth. Inclined load with fully developed passive earth pressure.

Fig. 10 shows first, for the case of weightless earth, the rupture-figure for a surface foundation (a). For foundations below the ground surface two statically possible rupture-figures are shown, one for a smooth side (b) and another for a rough side (c).

In the considered case of weightless (or frictionless) earth the stresses in the outermost rupture-line at base level are exactly the same as for a surface foundation. The reaction R on the base is, consequently, also the same, but in addition we have now an earth pressure P acting on the vertical side.

Therefore, due to the depth D , the foundation can actually take up a total load L , determined so as to be in equilibrium with the two forces R and P . The simplest way to take this into account is to add the earth pressure P (vectorially) to the main foundation load L . The resultant R (components V and H) can then be treated in the usual way, Q being calculated by means of the appropriate bearing capacity formula. It should be noted that depth factors must not be used, when the passive earth pressure is taken into account as proposed.

For load inclinations up to a certain value the rupture-figure shown in fig. 10c will occur. It indicates that in this case the passive earth pressure can be calculated as for a perfectly rough wall. For greater load inclinations the rupture-figure shown in fig. 10d will occur, and to this will correspond a slightly lower earth pressure. A limiting case occurs, when the base starts to slide horizontally. To this corresponds the passive earth pressure on a rough wall, which is translated horizontally.

As will appear from fig. 10, it is one and the same rupture-figure, which is responsible for both the bearing capacity and the passive earth pressure. Therefore, the passive earth pressure does not require other or greater movements of the foundation than does the bearing capacity.

On the other hand, we must limit the movements to allowable values. As regards the bearing capacity, this is usually ensured by dividing the ultimate bearing capacity with a safety factor of e.g. 2.0. For exactly the same reason we must also divide the ultimate passive earth pressure with a safety factor, e.g. 1.4.

I have already mentioned that for mainly vertical loads we use depth factors but assume no passive pressure on the side of the foundation, whereas for great horizontal loads we do the opposite.

In intermediate cases we may, as proposed by Bent Hansen, do the following. If the horizontal force H is smaller than the horizontal component of passive pressure on the whole height D , we calculate the height D_0 necessary to develop a passive pressure just balancing the horizontal force. We can then calculate the foundation for vertical load only, and with depth factors corresponding to the remaining height $D - D_0$.

13. Soil Parameters.

In case of $\varphi_u = 0$ (saturated clay) we must, of course, for c_u use the relevant undrained shear strength. For long term calculations we must use the effective parameters $\bar{\varphi}$ and $\bar{c}$ as found in drained triaxial tests.

For sand we can usually, on the safe side, assume $c = 0$. As regards φ , it has been common practice to use the friction angle φ_r found in ordinary triaxial tests. However, plate loading tests in several laboratories have shown that this leads to a severe underestimation of the bearing capacities. (Fig. 11).

Since the theoretical expressions for N_q and N_γ are developed for the case of plane strain, we must actually use the friction angle φ_p found in a plane

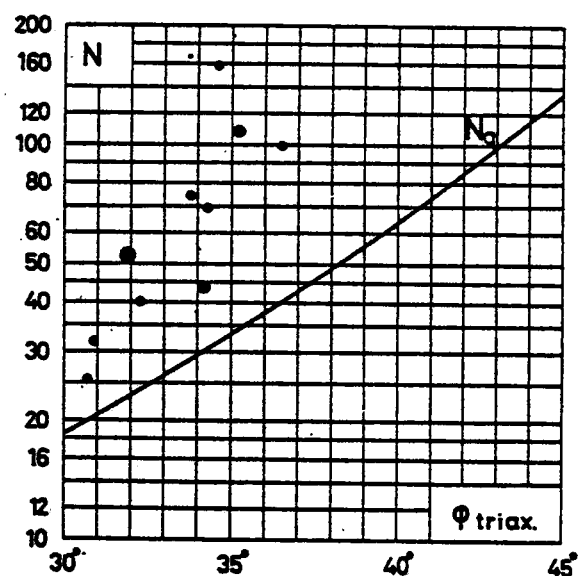


Fig. 11 Comparison of plate loading tests with calculated bearing capacities using friction angles from triaxial tests.

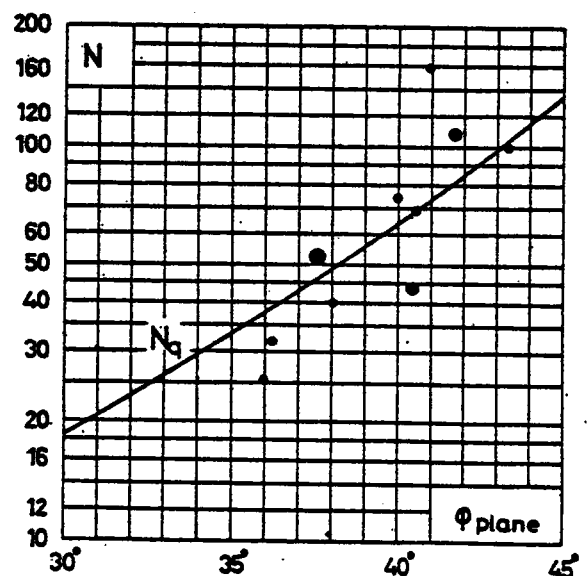


Fig. 12 Comparison of plate loading tests with calculated bearing capacities using friction angles corrected for effect of plane strain.

strain test. Such tests have been made in several laboratories with somewhat different results. In Denmark we use always in our bearing capacity calculations the following value:

$$\varphi_{\mu} = 1.1 \varphi_{\pi} \quad (38)$$

which should be on the safe side, since we have found factors up to 1.15.

This procedure has been found to give reasonable agreement between plate loading tests and the theoretical formulas. (Fig. 12).

14. Safety Factors.

The safety in bearing capacity problems is usually introduced as a total or overall safety factor F . This means that the bearing capacity Q should be calculated for the actual loads and actual soil parameters, and then the actual vertical load V on the foundation must not exceed Q/F .

In Denmark we prefer the system of partial safety factors. This means that we calculate a nominal bearing capacity Q_n for nominal loads and for nominal soil parameters defined by:

$$\begin{aligned} g_n &= g_a & p_n &= p_a \cdot f_p \\ c_n &= c_a / f_c & \tan \varphi_n &= \tan \varphi_a / f_\varphi \end{aligned} \quad (39)$$

The foundation should then be designed so that $V_n \leq Q_n$. The following values are used in Denmark:

$$f_p = 1.5 \quad f_c = 1.75 \quad f_\varphi = 1.25 \quad (40)$$

15. Example.

As a simple example we shall consider a couple of full-scale tests with foundation blocks, made by H. Muhs in Berlin.

The blocks had the dimensions $L = 2$ m, $B = 0.5$ m and $D = 0.5$ m. Base area $A = 1$ m². The soil was dense sand with $\bar{\gamma} = 0.95$ t/m³, and ground water level coincided with the ground surface. We can assume $c = 0$.

The first block was loaded centrally and vertically, and failed for $Q = 190$ t. If we estimate $\varphi_{\mu} = 47^\circ$ we find:

$$\begin{aligned} N_\gamma &= 300 & N_q &= 190 \\ d_{qB} &= 1 + 2 \cdot 1.07 \cdot (1 - 0.732)^2 \cdot \arctan(0.5/0.5) = 1.12 \\ s_{\gamma B} &= 1 - 0.4 \cdot 0.5/2.0 = 0.90 \\ s_{qB} &= 1 + 0.732 \cdot 0.5/2.0 = 1.18 \end{aligned}$$

We can now calculate the ultimate bearing capacity:

$$\begin{aligned} Q &= \frac{1}{2} \bar{\gamma} B N_\gamma d_{\gamma B} s_{\gamma B} + \bar{\gamma} D N_q d_{qB} s_{qB} = \\ &= \frac{1}{2} \cdot 0.95 \cdot 0.5 \cdot 300 \cdot 1 \cdot 0.90 + 0.95 \cdot 0.5 \cdot 190 \cdot 1.12 \cdot 1.18 = \\ &= 64 + 120 = 184 \text{ t } (\sim 190) \end{aligned}$$

If our formulas are correct, this shows that the plane friction angle was about 47° . To this corresponds a triaxial friction angle of 40° – 42° , which is quite realistic for dense sand. In fact, Muhs measured $\varphi_{\pi} = 40^\circ$.

A second block was also loaded centrally, but the load was inclined in the direction of the long sides L . At failure the forces $Q = V = 108$ t and $H_L = 39$ t. Passive pressure on the vertical side of the foundation was so small, that it could be neglected. Instead we reckon with depth factors.

Since $H_B = 0$ we find:

$$\begin{aligned} d_{qB} &= 1.12 & s_{qB} &= 1.18 & i_{\gamma B} &= i_{qB} = 1 & Bi_{\gamma B} &= 0.5 \\ d_{qB} s_{qB} i_{qB} &= 1.12 \cdot 1.18 \cdot 1 = 1.32. \end{aligned}$$

For the other direction we find:

$$\begin{aligned} d_{qL} &= 1 + 2 \cdot 1.07 (1 - 0.732)^2 \cdot \arctan(0.5/2.0) = 1.04 \\ i_{\gamma L} &= (1 - 0.7 \cdot 39/108)^5 = 0.235 & Li_{\gamma L} &= 0.47 \\ i_{qL} &= (1 - 0.5 \cdot 39/108)^5 = 0.370 & Li_{qL} &= 0.74 \end{aligned}$$

Since $Bi_{\gamma B} > Li_{\gamma L}$, we must use (25) and the lower γ -term in (36):

$$\begin{aligned} s_{\gamma L} &= 1 - 0.4 \cdot 0.47/0.5 = 0.625 \\ s_{qL} &= 1 + 0.732 \cdot 0.74/0.5 = 2.09 \\ d_{qL} s_{qL} i_{qL} &= 1.04 \cdot 2.09 \cdot 0.370 = 0.805 \end{aligned}$$

Since $d_{qL} s_{qL} i_{qL} < d_{qB} s_{qB} i_{qB}$, we must use the lower q -term in (36). Using the same values of N_γ and N_q as above, we get the ultimate, vertical bearing capacity:

$$\begin{aligned} Q &= \frac{1}{2} \bar{\gamma} L N_\gamma d_{\gamma L} s_{\gamma L} i_{\gamma L} + \bar{\gamma} D N_q d_{qL} s_{qL} i_{qL} = \\ &= \frac{1}{2} \cdot 0.95 \cdot 2.0 \cdot 300 \cdot 1 \cdot 0.625 \cdot 0.235 + \\ &= 0.95 \cdot 0.5 \cdot 190 \cdot 1.04 \cdot 2.09 \cdot 0.370 = \\ &= 42 + 72 = 114 \text{ t } (\sim 108) \end{aligned}$$

It will be seen that the new formulas explain the results quite well in this case.

	JOB TITLE:	CONTRACT NO. 52406-46	PAGE <u>VIII</u> - 0
	SUBJECT: The Computer Program CAP	SIGN. RAL	DATE 30/9-85
		CONTR.	DATE

APPENDIX VIII

SOME RECENT RESEARCH ON THE BEARING
CAPACITY OF FOUNDATIONS

G.G. MEYERHOF 1963

where B = width of foundation, D = depth of foundation, L = length of foundation, c = apparent cohesion of soil, γ = effective unit weight of soil, and N_c , N_q , and N_ϕ are bearing capacity factors which depend mainly on the angle of internal friction or shearing resistance ϕ of the soil.

The theoretical bearing capacity factors for a shallow horizontal strip foundation under a vertical load are (Prandtl, 1920):

$$\left\{ \begin{array}{l} N_c = (N_q - 1) \cot \phi \\ N_q = c^2 \tan \phi N_\phi \end{array} \right. \quad (2)$$

and, approximately, (Meyerhof, 1961b)

$$\left\{ \begin{array}{l} N_q = (N_c - 1) \tan (1.4\phi) \\ N_\phi = \tan^2 \left(\frac{1}{2}\pi + \frac{1}{2}\phi \right) \end{array} \right. \quad \text{where}$$

The bearing capacity factors for strip foundations and the partly theoretical and partly semi-empirical factors for circular or square footings (Meyerhof, 1951 and 1961a) are shown in Figure 1. Theoretical bearing capacity factors for rectangular foundations with a side ratio of B/L are not yet available, but they can be estimated by interpolating between the factors for strips ($B/L=0$) and circles ($B/L=1$) in direct proportion to the ratio B/L . Alternatively, the bearing capacity factors for rectangles can be obtained by multiplying the factors in equations (2) for strip foundations by the corresponding empirical shape factors

$$\left\{ \begin{array}{l} s_c = 1 + 0.2N_q B/L \\ s_q = s_\gamma = 1, \quad (\phi = 0) \\ s_c = s_\gamma = 1 + 0.1N_q B/L, \quad (\phi > 10^\circ) \end{array} \right. \quad (3)$$

with symbols as before.

It should be noted that for large values of the angle of internal friction ($\phi > 30^\circ$) the experimental factor N_q for circular or square foundations is smaller than that for strip foundations, whereas the opposite result is expected from bearing capacity theory. As shown previously (Meyerhof, 1961), this difference can be explained by the influence of the intermediate principal stress, which in the plane strain condition under strip foundations raises the experimental bearing capacity for large friction angles considerably above the theoretical value. While triaxial compression tests can, therefore, be used to determine the shear strength parameters for circular or square foundations, plane strain compression tests should be made to determine the corresponding parameters for strip foundations, when good agreement between the estimated and observed bearing capacities will be obtained.

For rectangular foundations the friction angles ϕ can be interpolated between the plane strain and triaxial values in proportion to the side ratio B/L of the foundation. Since the angle of internal friction in plane strain compression tests is roughly 10 per cent greater than in triaxial compression tests (Bishop, 1961; Bjerrum and Kunneneje, 1961) the friction angles for rectangular foundations are roughly given by

$$(4) \quad \phi_r = (1.1 - 0.1B/L)\phi,$$

where ϕ_r = angle of internal friction or shearing resistance from triaxial compression tests.

SOME RECENT RESEARCH ON THE BEARING CAPACITY OF FOUNDATIONS

GEORGE GEOFFREY MEYERHOF*

ABSTRACT

The first part of the paper summarizes the results of recent research on the bearing capacity of spread foundations of various shapes under a central vertical load and outlines the effects of foundation depth, eccentricity and inclination of the load. Simple formulae have been derived for use in practice and their application to the design of rigid and flexible foundations is briefly indicated.

The second part of the paper discusses the bearing capacity of single piles under vertical and inclined loads. The bearing capacity of piled foundations and free-standing pile groups is outlined, and the results of model tests on pile groups under central and eccentric loads are briefly analysed in relation to some problems in practice.

SOMMAIRE

La première partie de cet article présente un résumé des recherches récentes exécutées sur la force portante des fondations superficielles de différentes formes soumise à une charge centrée verticale; elle donne aussi un exposé des effets résultant de la profondeur des fondations, de l'excentricité et de l'inclinaison de la charge. Des formules simples y sont dérivées pour l'emploi dans la pratique, et l'application de ces formules au calcul des fondations rigides et flexibles est brièvement expliquée.

La deuxième partie discute de la force portante des pieux simples soumis à des charges verticales et obliques. La force portante des fondations sur pieux et des groupes de pieux isolés y est traitée; les résultats d'essais sur modèle réduit de groupes de pieux soumis à des charges centrées et excentrées sont analysés brièvement en rapport avec quelques problèmes pratiques.

Twenty years have passed since Terzaghi (1943) published his well-known bearing capacity theory, which with his earlier consolidation theory of clays and later settlement analysis of sands led to a better understanding of the fundamentals of foundation behaviour. Under this stimulus the study of the bearing capacity and settlement of foundations in the field amplified by laboratory research has resulted in a more rational approach to foundation design, and some research in the subsequent decade has been discussed previously (Meyerhof, 1953b). During the past decade much further progress has been made in foundation engineering, and within the limits of this paper only some research on the ultimate bearing capacity of foundations will be summarized. An attempt will be made to indicate the relationship of this research to earlier work and its application to some problems in the design of spread and pile foundations.

SPREAD FOUNDATIONS

The ultimate bearing capacity of spread foundations on uniform soils can generally be estimated with sufficient accuracy on the basis of plastic theory, and the theoretical relationship can be represented by (Terzaghi, 1943):

$$(1) \quad q = Q/BL = cN_c + \gamma DN_q + \gamma BN_\phi$$

*Head, Department of Civil Engineering, and Director, School of Graduate Studies, Nova Scotia Technical College, Halifax, Nova Scotia.

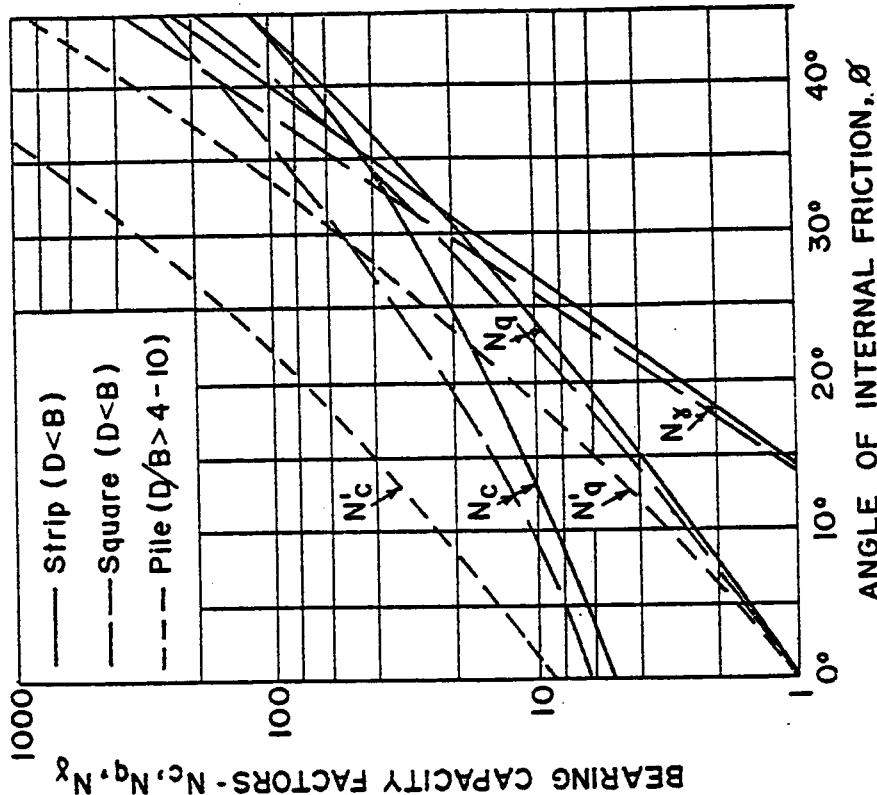


FIGURE 1. Bearing Capacity Factors for Spread and Pile Foundations

If the soil and ground-water conditions vary beneath a foundation, the average values of the effective unit weight, apparent cohesion, and friction angle of the soil within the failure zone should be used. This zone extends from ground level to a distance below the base of about the foundation width for circular or square foundations and up to about twice the foundation width for strip foundations. In the case of stratified materials having widely different soil properties, the bearing capacity can roughly be estimated from the maximum vertical stresses at the interfaces between the layers (Terzaghi and Peck, 1948) since more elaborate recent procedures on the basis of composite failure zones have not yet been sufficiently developed for practical use.

While the simple bearing capacity factors in equations (2) neglect the shearing resistance of the soil above foundation level, the corresponding

increase of the bearing capacity can be estimated from depth factors by which the individual factors have to be multiplied (Skempton, 1951; Hansen, 1955). For practical purposes the following expressions, which correspond to approximate failure surfaces and many test results (Meyerhof, 1951, 1955), are suggested for depths less than the width of the foundation:

$$(5) \quad \left\{ \begin{array}{l} d_c = 1 + 0.2 \sqrt{N_c D/B} \\ d_q = d_r = 1, \quad (\phi = 0) \\ d_q = d_r = 1 + 0.1 \sqrt{N_q D/B}, \quad (\phi > 10^\circ) \end{array} \right.$$

with symbols as before.

As the depth of the foundation increases, the depth factors increase at a decreasing rate and approach a maximum value which can be used for an estimate of the point resistance of piles, retaining walls and similar structures are

foundations subjected to eccentric and inclined loads, and these loading conditions can substantially reduce the bearing capacity of foundations. In order to allow for the eccentricity e of the resultant load R on the base of a foundation of width B (Figure 2), it was suggested (Meyerhof, 1953a) that an effective

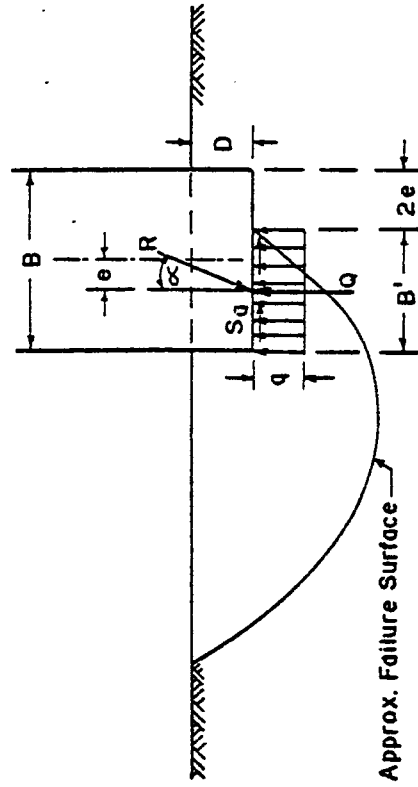


FIGURE 2. Base under Eccentric Inclined Load at Failure

foundation width B' is used in equation (1) and this width is twice the distance of the load from the toe of the foundation, i.e.

$$(6) \quad B' = B - 2e$$

Detailed model tests have shown that this procedure is on the safe side. The experimental effective width is between the above-mentioned value and the width estimated from the conventional contact pressure distribution (trapezium or triangle), which can be explained theoretically by more accurate failure zones (Hansen, 1955). For a double eccentricity of the load an effective contact area can be determined in such a way that its centre of gravity coin-

cides with the resultant of the load. In this connection a straight line of rotation across the base is required as the boundary of the effective contact area, but an approximate rectangular area of an effective width B' and effective length L' based on equation (6) in both directions is sufficiently accurate in practice.

The influence of inclined loads on the bearing capacity of spread foundations can readily be taken into account by using inclination factors, which have been deduced from more exact calculations (Schultz, 1952; Meyerhof, 1953a). In this connection the influence of the lateral soil resistance on foundations below the surface is particularly effective. For rough foundations the vertical component of the bearing capacity under a load inclined at an angle of α with the vertical (Figure 2) can be approximated by the following inclination factors (Meyerhof, 1956b):

$$(7) \quad \begin{cases} i_e = i_r = (1 - \alpha/90^\circ)^2 \\ i_r = (1 - \alpha/\phi)^2 \end{cases} \quad \text{and}$$

The vertical component of the bearing capacity in the general case of eccentric inclined loads is, therefore, approximately,

$$(8) \quad q = Q/B' L' = d_e i_e N_c + d_e i_r \gamma D N_s + d_e i_r \gamma B' N_s \frac{1}{3}$$

Model loading tests have shown that, as the inclination of the load increases, the bearing capacity of square foundations approaches that of strip foundations and the difference is negligible when failure occurs by sliding, as would be expected. Moreover, for a given inclination of the load an inclined foundation with a base normal to the resultant load gives a greater bearing capacity than a horizontal foundation (Meyerhof, 1953a), and inclined bases are therefore preferable for foundations of arch bridges. The theoretical relationships for the bearing capacity of inclined foundations are found to be similar to those for foundations located on the face of a slope or near the top edge of a slope (Meyerhof, 1957). The theory indicates that the bearing capacity of such foundations decreases with greater inclinations of the slope, especially for cohesionless soils, while for foundations on the top of clay slopes the bearing capacity decreases also considerably with greater height of the slope and is frequently governed by overall slope failure.

In order to resist an eccentric inclined load, the foundation will move and tilt sufficiently. At failure the necessary displacement in the direction of the load is approximately 5 to 20 per cent of the foundation width and the angle of rotation is about 1° to 5° according to the relative density or stiffness of the soil and the depth of the foundation. The influence of these relationships on the behaviour of structures is particularly important for statically indeterminate frames, bridges, and the lower storeys of multi-storey buildings. Model and full-scale tests on steel frames have given valuable indications and showed that the customary foundations can generally be considered as hinged bases and for fixity conditions either wide or deep foundations are required (Meyerhof, 1956b).

If the stiffness of the foundation is small in relation to the stiffness of the soil, then the problem of flexible elastic foundation slabs obtains; here calculation is difficult except under simple conditions. In practice one frequently

uses, therefore, a uniform contact pressure distribution or the method based on the modulus of subgrade reaction (Terzaghi, 1913). Recently the ultimate strength design has been extended to this case and a theory has been developed to estimate the load-bearing capacity of pavements and foundation slabs (Meyerhof, 1962a). In this way relatively simple solutions for the ultimate load under various loading conditions and slab sizes have been obtained. The proposed method can be used for both the assumption of a modulus of subgrade reaction and a modulus of deformation of the soil, and is confirmed by the results of loading tests on large slabs and model tests under concentrated loads.

PILE FOUNDATIONS

The bearing capacity of a single pile is the sum of point resistance and skin friction, both of which depend on the properties of the soil and the characteristics and method of installing the pile. Since the point resistance of vertical and inclined piles loaded in the direction of the pile is practically identical, equation (1) can be used if the bearing capacity factors include the limiting values of the depth factors, and the influence of the shape of the base and the compressibility of the soil (Meyerhof, 1959 and 1961a). For axially loaded single piles the ultimate bearing capacity is therefore, approximately,

$$(9) \quad Q = (c N_c' + \gamma D N_s') A_b + f_s A_s$$

where A_b = cross sectional area of base,

A_s = surface area of shaft,

f_s = unit skin friction, and other symbols as before.

The semi-empirical bearing capacity factors N_c' and N_s' for round or square driven piles with 60° points are shown in Figure 1. These factors apply only if the pile base is embedded in the load-bearing stratum near the base at least to a depth, which is, approximately

$$(10) \quad D = 4\sqrt{N_s' B}$$

In other cases the bearing capacity factors can be estimated by interpolating between the values for shallow foundations and piles roughly in direct proportion to the depth ratio D/B . If the soil properties vary near the base, the average values between about four times the pile diameter above the base and one pile diameter below the base should be used in cohesionless soils, while a somewhat smaller range is sufficient in clays, as indicated by the corresponding failure zone.

Although the skin friction in cohesionless soils is frequently neglected, it can be estimated for piles driven into sands by using an earth pressure coefficient of about 0.5 to 1.0 according to the relative density of the material (Meyerhof, 1951). The final value of the skin friction in clays after a few months, which governs the bearing capacity in practice, approximates the shearing strength for soft clays but hardly exceeds one-half of the shearing strength for stiff clays (Tomlinson, 1957) with an upper limit of about 1 ton/sq. ft. Detailed laboratory tests have shown (Meyerhof, 1962b) that the ratio of skin friction to shearing strength of soils depends mainly on the

REFERENCES

- BISHOP, A. W., 1961. "Discussions on 'Soil Properties and their Measurement.'" *Proc. Fifth Int. Conf. Soil Mech.* 3: 97.
- BJERRUM, L., and KJAMESSE, O., 1961. "Shearing Resistance of Sand Samples with Circular and Rectangular Cross Sections." *Norwegian Geotechnical Institute Publ.* no. 44: 1.
- EVANS, L. T., 1954. "Bearing Capacity of Piles Subjected to Horizontal Loads." *Am. Soc. Test. Mat., Spec. Tech. Publ.* no. 154: 30.
- HANSEN, J. B., 1955. "Simplified Borehole of Fundamentum Basecurve." *Ingeniøren* 64: 95.
- KERISIN, J. L., 1961. "Deep Foundations in Sands." *Proc. Fifth Int. Conf. Soil Mech.* 3: 73.
- KUZZI, A., 1957. "Bearing Capacity of Piles and Pile Groups." *Proc. Fourth Int. Conf. Soil Mech.* 2: 46.
- KISHIDA, H., 1962. Unpublished Data.
- MEYERHOF, G. G., 1951. "The Ultimate Bearing Capacity of Foundations." *Glotechnique* 2: 301.
- MEYERHOF, G. G., 1953a. "The Bearing Capacity of Foundations under Eccentric and Inclined Loads." *Proc. Third Int. Conf. Soil Mech.* 1: 440.
- MEYERHOF, G. G., 1953b. "Some Recent Foundation Research and Its Application to Design." *Struct. Engineer* 31: 151.
- MEYERHOF, G. G., 1955. "Influence of Roughness of Base and Ground-Water Conditions on the Ultimate Bearing Capacity of Foundations." *Glotechnique* 6: 227.
- MEYERHOF, G. G., 1956a. "Penetration Tests and Bearing Capacity of Cohesionless Soils." *Proc. Am. Soc. Civil Engrs.*, 82, no. SM 1: 800-1.
- MEYERHOF, G. G., 1956b. Discussion on "Rupture Surfaces in Sand under Oblique Loads." *Proc. Am. Soc. Civil Engrs.*, 82, no. SM 3: 1025-15.
- MEYERHOF, G. G., 1957. "The Ultimate Bearing Capacity of Foundations on Slopes." *Proc. Fourth Int. Conf. Soil Mech.* 1: 384.
- MEYERHOF, G. G., 1959. "Connection of Sands and Bearing Capacity of Piles." *Proc. Am. Soc. Civil Engrs.*, 85, no. SM 6: 1.
- MEYERHOF, G. G., 1960a. "The Design of Frankl Piles with Special Reference to Groups in Sands." *Proc. Symp. Pile Found.*, Stockholm: 105.
- MEYERHOF, G. G., 1960b. "Foundation Behaviour and Plastic Design of Frames." *Trans. Eng. Inst. Can.* 4: 58.
- MEYERHOF, G. G., 1961a. "The Ultimate Bearing Capacity of Wedge-shaped Foundations." *Proc. Fifth Int. Conf. Soil Mech.* 2: 105.
- MEYERHOF, G. G., 1961b. Discussion on "Foundations other than Piled Foundations." *Proc. Fifth Int. Conf. Soil Mech.* 3: 193.
- MEYERHOF, G. G., 1962a. "Load-Carrying Capacity of Concrete Pavements." *Proc. Am. Soc. Civil Engrs.*, 88, no. SM 3: 80.
- MEYERHOF, G. G., 1962b. "Some Problems in the Design of Rigid Retaining Walls." *Proc. Fifteenth Can. Soil Mech. Conf.*: 50.
- PRANDTL, L., 1920. "Über die Härte Plastischer Körper." *Proc. Kgl. Ges. Wiss. Göttingen, Math. Phys. Kl.* 74.
- SAFFERT, M., and TATE, A. P. K., 1961. "Model Tests on Pile Groups in Clay Soils." *Proc. Fifth Int. Conf. Soil Mech.* 2: 120.
- SCHULTZ, E., 1952. "Der Widerstand des Baugrundes gegen Schräge Sohldruckungen." *Ingentechnik* 29: 336.
- SKEMPTON, A. W., 1951. "The Bearing Capacity of Clays." *Proc. Build. Res. Congr., London* 1: 180.
- SOWERS, G. F., and MARTIN, C. B., et al., 1961. "The Bearing Capacity of Friction Pile Groups in Homogeneous Clay from Model Studies." *Proc. Fifth Int. Conf. Soil Mech.* 2: 155.
- STUART, J. C., HANNA, T. H., and NAVLOR, A. H., 1960. "Notes on the Behaviour of Model Pile Groups in Sand." *Proc. Symp. Pile Found.*, Stockholm: 97.
- TERZAGHI, K., 1943. *Theoretical Soil Mechanics* (New York: Wiley).
- TERZAGHI, K., and PECK, R. B., 1948. *Soil Mechanics in Engineering Practice* (New York: Wiley).
- TOMLINSON, M. J., 1957. "The Adhesion of Piles Driven in Clay Soils." *Proc. Fourth Int. Conf. Soil Mech.* 2: 60.
- TECHENITOFF, G. P., 1954. "The Resistance to Lateral Loading of Single Piles and of Pile Groups." *Am. Soc. Test. Mat., Spec. Tech. Publ.* no. 164: 38.
- WHITAKER, T., 1957. "Experiments with Model Piles in Groups." *Glotechnique* 7: 147.
- WHITAKER, T., 1960. "Some Experiments with Model Piled Foundations in Clay." *Proc. Symp. Pile Found.*, Stockholm: 124.

VOLUME CHANGES IN UNDISTURBED CLAY PROFILES IN WESTERN CANADA*

J. J. HAMILTON†

ABSTRACT

As part of an over-all study of the performance of building foundations in highly plastic soil areas of western Canada, the Division of Building Research has been measuring ground movements and changes in soil moisture conditions in grass-covered, undisturbed soil profiles under climatic conditions ranging from sub-humid to semi-arid. Results of measurements begun in 1951 in Winnipeg, Manitoba, and more recently those in Regina, Eston, and Tisdale, Saskatchewan, are reported. Results of a theoretical soil moisture depletion calculation, based on Thornthwaite's potential evapo-transpiration concept, are presented. It is proposed as a more rational way of measuring the vegetation-climate factor in humid to sub-humid climates than simply comparing air temperature and precipitation with long-term averages. Empirical relationships are suggested between calculated soil moisture depletion, the depth of free water table, and the depth of frost penetration under similar thermal conditions but different soil moisture conditions. Shrinkage *in situ* of undisturbed, unsaturated soils at temperatures well below 32° F has been observed and is attributed to thermal air-void volume change.

SOMMAIRE

Dans le cadre d'une étude d'ensemble concernant le comportement des fondations dans les régions de l'Ouest ayant des sols extrêmement plastiques, la Division des recherches en construction a effectué des mesures relatives aux mouvements du sol et aux changements des conditions d'humidité du sol dans des profils de sols non remués et recouverts d'herbe. Ces mesures sont faites dans des conditions climatiques allant de la sous-humidité à la semi-aridité. On donne dans l'étude les résultats de mesures effectuées en 1951 à Winnipeg au Manitoba et plus récemment à Regina, à Eston, et à Tisdale en Saskatchewan. On trouvera également les résultats d'un calcul théorique sur l'épuisement de l'humidité du sol fondé sur le concept de l'humidité du sol fondé sur l'évapotranspiration possible. Ce concept est suggéré comme une méthode plus rationnelle de mesure du facteur végétation-climat dans les zones climatiques de sous-humidité que la simple comparaison de la température de l'air et des précipitations avec les moyennes à long terme. Des relations empiriques sont suggérées entre l'épuisement calculé de l'humidité du sol, la profondeur de la pénétration du gel dans des conditions thermiques semblables unis dans des conditions d'humidité du sol différents. Une contraction *in situ* de sols non remués et non saturés à des températures bien inférieures à 32° F a été observée et elle a été attribuée à un changement de volume des espaces vides par voie thermique.

Movements of engineering structures founded near the surface of deposits of swelling and shrinking clay soils constitute a serious problem in large areas of western Canada. The severity of damage, and therefore public awareness, increases during periods of extremes in weather conditions and wanes during periods of more or less average weather conditions. Because these movements are closely linked with cycles in climate, engineers and architects cannot afford to overlook this important factor in the design of structures for these areas.

Although the problem has been recognized for many years, relatively little factual field evidence has been obtained to illustrate the magnitude of ground

*Presented at the Sixteenth Canadian Soil Mechanics Conference, September 12-14, 1962, Edmonton, Alberta.
†Research Officer, Prairie Regional Laboratory, Division of Building Research, National Research Council, Saskatoon, Saskatchewan.

can be explained by the overlapping of the individual zones of shearing deformation in the soil leading to local and progressive shearing failure at customary pile spacings, so that for a given spacing the bearing capacity per pile in a group decreases as the size of the group (number and length of piles) increases.

On the other hand, tests on free-standing model pile groups in loose sand (Kedzi, 1957; Stuart *et al.*, 1960) showed that for a pile spacing of about two pile diameters the total bearing capacity is more than twice the full bearing capacity, which is approached by about six or seven pile diameters (Figure 5). This result is supported by theoretical estimates of the compaction and deformation of loose sand by pile driving to which the effect of an overlapping of the individual soil failure zones has to be added at small pile spacings (Meyerhof, 1959). Thus it was found that for a given pile spacing the bearing capacity per pile in a free-standing group in loose sand increases as the size of the group increases.

In the absence of test results for eccentric and inclined loads the limited data on pile groups summarized above indicate that the bearing capacity of free-standing groups with a small pile spacing can be estimated as for an equivalent pier foundation. For a greater pile spacing, however, the total bearing capacity depends both on the compression and tension piles and on the lateral earth pressure (Figure 4) for which simple methods of analysis have recently been suggested for groups of vertical and inclined piles (Meyerhof, 1960a).

CONCLUSIONS

Although the ultimate bearing capacity of spread foundations can usually be estimated from customary theory and soil tests with sufficient accuracy, recent research has led to a better agreement with observations by improvements of both theory and soil testing. Thus, for strip foundations the theory should be used in conjunction with the shearing strength of the soil obtained from plane strain compression tests, while for circular or square foundations the results of triaxial compression tests apply. The influence of foundation shape and depth, the eccentricity and inclination of the load, the effect of ground-water conditions and sloping ground can readily be taken into account by recent extensions of bearing capacity theory. However, further research is required to deal with variable soil conditions.

The ultimate bearing capacity of single piles can be estimated theoretically for simple cases of loading and uniform soils. Eccentric and inclined loads and variable soil conditions do not yet permit accurate estimates. Moreover, for pile groups only limited information of the ultimate load is available at present from full-scale tests, although the results of model tests give some indication of their behaviour. These tests indicate that piled foundations can generally be treated as piers in practice, but further research is required for eccentric and inclined loads as well as variable soil conditions. More difficult is the problem of free-standing pile groups since the effect of overlapping zones of shearing deformation and volume changes of the soil in the field cannot be deduced from model tests and more full-scale observations are required.

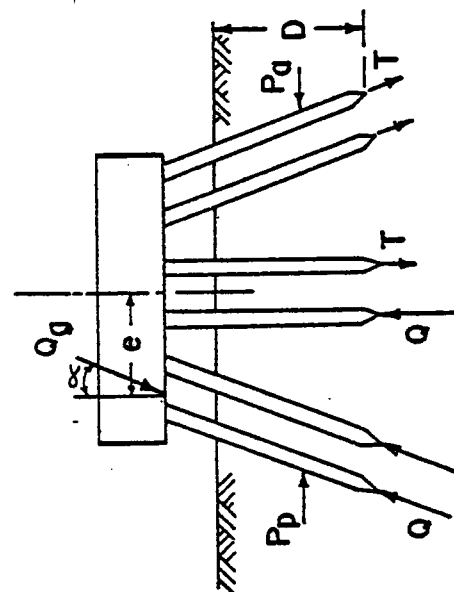


FIGURE 4. Free-Standing Pile Group at Failure

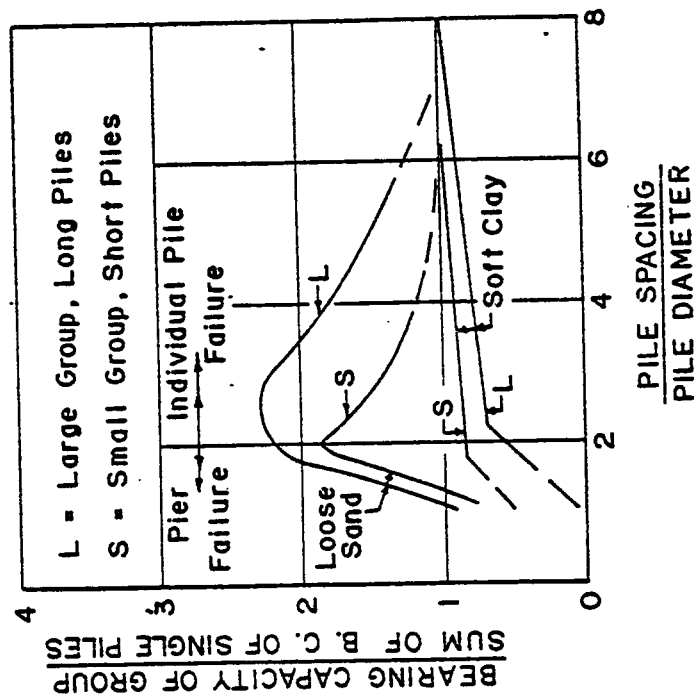


FIGURE 5. Typical Results of Model Tests on Free-Standing Pile Groups

type and roughness of the pile material, and this ratio is about 0.6 to 0.8 for steel, 0.7 to 0.9 for wood (parallel to the grain) and 0.8 to 1.0 for concrete.

For piles driven into cohesionless soils, static cone penetration tests are particularly suitable for an estimate of the bearing capacity, although for large pile diameters the point resistance is more in loose sand and less in dense sand than the cone resistance, which seems to be due to the volume changes associated with the deformation of the soil in the failure zone (Kerisel, 1961). A rough estimate of the bearing capacity of piles in such soils can also be obtained from the results of standard penetration tests (Meyerhof, 1958a).

More difficult is an estimate of the bearing capacity of piles which are not axially loaded. While a pile with transverse load can be designed similarly to a wall but with the additional soil resistance on the sides parallel to the load (Terzaghi, 1943), the bearing capacity of an obliquely loaded pile is between the transverse and longitudinal value. By the inclined loading the lateral earth pressure governing the skin friction is increased and the axial component of the bearing capacity governing the point resistance is reduced, as for a deep foundation. Hence the bearing capacity of batter piles is smaller if the load is more inclined than the batter of the pile, although the bearing capacity can be increased by fixity at the pile head, as shown by the results of tests in sand (Evans, 1954; Tschobortoff, 1954). For the purpose of estimating the bearing capacity of such piles the relationships can conveniently be represented by polar bearing capacity diagrams (Meyerhof, 1960a).

Recent investigations of the group action of piles and ultimate bearing capacity of pile foundations have shown that two types of pile groups have to be distinguished, namely pile groups in which the caps or foundations rest on load-bearing soil (piled foundations) and pile groups with caps clear of the ground or with piles driven through soft strata into dense soil (free-standing pile groups).

A piled foundation (Figure 3) can be considered as a pier foundation with its base at the depth of the pile points, and the total bearing capacity is practically independent of the pile spacing. Although at present only model tests with a central vertical load on piled foundations in clay are available (Whitaker, 1960), some exploratory loading tests on vertical pile groups with a close pile spacing in sand (Kishida, 1962) indicate that similar conditions probably obtain generally for piled foundations in uniform soils. For a central load on such foundations with vertical piles at customary spacings up to about four or five pile diameters the total bearing capacity is thus the sum of the base resistance of the equivalent pier and the shearing strength of the soil along the perimeter of the group less the weight of the enclosed soil (Terzaghi and Peck, 1948).

For eccentric or inclined load the foundation tilts and the earth pressure on the sides of the group and the base shear should generally be taken into account (Figure 3). Exploratory loading tests on model piled foundations under vertical load with eccentricities up to one-half the width of the group indicate that the reduced vertical base resistance is offset by an increased skin friction. As a result the total bearing capacity of the group is practically unaffected by the eccentricity of the load, within the above experimental limit, and can be estimated as for piled foundations under a central load.

However, if the foundation width is of the same order of magnitude as the pile length, the lateral earth pressure is relatively small and can be neglected. The calculation is then similar to that of a spread foundation in which only compression piles are loaded and tension piles are ignored by using an effective base area (Meyerhof, 1960a). No loading tests to failure appear to have been made on piled foundations with inclined piles under various loading and soil conditions. An estimate of the bearing capacity can be made in a similar way as for groups with vertical piles but allowing for the horizontal component of the pile resistance, and an approximate procedure has been suggested before (Meyerhof, 1960a).

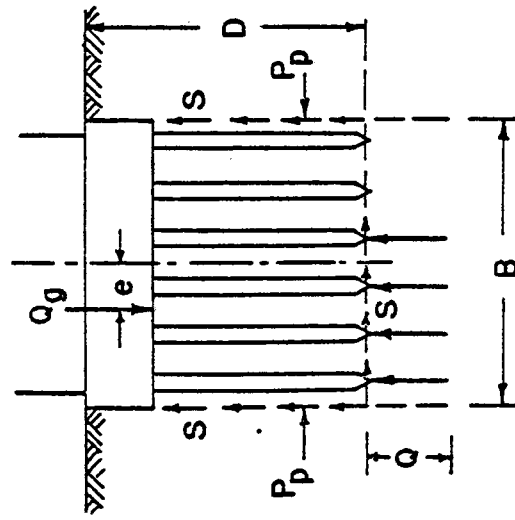


FIGURE 3. Piled Foundation at Failure

The total bearing capacity of a free-standing pile group (Figure 4) is the smaller amount of either the bearing capacity of an equivalent pier foundation or the sum of the bearing capacities of the individual piles. It is found that for a pile spacing of less than about two or three times the pile diameter the individual failure zones in the soil around the piles intersect. This overlap produces soil arching between the piles and leads to pier action of the group so that its bearing capacity is the same as that of a piled foundation. For a greater pile spacing the individual action of the piles governs, although the deformation and any volume changes of the soil near the piles have to be considered in estimating the total bearing capacity.

Model tests on free-standing vertical pile groups in clay with central vertical load (Whitaker, 1957; Sowers and Martin, 1961) and with small eccentricities of the load (Saffery and Tate, 1961), which did not affect the total bearing capacity, showed that for a pile spacing of about two pile diameters the group capacity is only about two-thirds of the full bearing capacity reached at a spacing about seven or eight pile diameters (Figure 5). This difference